THREE ESSAYS IN ENVIRONMENTAL ECONOMICS: AGRICULTURAL WAGES, CROP YIELDS, AND RECREATIONAL SHELLFISHING

by

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"Recite in the name of your Lord who created—Created man from a clinging substance."

Qur'an [96:1–2]

First, I thank Allah, the Creator of the universe, for granting me the strength and opportunity to pursue my PhD. I dedicate this dissertation to peace-loving people around the world, especially those deprived of their homeland in the name of establishing peace.

I am deeply grateful to my advisor, Dr. James Rising, for his guidance, patience, and support throughout this journey. Whenever I felt stuck, he helped me look in another direction and find a way forward.

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May Allah accept this work and guide me to use my knowledge in service of humanity. I pray for a life pleasing to Him and, when my journey ends, the welcome promised in these verses:

[To the righteous it will be said], "O reassured soul, Return to your Lord, well-pleased and pleasing [to Him], And enter among My [righteous] servants, And enter My Paradise." Qur'an [89:27–30]

Until that day, there is work to do and promises to keep. I carry Robert Frost's words with me:

But I have promises to keep,
And miles to go before I sleep.

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ABSTRACT

This dissertation presents three essays in environmental economics on agricultural wages, crop-yield losses under climate change, and the value of recreational shellfishing.

The first essay asks how agricultural cash wages adjust to crop conditions, and whether that adjustment differs between jobs that provide meals and jobs that do not. Monthly district wage records from Bangladesh for 2017–2025 allow a comparison of jobs providing no meals with jobs providing three meals per workday. In the pooled baseline specification that instruments rice yield with growing-season weather, a 10 percent higher yield is associated with cash wages about 2 percent lower. One possible explanation is that greater crop supply lowers the price farmers receive without raising the output of an additional worker-day. Weather predicts yield weakly and may also affect work directly, and the estimate is sensitive to seasonal controls and the inclusion of particular years. Descriptively, cash wages in harvest months are about 6 percent higher than in other months in no-meal jobs and about 5 percent higher in three-meal jobs. Three-meal jobs pay less cash, yet their cash pay rises at harvest nearly as much: lower cash pay does not imply smaller seasonal changes in that pay.

The second essay estimates a crop-specific hybrid model for projecting climate-related yield losses: a weighted average of one fitted empirical weather–yield component and three calibrated process-based crop components. Nonnegative weights estimated on matched subnational data determine the combination for maize, rice,

and wheat. In regions omitted from estimation, the hybrid predicts better than the empirical model alone for every crop, but it is not the most accurate predictor for any crop. Under the higher-forcing climate scenario SSP3-7.0, median hybrid yield changes for 2096–2100 are −16.6% for maize, −47.8% for rice, and −10.7% for wheat relative to 2015. These projections use three climate models. The empirical component applies its 1978–2016 weather response; the crop simulations fix management inputs and atmospheric CO_2 at 2015 levels. Short-run responses are assumed to apply over the long run, with no additional adaptation or future yield or technology trend. The weights and calibration reduce projected losses relative to the empirical model alone most for maize and wheat; the wheat difference disappears without calibration's rescaling of the simulated responses.

The third essay values recreational crabbing and clamming in the broader Mid-Atlantic region. A 2026 survey of Delaware fishing-license holders assigns hypothetical cost increases at random and asks whether participants would still have taken a reported 2025 trip. Nonparametric estimates give lower-bound mean willingness to pay of about $66 per crabbing day and $58 per clamming day. The estimates change when reported confidence and exposure to an earlier valuation question are taken into account. These activity-specific day values describe participants reached through the license-holder survey and provide local evidence for decisions about recreational shellfishing access.

Chapter 1

IN-KIND PAYMENT AND AGRICULTURAL CASH WAGES: SEASONAL AND YIELD VARIATION IN BANGLADESH

Muhammad Taky Tahmid

Abstract

Higher crop yields can change the amount of agricultural work available and the value of a day's labor. Workers may also receive in-kind pay—goods or services, such as meals, alongside cash wages—which may accompany longer employment relationships. This chapter asks how agricultural cash wages vary with rice yields and whether that relationship differs between jobs with and without meals. Using Bangladesh district records for 2017–2025, we compare jobs providing no meals with jobs providing three meals per workday and estimate the yield–wage relationship using growing-season temperature and rainfall as instruments for rice yield. In the pooled baseline specification, a 10 percent higher yield is associated with cash wages about 2 percent lower. One possible explanation is that greater crop supply lowers the price farmers receive without raising the output of an additional worker-day. The estimate comes with three caveats: weather predicts yield weakly after accounting for district and year differences; it may affect wages through channels other than yield; and the result is sensitive to more detailed seasonal controls and the inclusion of particular years. A separate descriptive comparison shows that cash wages are about 6 percent higher during harvest months than in other months in no-meal jobs and

about 5 percent higher in three-meal jobs. Lower cash pay in meal jobs therefore does not come with markedly smaller seasonal changes in that pay.

JEL codes: Q54, J41, J31, J43, O13

Keywords: agricultural wages, in-kind pay, rice yields, Bangladesh

1.1 Introduction

Workers may receive part of their pay as goods or services rather than cash. This in-kind payment can include meals or food rations, housing, clothing, fuel, transport, utilities, childcare, or subsidized loans (International Labour Organization, 2015). Such payments change what workers receive, but they do not by themselves tell us how the cash part of pay adjusts when work becomes more or less valuable.

Agriculture in developing economies is a natural setting for this question, because the value of farm work changes with the crop calendar and with crop conditions. Research on agricultural wages links wage changes to weather, production, and workers' employment alternatives (Jayachandran, 2006; M. R. Rosenzweig & Udry, 2014; Emran & Shilpi, 2018; Kaur, 2019). Payment arrangements also differ: some agricultural workers are hired by the day, while others maintain longer relationships with the same employer (Bardhan, 1983; Eswaran & Kotwal, 1985). Meals may accompany these longer relationships, but receiving meals alone does not show whether a job provides continuing employment or smaller changes in daily cash pay. The wage studies cited here measure an average or representative wage, so they cannot show whether cash wages adjust in the same way in jobs with and without meals.

This chapter asks how agricultural cash wages adjust to crop conditions, and whether that adjustment differs when part of pay is provided as meals. The first part of the question matters on its own. A larger crop is often expected to raise farm wages, but higher output per unit of land need not raise the value of an additional

day of work: if the extra crop mainly requires more worker-days, and the larger supply lowers the price farmers receive, that day can be worth less. The second part concerns payment arrangements. If meals accompany arrangements that keep cash pay steadier, or if meal values move with food prices, cash wages in meal jobs could adjust differently from cash wages in jobs paid only in cash.

We study the question in Bangladesh, where the Bangladesh Bureau of Statistics (BBS) reports monthly district cash wages by gender and by whether employers provide zero, one, two, or three meals per workday. Using records for 64 districts from 2017 to 2025, we compare jobs paid only in cash with jobs that also provide three meals a day, the clearest contrast in meal provision. The main analysis relates crop-season average cash wages to district rice yield, using growing-season weather as an instrument for yield. It first estimates a pooled benchmark that imposes one yield slope on both meal categories, and then allows the slope to differ between them. A simpler, descriptive comparison of the harvest months with other months of the same district, year, and gender shows how cash pay moves in the predictably busy months of each year. The calendar comparison captures recurring seasonal wage differences; the yield comparison relates crop-season average wages to output per unit of land across seasons and years. Their estimates need not have the same sign, and the harvest comparison does not identify the effect of yield.

In the pooled benchmark, higher rice yields are associated with lower cash wages: the estimate of -0.200 implies that a 10 percent higher yield goes with cash wages about 2 percent lower (Table 1.2). This pattern is consistent with the price account above, although these data cannot test that account. The estimate comes with specific caveats: weather predicts yield weakly after the controls and may also affect farm work directly, and the negative estimate is sensitive to more detailed seasonal controls and to excluding 2024 (Section 1.7.1). When the slope is allowed

to differ between meal categories, the estimated difference is imprecise, so the data cannot tell whether meal jobs respond to yield differently, and the pooled benchmark does not show that their responses are the same.

The harvest comparison gives a descriptive result. Cash wages in harvest months are about 6 percent higher than in other months in no-meal jobs and about 5 percent higher in three-meal jobs (Table 1.3); one- and two-meal jobs show a similar increase. The estimated gap is small, although its confidence interval allows differences in either direction, and the pattern is similar when we allow for a trend in relative wages over the year or use only records that cover all twelve months (Section 1.7.3). Three-meal jobs pay less cash, but their harvest-related cash-wage increase is nearly as large. This does not explain why the increases are similar, show that they are equal, or show that providing meals has any effect on wages.

The chapter adds to work on agricultural wages in two ways. It finds that, in the baseline specification, higher rice yields go with lower rather than higher cash wages, a pattern that a price-based account could explain, although the data cannot test that account. And using records that separate payment categories, it compares this relationship across meal categories and shows that lower cash pay in meal jobs does not come with a markedly smaller seasonal rise in that pay. The evidence concerns recorded daily cash wages. Because the records do not value meals or follow individual workers, they cannot establish differences in employment contracts, total compensation, or annual earnings.

Section 1.2 reviews related work, and Section 1.3 describes Bangladesh's rice seasons and wage records. Section 1.4 sets out the conceptual framework, including how rice yield could raise or lower the value of harvest work. Sections 1.5 and 1.6 describe the data and the two comparisons. Section 1.7 reports the yield analysis and

then the descriptive harvest comparison, Section 1.8 discusses what the comparisons and the records cannot show, and Section 1.9 concludes.

1.2 Literature Review

Existing research links agricultural productivity to wages, while showing that workers' employment alternatives shape the response. Jayachandran (2006) finds that rainfall-driven productivity raises wages in Indian districts, with larger responses where workers have less access to banking or migration. In Bangladesh, Emran and Shilpi (2018) find that favorable rainfall raises crop productivity and wages even as the share of hired labor falls. Wage rigidity also matters: Kaur (2019) finds that Indian village wages rise after positive rainfall shocks but resist falling after droughts. Bangladesh studies model wage determination and examine its links to rice production and prices (Boyce & Ravallion, 1991; Palmer-Jones & Parikh, 1998). Those price links need not be strong in every period. Using monthly division-level data for 1994–2014, Hassan and Kornher (2022) find no short-run transmission of rice-price shocks to farm wages in six of eight divisions after a break around 2007–2009. This cautions against assuming ready price transmission, although their study does not cover our later period.

These findings help answer how agricultural wages respond to crop conditions, but leave open whether the yield–wage relationship differs across payment arrangements. The empirical studies cited above examine an average or representative wage series rather than compare yield responses in jobs paid only in cash and jobs that also provide meals. A response estimated for one wage series need not apply to both categories. Two questions follow: how cash wages respond to rice yields in this setting, and whether that response differs when part of pay is provided in kind.

Employment models and research on in-kind pay give reasons to ask this question. Models of agrarian labor markets distinguish workers hired by the day from workers with longer relationships with an employer (Eswaran & Kotwal, 1985; Bardhan, 1983). Employers may secure workers for the peak season in return for steadier employment, although steadier employment need not mean a steadier daily wage. In Bardhan's model, labor tying may increase with yield-raising improvements and tighter labor markets. Payment in kind can also respond to risk: Datta, Nugent, and Tishler (2004) model the cash–kind mix in casual agricultural wages as a response to risk, and Ito and Kurosaki (2009) find that Indian farm households increase their labor supply more in in-kind wage work than in cash wage work when weather risk rises. This work establishes why payment arrangements matter, but does not show whether cash wages in meal jobs respond differently to yields. Nor does it establish that meal provision identifies a longer employment contract.

This chapter examines these questions using Bangladesh wage records that distinguish jobs providing no meals from jobs providing three meals per workday. Its negative baseline yield–wage association contrasts with the positive productivity–wage relationships reported above and is consistent with the possible price explanation in Section 1.4.2. Our contribution is to examine yield-related and recurring seasonal wage changes across recorded payment arrangements, with the pooled yield relationship as the main investigation and the harvest comparison as supporting descriptive evidence. The meal categories distinguish forms of payment, not observed contract types or an assigned treatment.

1.3 Background: Rice Seasons and Agricultural Pay in Bangladesh

The setting must provide recurring wage variation and distinguish the cash component of different payment arrangements. Bangladesh's rice calendar and

recorded meal categories provide both. The calendar defines the harvest comparison, the separate rice seasons define the units of the yield and weather comparison, and the meal categories identify the observed form of payment.

Bangladesh has three partly overlapping rice crops. In the crop windows used here, irrigated dry-season *Boro* runs from January to May, *Aus* from April to August, and *Aman* from June to November. Boro accounts for roughly 55 percent of annual rice output (Bangladesh Bureau of Statistics, 2024). A district can therefore go through several production cycles in one calendar year, each with different weather and periods of farm work.

Aus has the highest average extreme heat and seasonal rainfall in the panel, whereas Boro is much drier and relies on irrigation (Table 1.1, Panel D). Aman grows through the monsoon and faces a different combination of heat and rain. Figure 1.1 shows these climate differences alongside rice yields. Boro yields are consistently higher, and yields change over time in all three seasons. An annual district average would combine crops grown under different conditions and obscure the timing of changes in output. We therefore keep district–season observations separate.

Hired farm work follows the same calendar. Earlier Bangladesh evidence describes high demand for hired labor around the Aman harvest and Boro transplanting, with a lean period before the Aman harvest (Khandker, Samad, & Badruddoza, 2017). Dates and intensity vary by place and year, and workers may seek other employment between the busy periods. In the BBS wage records, cash wages are highest in April and May, when Boro is harvested, and in November and December, when Aman is harvested (Figure 1.3). For the harvest comparison, we use these fixed calendar months rather than weather-dependent harvest dates.

The BBS records monthly district *cash* wages by gender and by zero, one, two, or three employer-provided meals per workday (Bangladesh Bureau of Statistics,

2025). These rows show whether food accompanies cash pay, but they do not give meal values or the duration of employment. We focus on no-meal and three-meal jobs because their recorded payment arrangements differ most clearly, and we compare how their cash wages move over the agricultural year and with seasonal rice yields.

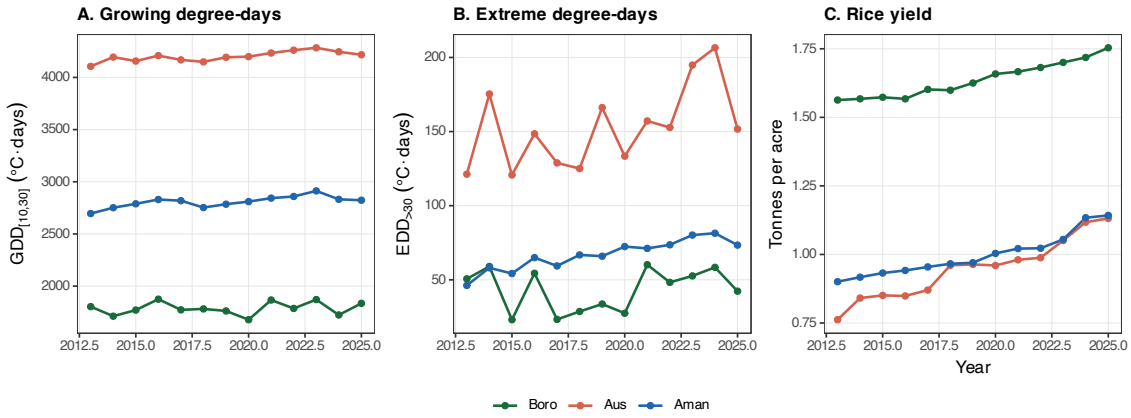


Figure 1.1: Climate and Rice Yields by Growing Season, 2013–2025. District-mean growing degree-days ($GDD_{[10,30]}$), extreme degree-days ($EDD_{>30}$), and rice yield (tonnes per acre) by year. Boro is the irrigated dry season; Aus faces pre-monsoon heat; Aman is monsoon rice. The series show the different growing conditions behind the season-specific weather–yield regressions.

1.4 Conceptual Framework

This section uses a simple pay benchmark and conditional predictions to interpret the wage estimates. Crop conditions can change the value of farm work (Jayachandran, 2006; Emran & Shilpi, 2018); we consider how that change reaches cash pay, first through yield variation and then through the recurring harvest season.

1.4.1 The cash component of an in-kind wage

Let V be the value of a worker's day of labor to the employer, q the value of the in-kind component per workday, and w the cash wage. In a competitive spot-market benchmark, total daily pay equals the value of labor, so

$$w = V - q. \tag{1.1}$$

In a no-meal job $q = 0$ and $w^0 = V$. In a three-meal job q is the total value of all three meals served in a workday, and $w^3 = V - q$. This benchmark assumes comparable jobs face the same value of labor; the records do not establish that assumption. Changes in cash pay are

$$\Delta w^0 = \Delta V, \qquad \Delta w^3 = \Delta V - \Delta q, \tag{1.2}$$

so the three-meal response differs from the no-meal response by $-\Delta q$.

With fixed meal values, a rise in V raises both cash wages by the same amount in taka, giving the lower-paid three-meal job a larger percentage increase. A smaller meal-job increase requires q to rise, for example through larger or more expensive meals; fixed or falling meal values cannot explain it under this benchmark.

The source of the change in V matters for q. Holding meal quantity and quality fixed, an external rice-price increase that raises both the value of farm work and meal costs makes three-meal cash pay rise by less in taka. If increased crop supply instead lowers food prices, meal costs fall. The change in three-meal cash pay is then more positive in taka than the change in no-meal pay, although either cash wage may still fall. Meals contain more than rice, so their value need not follow rice prices one for one.

1.4.2 Rice yield and the value of harvest labor

Yield is rice output per unit of land (tonnes per acre in the data), not total production or output per worker. Write $V = P\,a$, where P is the price farmers receive and a is the extra rice harvested with an additional worker-day. Holding the workday fixed because hours are unobserved, this expression separates two channels into the value of labor.

A denser crop may allow an additional worker-day to bring in more rice, raising its value at a fixed price. But harvesting the larger crop may mainly require more worker-days without increasing that extra output. Where workers are scarce, an additional day can also prevent losses from late harvesting; where workers are readily available, its contribution may change little. Higher yield per acre therefore need not raise the value of an additional worker-day.

A price decline provides a second channel, conditional on higher yields expanding supply enough in the relevant market to lower the price farmers receive. If the extra output from an additional worker-day changes little while the price falls, V falls and the benchmark predicts a lower cash wage. Cheaper food may also reduce q, offsetting part or all of this decline in three-meal jobs. Their cash wages may then fall less than no-meal wages or may rise. This is why an external price increase and a yield increase that lowers prices need not produce the same wage responses.

Changes in V concern the pooled yield–wage relationship; changes in q help interpret differences across meal categories. Pooling imposes, rather than establishes, equal slopes.

The channels interpret crop-window average wages, not identified harvest workers' pay. Additional worker-day output and producer prices are unobserved, so the channels cannot be tested here. A price account must also explain wage variation remaining after year–meal effects; the relevant price variation could be

local or reflect exposure to broader price changes. The empirical strategy discusses anticipated yields and direct weather effects.

1.4.3 Payment arrangements and harvest wage predictions

Agrarian labor markets combine daily hiring with longer relationships between workers and employers (Eswaran & Kotwal, 1985). Labor-tying models explain how employers secure workers for the peak season while workers obtain steadier employment (Bardhan, 1983). If meals accompany such arrangements and those arrangements steady daily cash pay, meal-job wages should rise less when harvest work becomes more valuable. This prediction is conditional: meals do not identify contracts, and secure employment can steady annual income without steadying the daily wage. Changes in the workers or jobs represented in each published wage row could also produce different wage patterns.

Insurance against unexpected shocks in implicit-contract models (Baily, 1974; Azariadis, 1975) is distinct from this predictable seasonal pattern; in-kind pay can also respond to risk (Datta et al., 2004).

The harvest comparison uses the Boro and Aman harvest months, whose timing follows the crop calendar and which earlier Bangladesh evidence links to high demand for hired labor (Khandker et al., 2017). Seasonal smoothing predicts a smaller meal-job increase; fixed meal values predict equal increases in taka and a larger percentage increase. Unobserved meal values mean similar percentage increases cannot distinguish these accounts. The comparison describes cash-wage changes, not labor demand, contract type, or insurance. In both the yield and harvest analyses, the *meal gap* denotes the three-meal response minus the no-meal response.

1.5 Data and Summary Statistics

Both comparisons use BBS monthly district cash wages for 2017–2025, reported by gender and by the number of meals provided. The harvest comparison uses them at the district–month–year level. The yield comparison averages them over the rice seasons and matches them to BBS district rice yield and to ERA5 weather. We deflate wages with the BBS consumer price index. The appendix documents the construction and coverage of these data.

1.5.1 Agricultural cash wages

The BBS *Yearbook of Agricultural Statistics* reports nominal daily agricultural cash wages by district, month, and gender (Bangladesh Bureau of Statistics, 2025). Each row records whether employers provide zero, one, two, or three meals per workday. The meal category describes how a recorded job is paid; the yearbooks do not record employment contracts, hours, or the value of the meals. We compiled the 2017–2025 series from these tables and checked each district and month against the published values. We deflate wages using the BBS national annual CPI (base 2021/22). This adjustment does not account for differences in prices across districts.

The monthly harvest sample contains 16,663 wage records by district, month, year, gender, and meal category, restricted to no-meal and three-meal jobs in 64 districts. The baseline regression uses 16,605 observations after removing 58 records that are the only observation in their district–year–gender–meal group. For the yield comparison, we average monthly wages within each district's Boro, Aus, and Aman crop windows. The resulting no-meal and three-meal seasonal sample contains 4,140 cells after removing fixed-effect singletons.

Coverage differs between the monthly wage records and the seasonal yield panel. In the seasonal panel, no-meal records begin in 2017, while the one-, two-, and three-meal categories begin in 2018; the monthly file already contains some meal-category records in late 2017. The seasonal panel records three-meal wages in 46 districts and no-meal wages in 64, and female wages are missing from it in three districts. Each wage cell is a published district average, and the unbalanced panel does not follow individual workers. The appendix maps the distribution of three-meal records. Its shares measure the coverage of wage records, rather than employment shares.

Within a year, three-meal wages are reported in fewer months than no-meal wages. A district–year–gender cell has a three-meal wage in 8.8 months on average, compared with 11.4 months for zero meals, and 53 percent of three-meal cells cover all twelve months, compared with 87 percent of no-meal cells. Some consecutive monthly values are unchanged: 23 percent of month-to-month changes in three-meal wages are exactly zero, compared with 18 percent for zero meals. The harvest results are similar when we restrict the sample to cells observed in all twelve months.

1.5.2 Rice yield and weather

Rice yield is production divided by area, measured in tonnes per acre using seasonal district data from the BBS yearbooks (Bangladesh Bureau of Statistics, 2024). It is output per unit of land, not total production or output per worker; the data contain no measure of how much rice a worker harvests in a day. ERA5 supplies hourly temperature and precipitation for the same districts, seasons, and years (Hersbach, Bell, Berrisford, et al., 2020). Following Schlenker and Roberts (2009a), we construct growing degree-days between 10 and 30°C and extreme degree-days above 30°C within each crop window. We approximate within-day temperature with a sinusoidal curve

and area-weight the indices to districts. Precipitation is the seasonal total. Yield and weather data cover 2013 to 2025. Each district's rice area averaged over 2013–2016, before the wage sample begins, measures its rice intensity. For the price association in Appendix Table A22, we use WFP/HDX district retail coarse-rice prices averaged over harvest windows (World Food Programme, 2026). These are retail market prices, not the prices farmers receive. The exploratory monthly check in Appendix Table A21 uses the same source, matched by district and month.

1.5.3 Summary statistics

Table 1.1 describes the four-category seasonal wage panel and the yield and weather data. Mean real cash pay is 417 taka per day: 434 taka in no-meal cells and 403 taka in three-meal cells. The female and male means are 347 and 465 taka. Within the same district, month, and gender, three-meal cash wages average about 83 percent of no-meal wages.

The national means in Figure 1.2 show rising rice yields in each season over 2017–2025. Real cash wages fall in 2020, recover through 2022–2023, and fall again in 2024–2025. These movements are similar across seasons and meal categories, but parallel national trends do not show how cash wages in different payment categories move with the harvest or with yields. The figure describes unadjusted averages; the regressions use variation within districts and years.

Table 1.1: Summary Statistics: Wage Panel and Climate (Bangladesh, 2017–2025)

	N	Districts	Mean	SD	Mean log
A. Real daily wage by gender					
Female	3,178	61	347.0	66.6	5.83
Male	4,530	64	465.2	76.8	6.13
B. Real daily wage by meal provision					
Zero meals	3,096	64	434.3	96.7	6.05
One meal	2,140	63	406.3	90.9	5.98
Two meals	1,394	52	403.2	88.1	5.97
Three meals	1,078	46	402.7	85.1	5.97
C. Real daily wage by growing season					
Boro	2,656	64	405.7	90.2	5.98
Aus	2,438	64	429.0	96.7	6.04
Aman	2,614	64	415.6	91.4	6.00
All cells	7,708	64	416.5	93.2	6.01

D. Climate and yield (district×season×year)	Boro Mean (SD)	Aus Mean (SD)	Aman Mean (SD)
$\text{GDD}_{[10,30]}$	1,787 (734)	4,208 (1,263)	2,825 (643)
$\text{EDD}_{>30}$	39.2 (26.0)	151.1 (57.6)	69.8 (31.8)
Precipitation (mm)	224 (179)	1,666 (685)	862 (412)
Observed log yield	0.50 (0.11)	-0.05 (0.21)	-0.02 (0.23)
Fitted $\log \widehat{y}$	0.52 (0.09)	0.00 (0.20)	0.01 (0.23)
N cells	445	445	445
Districts	64	64	64

Sources. A–C: BBS wages, CPI-deflated (base 2021/22). D: ERA5 (Hersbach et al., 2020); BBS yields; Stage 1 fitted log yield.

Notes. A–C: Stage 2 cells (district×season×gender×meal×year; N = 7,708 in 64 districts). Wages in real BDT/day. D: climate/yield entries are means (SD); N cells and Districts are counts.

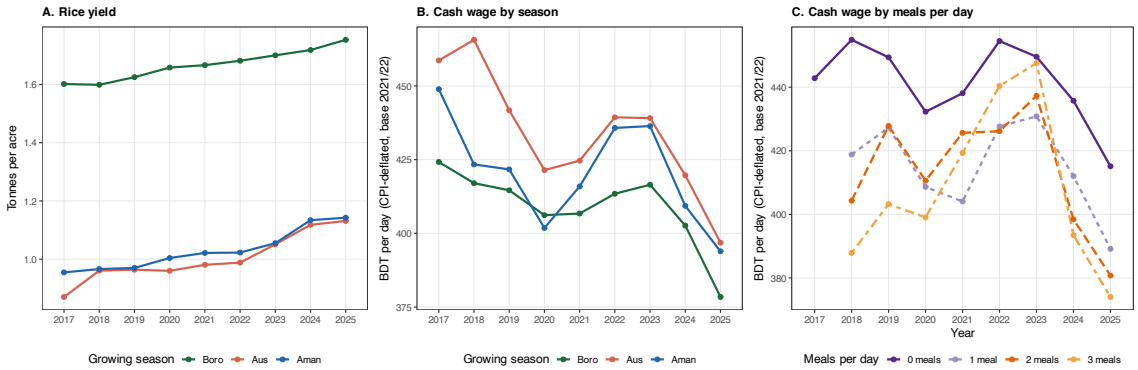


Figure 1.2: National Mean Rice Yield and Real Cash Wages, 2017–2025. Panel A: mean rice yield by season (tonnes per acre). Panels B–C: mean real cash wage (BDT per day, CPI-deflated) by season and by meals per workday, gender cells pooled within each group. Lines are descriptive averages across available districts, calculated by first averaging observations within each district. District coverage varies by year and meal category. In 2017 the seasonal panel plotted here records only no-meal wages. No fixed effects.

1.6 Empirical Strategy

The empirical questions are how cash wages vary with rice yield, and whether that relationship and the seasonal pattern of cash pay differ between jobs with and without meals. We compare cash wages across payment categories; we do not estimate the causal effect of adding meals to a job. The main analysis relates crop-season average cash wages to rice yield, with observed yield instrumented by growing-season weather. It first estimates a pooled benchmark with a single yield slope for both meal categories, and then a model that allows the slope to differ. Its interpretation depends on whether the instruments are strong and valid, which we assess separately. We describe the simpler harvest specification first and then the yield models; the Results section presents the yield analysis first. The harvest comparison measures how much the recorded cash wage of each job type rises in the harvest months, and because labor demand and supply are not observed, it does not identify changes in

either. Both comparisons contrast no-meal and three-meal cash wages within the same district, season or month, and gender.

1.6.1 Harvest peaks

For district d, month τ, year t, gender g, and meal category $m \in \{0, 3\}$, the monthly wage equation is

$$\log w_{d\tau tgm} = \pi\, H_\tau + \delta\big[H_\tau \times \mathbf{1}(m = 3)\big] + \alpha_{dtgm} + u_{d\tau tgm}. \tag{1.3}$$

The harvest indicator H_τ equals one in April and May for Boro and in November and December for Aman. The fixed effects α_{dtgm} compare months within the same district, year, gender, and meal category. Thus π is the no-meal harvest increase, and the meal gap δ is the difference between the three-meal and no-meal increases. The indicator follows the crop calendar, so its value does not depend on the year's weather or on local wage conditions. We also estimate equation (1.3) with the real daily wage in taka as the outcome, which gives the taka version of the comparison.

The comparison between meal categories has the algebraic form of a difference-in-differences: it removes wage movements common to both job types in harvest months. It remains descriptive. Here, δ compares published wage series; it does not identify the effect of changing a job's meal provision. Seasonal conditions and the jobs represented in each category may differ. The checks below assess the sensitivity of the observed contrast; they do not resolve selection into meal provision. The main concern for this description is a seasonal pattern that differs between the job types for reasons other than the harvest. We examine this with placebo "harvests" in adjacent non-harvest months, a month-by-month version of equation (1.3) that replaces H_τ with an indicator for each month (equation (1.5), Figure 1.3), and a specification that adds a common linear month trend and its interaction with three

meals. Because three-meal wages are reported in fewer months, we also re-estimate the model on cells observed in all twelve months.

To explore whether the meal gap depends on how important rice is locally, we interact H_τ and $H_\tau \times \mathbf{1}(m = 3)$ with R_d, the standardized log of district rice area averaged over 2013–2016, before the wage sample begins. We also estimate equation (1.3) separately for districts in each third of R_d.

1.6.2 Yield variation: instrumental variables

Let $y_{dst} = \log Y_{dst}$ be observed log rice yield in district d, season s (Boro, Aus, or Aman), and year t. For gender g and meal category $m \in \{0, 3\}$, the wage equation is

$$\log w_{dsgmt} = \beta_0 \, y_{dst} + \beta_3 \big[y_{dst} \times \mathbf{1}(m = 3) \big] + \lambda_{tm} + \mu_{dsgm} + \varepsilon_{dsgmt}. \tag{1.4}$$

The cell effects μ_{dsgm} absorb persistent cash-wage differences across district–season–gender–meal categories. The year×meal effects λ_{tm} allow national wage movements to differ by meal category. We use two versions of this equation. The pooled benchmark sets $\beta_3 = 0$, so that a single yield coefficient applies to both meal categories. It summarizes the overall relationship between yield and cash wages in the two series and is reported in Panel A of Table 1.2, including the baseline two-stage least-squares estimate discussed in Section 1.7.1. It is not by construction an average causal effect. The pooled model imposes a common slope; it does not establish that the underlying relationships are equal. The full equation (1.4) then allows the slopes to differ: the no-meal response is β_0, and the meal gap is β_3 (Panel B).

Observed yield may be correlated with local conditions that also affect wages, including irrigation, input use, and labor supply. We instrument y_{dst} with growing degree-days between 10 and 30°C, extreme degree-days above 30°C, and precipitation. Interacting each variable with the season gives the nine instruments of the

pooled benchmark. The meal-gap model has two endogenous variables, yield and its interaction with three meals, and uses the nine instruments together with their interactions with the three-meal indicator, eighteen in all. In the richest specification, we replace λ_{tm} with season×year×meal effects, so that only weather differences across districts within the same season and year identify the estimate.

A causal interpretation rests on the exclusion restriction: weather must affect cash wages only through rice yield. Heat and rain could also change working days, workers' productivity, or other employment opportunities, which would violate it. Instrument strength is a second condition, since growing-season weather must predict enough of the yield variation that remains after district and year effects (Appendix Table A6). To assess these conditions we report the first-stage F statistic, the reduced form, a Sargan overidentification test, and, for the pooled benchmark, estimates that omit one year at a time. For the common slope of the pooled benchmark we also report an Anderson–Rubin confidence set. It remains valid when instruments are weak, provided the instruments are valid, including the exclusion restriction, but it does not test that restriction.

1.6.3 The two-step fitted-yield estimates

The appendix reports an earlier estimator that replaces observed yield in equation (1.4) with weather-predicted yield and estimates the wage equation by ordinary least squares. It uses the same weather variation as the instrumental-variables estimator, but its ordinary standard errors treat the first-stage prediction as known. We estimate the first stage separately by season on 2013–2025 data, using all three weather variables. For every two-step coefficient, a joint district bootstrap resamples

districts and re-estimates both stages in each of 999 draws. Stars and reported p-values use the estimate divided by its joint bootstrap standard error, with a t reference distribution on 63 degrees of freedom.

1.6.4 Inference

In both comparisons, we cluster standard errors by district (64 districts). Wage cells within a district share local shocks across seasons, months, genders, and meal categories. For the rice-intensity interaction, which varies only across districts, we also report a permutation test that reassigns rice intensity across districts.

1.7 Results

We first examine the relationship between rice yield and crop-season average cash wages, using the pooled IV benchmark and then allowing the relationship to differ by meal category. We next report the companion two-step estimates. The final subsection describes harvest-month wage changes, which use a different source of variation.

1.7.1 Rice yield and cash wages

In the pooled baseline IV specification, a 10 percent higher rice yield is associated with crop-season average cash wages about 2 percent lower: the coefficient is -0.200 (standard error 0.084) in column (4) of Table 1.2. The model imposes a common slope on both meal categories. Column (1) gives the ordinary least-squares benchmark, 0.001 (0.029): once the wage-cell and year×meal effects are removed, observed yield has almost no association with cash wages. Because observed yield may move with irrigation, input use, and local labor supply, column (4) instruments it with growing-season weather.

Table 1.2: IV Estimates: Observed Rice Yield Instrumented by Growing-Season Weather (Bangladesh, 2017–2025)

	(1) OLS	(2) First stage	(3) Reduced form	(4) 2SLS	(5) 2SLS
Dependent variable	Log wage	Log yield	Log wage	Log wage	Log wage
Panel A: Common yield slope					
Log rice yield	0.001			−0.200**	0.076
	(0.029)			(0.084)	(0.287)
First-stage F		6.63			1.26
Instruments jointly, p			0.009		
Anderson–Rubin 95% set				[−0.30, −0.10]	$(-\infty, 0.53] \cup [1.28, \infty)$
Sargan overidentification p				0.002	0.007
Panel B: Three-meal minus no-meal gap					
Log yield × three meals	0.028			0.093	0.322
	(0.049)			(0.176)	(0.509)
First-stage F (yield; × 3 meals)		8.0; 4.9			1.2; 0.8
Instruments jointly, p			<0.001		
Year–meal effects	Yes	Yes	Yes	Yes	No
Season–year–meal effects	No	No	No	No	Yes
Wage-cell effects	Yes	Yes	Yes	Yes	Yes
N	4,140	4,140	4,140	4,140	4,140

Notes. Same cells, no-meal and three-meal sample and fixed effects as Table A7, with observed log rice yield in place of fitted yield. Wage-cell effects are specific to each district, season, gender and meal category. Year–meal effects allow national wage changes to differ by meal category. Instruments: growing-degree days (10–30°C), extreme-degree days (above 30°C) and precipitation for each rice season (nine instruments); Panel B adds their three-meal interactions (eighteen). Column (2) reports the cluster-robust F statistic of the excluded instruments and column (3) the joint p-value of the instruments in the reduced form; in Panel B column (3) tests all eighteen. Column (5) adds year×season×meal fixed effects, so only weather differences across districts within a season and year identify the effect. The Anderson–Rubin 95% set for the common slope is obtained by numerically inverting the cluster-robust Wald test of the instruments, with refined boundaries and separately evaluated tail limits; it is robust to weak instruments if the instruments are valid. In column (5) it consists of two unbounded rays (Appendix Table A3). The Sargan statistic assumes homoskedastic errors and does not account for district clustering. In Panel B, the two first-stage F statistics are separate joint tests of the excluded instruments for each endogenous variable; they are not conditional-strength or rank tests and do not fully assess joint identification of the two slopes. District-clustered standard errors in parentheses.
* Significant at 10%; ** at 5%; *** at 1%.

A negative slope runs against the simple expectation that a larger crop raises the value of harvest labor, but it is consistent with one of the channels in Section 1.4.2. If the extra rice from an additional day of work changes little with yield, and larger harvests expand supply enough to lower the price farmers receive, an additional day's work is worth less. The estimated association between weather-predicted yield and district retail rice prices is negative but imprecise: -0.090, with a bootstrap standard error of 0.093 (Appendix Table A22). This is a possible explanation, not a tested mechanism: the price series measures retail rather than producer prices, the wage is a crop-season average rather than the pay of identified harvest workers, and the output of an additional worker-day is unobserved.

The estimate comes with three caveats. First, the relationship between growing-season weather and rice yield is noisy once district and year effects are removed (Appendix Table A6), so the instruments are weak: the cluster-robust first-stage F statistic for the nine instruments is 6.6. In the baseline first stage, the Aus and Aman extreme-degree-day coefficients and the Boro precipitation coefficient are positive and statistically detectable; these associations weaken with season–year–meal controls (Appendix Table A2). The instruments are jointly significant in the reduced-form wage equation (column 3, $p = .009$), and the Anderson–Rubin 95 percent set, $[-0.30, -0.10]$, excludes zero. That set is robust to weak instruments, but only if the instruments are valid.

Second, weather may affect cash wages through channels other than rice yield, such as working days or worker productivity. Such effects would violate the exclusion restriction. The conventional Sargan test rejects the overidentifying restrictions ($p = .002$), although it assumes homoskedastic errors and does not account for district clustering.

Third, the estimate is sensitive to the specification and the sample. Column (5) replaces the year×meal effects with season×year×meal effects, so that only weather differences across districts within a season and year identify the slope. The first-stage F falls to 1.3, the slope becomes 0.076 (0.287), and the Anderson–Rubin set is unbounded, excluding only values between 0.53 and 1.28 (Appendix Table A3). The estimate is also sensitive to which years are included. Omitting 2024 moves the pooled IV coefficient from −0.200 to −0.008 (standard error 0.104), while the first-stage F statistic falls from 6.6 to 2.8; Appendix Table A4 reports all nine omissions. The negative baseline estimate is sensitive to including 2024, and the first stage is weaker when that year is omitted. This does not show that the 2024 data are in error or that the true relationship is zero. These diagnostics limit a causal interpretation of the negative baseline estimate.

Allowing the slope to differ between meal categories gives an instrumented meal gap of 0.093 (0.176) in column (4), where the first-stage F statistics for the two endogenous variables are 8.0 and 4.9, and 0.322 (0.509) in column (5); the observed-yield gap in column (1) is 0.028 (0.049) (Panel B of Table 1.2). Both instrumented intervals include large gaps of either sign. The estimates therefore neither establish different yield responses across meal categories nor show that the responses are equal, and they do not test the framework's conditional prediction about meal values.

1.7.2 The two-step fitted-yield estimates

The two-step common-slope estimate is −0.082, with a district-clustered standard error of 0.043 and a joint bootstrap standard error of 0.053, which accounts for the estimated first stage. It has the same sign as the IV estimate but is not statistically distinguishable from zero using the bootstrap standard error. Appendix Table A7 estimates equation (1.4) by ordinary least squares with weather-predicted

rather than observed yield, using growing-degree days, extreme-degree days, and precipitation in every season. Because it relies on the same weather, it is not independent confirmation. The meal gap is 0.117, with standard errors of 0.120 and 0.138, and its 95 percent bootstrap interval runs from −0.153 to 0.415. The appendix also reports gender estimates and a compact comparison of alternative rice measures, lagged yield, and simpler fixed effects, all with bootstrap standard errors (Appendix Table A12). At the 5 percent level, only the first-difference specification with common year effects gives a meal gap that differs from zero, and it loses significance when national year effects vary by meal category.

1.7.3 Harvest-month cash wages

The harvest comparison uses a different source of variation, the recurring busy months of the crop calendar, and describes how cash pay moves in those months; it does not estimate the effect of yield.

Cash wages rise substantially at harvest in both no-meal and three-meal jobs, by similar percentages (Table 1.3). In column (1), no-meal cash wages are 5.7 percent higher in the Boro and Aman harvest months than in other months of the same district, year, and gender. The three-meal increase is about 5.1 percent. The difference from the no-meal increase is −0.6 percentage points (standard error 0.6), with a 95 percent confidence interval for the meal gap from −0.019 to 0.006. The difference is not statistically significant, but significance alone does not say how large a difference the data allow. Holding the estimated no-meal increase fixed, the lower end of this interval corresponds to a three-meal increase roughly two-thirds as large. If meals came with arrangements that held daily cash pay steady through the season, the three-meal increase would be much smaller. This rules out large differences under that comparison, but does not establish equal increases or an effect of meal provision.

Table 1.3: Harvest Wage Peaks in No-Meal and Three-Meal Jobs (Bangladesh, 2017–2025, Monthly)

Dependent variable: log(real daily wage); no-meal and three-meal cells only					
	(1)	(2)	(3)	(4)	(5)
	All	Rice-intensity	Districts by rice intensity		
	districts	interaction	Low	Middle	High
Harvest month	0.057***	0.057***	0.045***	0.050***	0.077***
	(0.005)	(0.005)	(0.008)	(0.006)	(0.009)
Harvest × three meals	−0.006	−0.006	0.010	0.010	−0.038***
	(0.006)	(0.006)	(0.008)	(0.009)	(0.012)
Harvest × rice intensity		0.014***			
		(0.003)			
Harvest × three meals × rice intensity		−0.014**			
		(0.007)			
Wage-cell effects	Yes	Yes	Yes	Yes	Yes
Districts	64	64	22	21	21
N	16,605	16,605	5,861	5,356	5,388

Notes. Unit: district×month×year×gender×meal. Harvest months: April and May (Boro) and November and December (Aman). The harvest coefficient compares harvest months with other months of the same district, year, gender and meal category; harvest×three meals is the difference between the three-meal and no-meal harvest premiums. Rice intensity: log district rice area (all seasons), 2013–2016 average, standardized; columns (3)–(5) split districts into thirds by it. The absolute magnitude of the lower 95% confidence bound on harvest×three meals is about 32% of the estimated no-meal harvest increase in column (1). District-clustered standard errors in parentheses.

* Significant at 10%; ** at 5%; *** at 1%.

Columns (2)–(5) ask whether this pattern depends on how important rice is locally. The no-meal harvest increase rises with district rice intensity, from 4.5 percent in the lowest third of districts to 7.7 percent in the highest (columns 3–5), and the continuous interaction in column (2) is 0.014 (0.003). This is what one would expect if the demand for harvest labor peaks more sharply where rice dominates. The meal gap is 0.010 in each of the lower two thirds and −0.038 (0.012) in the highest third, where the three-meal increase is about half the no-meal increase. In column (2), the meal gap falls by 0.014 (0.007) for each standard deviation of rice intensity, and a permutation test that reassigns rice intensity across districts gives $p = .036$.

The rice-intensity pattern suggests a possible dynamic. Where rice work is concentrated and the harvest peak is sharpest, employers may have the most reason to secure workers for the season, and the continuing arrangements described in Section 1.4.3 could come with meals and a smaller harvest increase in cash pay. The evidence for this is limited. The pattern comes from women's wages, and it is not significant for men or when the five largest rice districts are dropped (Appendix Table A20). The records do not observe contracts, so differences in the jobs recorded as providing three meals in these districts could produce it as well. We therefore treat it as exploratory.

Figure 1.3 shows the monthly pattern behind the pooled comparison. It plots the coefficients of

$$\log w_{d\tau tgm} = \sum_{k \neq \text{Jan}} \theta_k \, \mathbf{1}(\tau = k) + \sum_{k \neq \text{Jan}} \phi_k \left[\mathbf{1}(\tau = k) \times \mathbf{1}(m = 3) \right] + \alpha_{dtgm} + u_{d\tau tgm}, \quad (1.5)$$

which replaces the harvest indicator in equation (1.3) with a separate indicator for each month and keeps the same district×year×gender×meal fixed effects. Panel (a) plots θ_k, the no-meal wage in each month relative to January. It peaks in April–May and November–December and is lowest in January–March, so the harvest increase

is not driven by a single month. Panel (b) plots ϕ_k, the three-meal minus no-meal difference relative to January. That difference drifts down through the year, to about -0.045 log points in December, and the harvest months do not depart clearly from the drift.

The drift matters for the pooled comparison because the harvest months fall later in the year, on average, than the other months. A downward drift therefore makes the pooled meal gap more negative and leans toward finding smoothing, not against it. This drift can help account for why the Boro and Aman harvests taken one at a time give gaps of opposite sign, and why placebo "harvests" at the two ends of the year do the same (Appendix Table A18). Adding a common linear month trend and its interaction with three meals moves the meal gap to 0.001 (0.007), and the magnitude of the lower confidence bound falls to 24 percent of the harvest increase. The trend-adjusted comparison assumes a linear relative-wage drift. We therefore report it alongside the baseline; both give a small estimated meal gap.

In taka, no-meal wages rise by 25.6 at harvest, and three-meal wages by 3.6 less, with a standard error of 2.5. The 95 percent interval for that difference runs from -8.6 to 1.4 taka (Appendix Table A18), and with month trends the difference is -0.5 (2.9). In the main sample, these estimates allow equal increases in taka, as the fixed-meal-value benchmark predicts. This result is sensitive to matching: when both wage categories must be observed in the same district, month, and gender, the three-meal increase is smaller in taka (Appendix Table A21). The log estimates tell a slightly different story. Under the fixed-meal-value benchmark for comparable jobs, lower starting cash wages imply a larger percentage increase in meal jobs. The baseline estimate instead gives a slightly smaller increase, but this comparison does not test that benchmark. The taka estimates concern absolute wage increases, while the log estimates concern proportional increases from different starting wages.

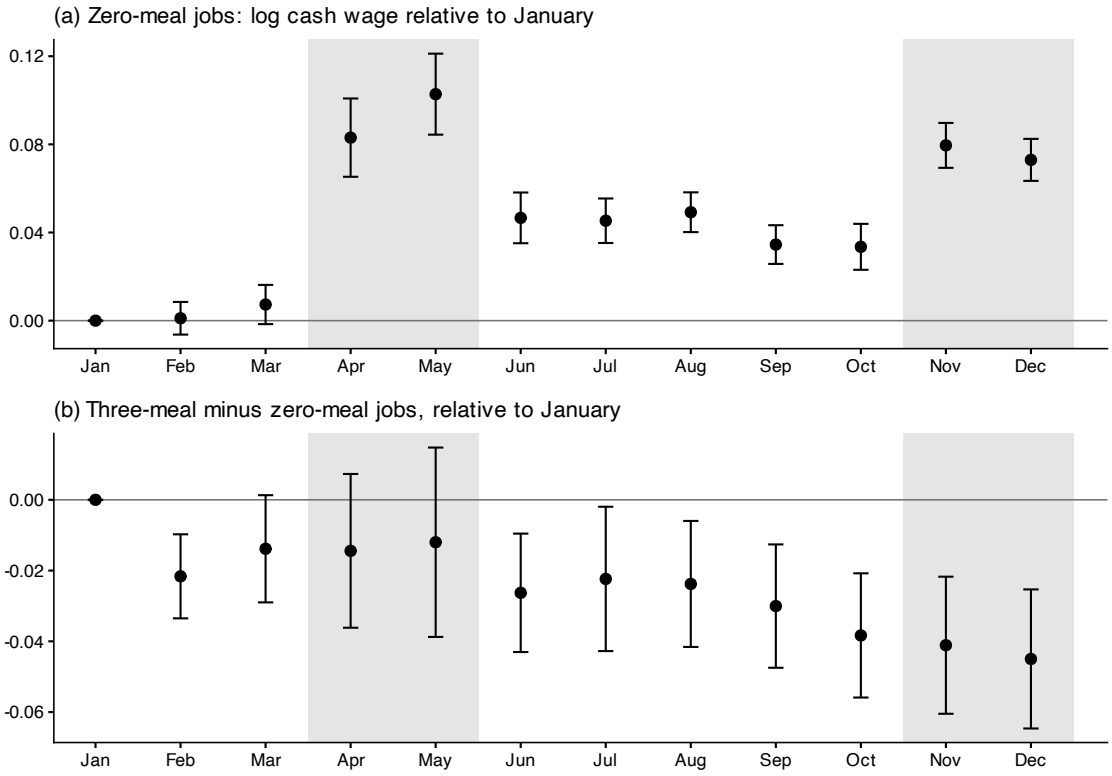


Figure 1.3: Monthly Cash Wages in No-Meal and Three-Meal Jobs, 2017–2025. Coefficients from equation (1.5). Panel (a): log real cash wage in no-meal jobs by month, relative to January (θ_k). Panel (b): three-meal minus no-meal log cash wage by month, relative to January (ϕ_k). Both panels include district×year×gender×meal fixed effects. Bars are 95% confidence intervals with district-clustered standard errors. Shading marks the Boro (April–May) and Aman (November–December) harvests.

The meal gap stays small in the other harvest checks. It is close to zero for men and somewhat negative for women, and it changes little when any one year is dropped or the Aus harvest months are added. Three-meal wages are reported in fewer months than no-meal wages (Section 1.5.1). Restricting the sample to cells observed in all twelve months gives a gap of −0.005 (0.007), and restricting it further to district–year–gender cells in which both job types are observed all year gives −0.008 (0.005). The harvest increase is similar across all four meal categories, from 4.7 percent with one meal to 5.7 percent with none, and within a year the monthly cash wage is slightly *more* variable in three-meal jobs than in no-meal jobs (Appendix Table A19).

One possible interpretation is that meals help workers meet food needs without keeping daily cash wages steady. Research on agricultural households in India links work paid in kind to food-security concerns (Ito & Kurosaki, 2009); it does not establish that such jobs have smaller seasonal cash-wage increases. In our exploratory price check, higher rice prices are associated with a wider cash-pay difference between meal and no-meal jobs, and adding the observed retail-price control changes the matched-sample harvest gap little (Appendix Table A21). This check does not establish why the two wage series have similar harvest increases. We cannot test the food-security interpretation because meal values and workers' food consumption are not observed.

1.8 Limitations

Both comparisons describe recorded daily cash wages in Bangladesh; they do not establish how in-kind pay behaves in other labor markets.

Interpreting the yield estimate causally requires that weather affect cash wages only through rice yield. The instruments also predict yield weakly, and the estimate is sensitive to seasonal controls and to including 2024 (Section 1.7.1). Anderson–Rubin

inference addresses weak instruments only if the instruments are valid. The imprecise meal-gap estimates also do not establish equal responses across payment categories.

The fitted-yield estimates use the same weather variation and share these limits. Their ordinary district-clustered standard errors treat fitted yield as known, so we base their inference on a joint district bootstrap that repeats both stages. This leaves the meal-gap conclusion unchanged but matters for some appendix coefficients: the two-step common slope in Appendix Table A7 is significant at the 10 percent level with clustered standard errors but not with bootstrap ones.

The harvest comparison measures how cash wages differ between harvest and other months. It does not measure labor demand or supply, and it concerns predictable seasonal changes rather than protection across good and bad years. Interpreting the gap as a difference in harvest-related adjustment requires separating it from other seasonal differences between the two series. The relative-wage drift in Figure 1.3 shows that this holds only approximately. The baseline comparison leaves the drift in place, and the trend-adjusted comparison removes it only if it is linear in the month. Neither specification is free of assumptions, although both give a small meal gap.

Finally, the BBS records report cash wages and meal counts but not meal values, hours, days worked, tasks, contracts, or worker identities. Three-meal wages are reported in fewer months than no-meal wages, and some consecutive monthly values repeat. A change in a wage row can therefore reflect a change in pay for the same job or a change in the jobs and workers the row covers. The records cannot separate these, and they do not measure total compensation, employment, or annual earnings. Nor do they measure the extra rice from an additional worker-day or the prices farmers receive, which the price account of the yield estimate would require.

1.9 Conclusion

This chapter asks how agricultural cash wages adjust to crop conditions, and whether that adjustment differs when part of pay is provided as meals. Using Bangladesh district wage records that separate jobs by the meals employers provide, it draws on two sources of variation that need not move the value of farm work in the same way: differences in rice yield across seasons and years, and the recurring harvest season.

In the pooled baseline IV specification, higher yields go with lower crop-season average cash wages, about 2 percent lower for a 10 percent higher yield. One explanation is that a larger crop adds little to what an additional day of work brings in while lowering the price farmers receive. The evidence cannot settle this: weather predicts yield weakly after the controls and may also affect farm work directly, the estimate is sensitive to more detailed seasonal controls and to excluding 2024, and the data do not identify the price account. Nor can the estimates tell whether meal jobs respond to yield differently from jobs paid only in cash.

The separate harvest comparison shows how cash pay moves in the busy season. Cash wages in harvest months are about 6 percent higher than in other months in no-meal jobs and about 5 percent higher in three-meal jobs. Jobs that provide meals pay less cash, but the cash they pay still rises substantially at harvest. The two results do not conflict: one compares predictably busy months with other months of the same year, the other compares seasons with more and less output per unit of land.

Three kinds of data would sharpen these lessons. Monthly records of meal values would show whether total daily pay moves with the cash wage. Measures of the extra output from an additional worker-day, together with the prices farmers receive, would help test the price account, and panels that follow workers and employers would

help assess whether longer relationships steady earnings through employment rather than through the daily wage.

Together, the evidence separates two features that in-kind pay is sometimes assumed to bundle. In these records, a lower cash wage in meal jobs does not come with a markedly smaller seasonal rise in that wage, so meal provision by itself does not indicate steadier daily cash pay. Whether meal jobs offer more secure employment, higher total compensation, or smoother annual income is a different question that the records cannot answer, and the results need not hold for other forms of in-kind pay or other labor markets.

Chapter 2

HYBRID MODELS FOR CROP YIELD PROJECTIONS: CROP-SPECIFIC WEIGHTING OF EMPIRICAL AND PROCESS-BASED MODELS

Muhammad Taky Tahmid, James Rising, Mark Cane

Abstract

Climate change affects crop yields through both crop growth and farmers' responses to changing conditions. Empirical models estimate weather responses from observed yields under existing farming practices, while process-based models simulate crop growth under prescribed management. We examine whether combining them improves prediction across regions and how it changes projected climate losses. We estimate a hybrid model for maize, rice, and wheat: a weighted average of one fitted empirical weather–yield component and three calibrated process-based components. Using matched subnational yield data, we estimate crop-specific weights and evaluate predictions in regions excluded from estimation. The empirical model receives nearly all the historical weight for rice, about a third for maize, and a quarter for wheat. In the excluded regions, the combination predicts better than the empirical model alone for every crop, but it is not the most accurate predictor for any crop. Nevertheless, combining the approaches changes projected losses substantially: under SSP3-7.0, median losses are about 13 percentage points smaller for maize and 6 points smaller for wheat than in the empirical model alone, with little change for rice. Without the scale adjustments used to calibrate the simulations, the wheat difference disappears. These projections compare 2096–2100 with 2015. The empirical component carries

forward its 1978–2016 weather response, and the simulations hold management inputs and atmospheric CO_2 at 2015 levels. We apply short-run responses over the long run, with no additional adaptation or future yield trend.

JEL codes: Q54, Q10, C53, C52

Keywords: crop yields, climate change, hybrid models, process-based models, adaptation

2.1 Introduction

Projections of crop yields under climate change feed estimates of future food supply, farm incomes, and the economic cost of warming. Weather already accounts for about a third of the year-to-year variation in global crop yields (Ray, Gerber, MacDonald, & West, 2015), and climate trends since 1980 have reduced global maize and wheat output (Lobell, Schlenker, & Costa-Roberts, 2011). How much larger future losses will be depends on two responses: how crops respond to heat and water stress, and how farmers change irrigation, planting dates, varieties, and inputs as conditions change.

Two approaches construct weather responses differently. Empirical models estimate weather–yield relationships from yields realized in farmer-managed fields. Their coefficients estimate the net response under existing farming systems within the range of historical weather. Process-based crop models (PBMs) simulate plant growth using crop-growth parameters and management rules specified by their developers and simulation protocols. They can be run under climates outside the historical range, and they often disagree with observed responses and with one another (Schauberger, Archontoulis, Arneth, et al., 2017; Müller, Franke, Kollas, et al., 2021). The two approaches can therefore project quite different losses from the same future climate.

A growing literature combines them. Roberts, Braun, Sinclair, Lobell, and Schlenker (2017) show that a combination of statistical and process-based models predicts US Corn Belt maize yields better than either alone. Other studies adjust ensembles of crop simulations with yield observations (Makowski, 2017), use simulated crop variables as inputs to data-driven predictors (Shahhosseini, Hu, Huber, & Archontoulis, 2021; von Bloh, Lobell, & Asseng, 2024), or show that the rule used to average crop models can change global projections (Yin et al., 2025). These studies show that combining observed and simulated information can improve prediction in particular settings. We extend this approach by estimating explicit weights for an empirical weather response and several calibrated PBMs, crop by crop on common subnational data from several continents. We then assess prediction in regions left out of estimation and apply the combination to climate projections.

We propose a hybrid model that does this for maize, rice, and wheat. It combines an empirical weather–yield response with three PBMs (EPIC, pDSSAT, and LPJmL), each calibrated to the same subnational yield observations, using crop-specific weights that are nonnegative and sum to one. We estimate the weights on historical data, re-estimate the full model with each major region held out, and apply the fitted combination to projected changes in weather and simulated yields under the lower- and higher-forcing climate scenarios SSP1-2.6 and SSP3-7.0, using PBM simulations with management inputs and atmospheric CO_2 fixed at 2015 levels. The empirical component applies the weather response estimated over 1978–2016. These climate-only projections assume that the short-run response applies over the long run; they add neither future adaptation nor a secular yield or technology trend.

The historical weights differ sharply by crop. The empirical component receives a weight of 0.33 for maize, 0.94 for rice, and 0.25 for wheat, with the remaining weight

spread across the calibrated PBMs. These weights describe each component's contribution to historical fit. They are also sensitive to the estimation sample: excluding Brazil lowers the maize empirical weight to zero.

Prediction in held-out regions gives a mixed picture. After removing one overall mean from each held-out region, the hybrid predicts more accurately than the empirical component alone for all three crops, but it is not the most accurate predictor for any crop. Calibrated pDSSAT predicts maize better; for rice and wheat, pDSSAT and the hybrid are within 0.001 log points, while calibrated EPIC predicts rice best and an unconstrained combination predicts wheat best. The nonnegative, sum-to-one weights define a weighted average of the fitted components; they do not establish that this combination will predict future climate responses more accurately. We use it to examine how carrying both sources of evidence into projections changes estimated losses.

The combination matters for projected losses. Under SSP3-7.0, pooling projection records for 2096–2100 from three climate models, the hybrid gives median subnational yield changes relative to 2015 of -16.6% for maize, -47.8% for rice, and -10.7% for wheat. The empirical model alone gives -30.1%, -50.6%, and -16.8%. The difference depends on both the weights and the calibration slopes, which rescale the PBMs' projected changes. For wheat, the smaller hybrid loss disappears without that rescaling. Rice stays close to the empirical projection because its empirical weight is high. These figures are medians across projection records under the stated farming-system assumptions; they are not forecasts of production.

The weights summarize historical fit and do not identify adaptation. Both sources describe weather responses under existing or specified farming systems, and neither estimates the additional response to persistent warming. Section 2.3 develops this distinction with a common management framework. Section 2.4 describes the

matched data, Section 2.5 the estimator and validation, Section 2.6 the weights, prediction comparisons, and projections, and Section 2.7 what the results imply for projecting climate damage to agriculture.

2.2 Literature Review

2.2.1 Climate change and crop yields

Historical studies show that climate change has already affected agriculture. Using production records for 1980–2008, Lobell et al. (2011) estimate that climate trends reduced global maize and wheat output by 3.8% and 5.5%, respectively, relative to a no-trend counterfactual. Ortiz-Bobea et al. (2021) find that anthropogenic climate change has slowed growth in agricultural total factor productivity. These estimates describe realized outcomes under the crops and management systems observed to date, not how yields will respond when future conditions move beyond the historical record.

Both empirical and process-based studies project losses under warming, but they answer different questions. Zhao et al. (2017) synthesize yield responses in studies that exclude adaptation, while the coordinated crop-model ensemble of Jägermeyr et al. (2021) projects substantial changes under future emissions scenarios. In contrast, the dynamic empirical model of Wing, De Cian, and Mistry (2021) distinguishes farmers' short-run and longer-run responses to weather. Differences among these projections partly reflect their different assumptions about crop processes, management, and adaptation. They motivate a comparison of what each model class contributes rather than a direct comparison of headline losses.

2.2.2 Process-based and empirical crop models

Process-based crop models (PBMs) represent growth processes such as phenology, water stress, and nutrient uptake. This lets them simulate weather combinations absent from observed yield records, conditional on their model structure and management rules. Coordinated exercises provide common climate inputs and protocols for comparing the models (C. Rosenzweig et al., 2018; Franke et al., 2020). The three PBMs used in this chapter differ in their representations of crop and soil processes: EPIC emphasizes soil and nutrient dynamics (Williams, 1995; Balkovič et al., 2014), LPJmL links crops to a dynamic vegetation model (Bondeau et al., 2007), and pDSSAT implements DSSAT on a global grid (Hoogenboom et al., 2024).

A detailed crop model need not reproduce the yield response observed in fields. Schauberger et al. (2017) compare observed and simulated US crop responses to high temperatures and find that simulated irrigation protects yields more than observed irrigation. In experiments testing 14 rice models, Sun et al. (2021) find that the models understate losses from short episodes of extreme heat. Management choices and harmonization can also alter model results (Folberth, Baklanov, Balkovič, et al., 2019). Large ensembles consequently reveal substantial disagreement across crop models as well as across climate inputs (Müller et al., 2021). These findings identify reasons to examine the simulated response against observed yields; they do not imply that every PBM has the same bias.

Empirical models instead estimate weather responses from realized yields. Fixed-effects panels compare weather variation within locations while absorbing persistent differences across them. Schlenker and Roberts (2009b) find sharply nonlinear US yield losses above crop-specific heat thresholds; Lobell, Hammer, McLean, et al. (2013) emphasize the role of extreme heat for US maize. These responses are observed under existing farming systems and can include farmers' adjustments within

a season. They cannot separately identify the physiological and management contributions to the estimated weather effect. Wing et al. (2021) show why the distinction between short-run shocks and longer-run adjustment matters for projection. Nor does historical fit establish which response will extrapolate best to unfamiliar climates.

2.2.3 Hybrid models and ensemble weighting

One line of work uses observed yields to adjust an ensemble of process-based projections. Makowski (2017) develops a Bayesian adjustment of crop-model outputs using yield observations. Yin et al. (2025) show that the method used to average crop models can itself materially change global maize projections. These studies discipline or combine PBM outputs, but they do not carry an empirical weather-yield response as a separately weighted projection component.

Another line uses simulations as inputs to data-driven predictors. Shahhosseini et al. (2021) add APSIM variables to machine-learning forecasts of US Corn Belt maize yields. von Bloh et al. (2024) transfer DSSAT-based crop knowledge into machine-learning predictions of wheat yields. Both demonstrate that simulated crop processes can help predict observed yields. Their flexible predictors, however, do not assign an explicit, comparable weight to an empirical climate response alongside calibrated PBM responses.

The closest precedent is Roberts et al. (2017). They compare a statistical model, a process model, and their combination against yields from farmer-managed maize fields in the US Corn Belt. The combination predicts better than either component, and its response to extreme heat is more negative than that of the process model. This establishes that observed and simulated information can complement one another. It leaves open how an explicitly constrained combination behaves across crops and regions, and how its historical balance changes projected climate responses.

2.2.4 Research gap and chapter contribution

The literature has established historical climate damage, documented both the reach and the limits of crop simulations, and shown that combining empirical and process-based information can improve prediction. This chapter builds on those results with a common weighting framework across several staple crops, evaluated in regions outside the estimation sample and carried into climate projections. It constructs a weighted average of one fitted empirical component and three statistically calibrated PBM components on matched observations for maize, rice, and wheat, estimating crop-specific nonnegative weights, testing them in held-out regions, and applying them to projected changes in weather and simulated yields.

2.3 Conceptual Framework

The hybrid combines two kinds of evidence about how yields respond to weather. Realized yields reflect crop physiology and the management choices of farmers. Process-based crop models (PBMs) simulate crop physiology under management that their protocols prescribe. Both can be written with the same management variable. Doing so shows what each source of evidence contains, how the hybrid combines them, and what the combination can and cannot reveal about adaptation.

2.3.1 Weather, management, and adaptation

A weather shock affects yield through plant growth and through the changes farmers make in response. For a given crop, write realized log yield in location i and year t as

$$\log Y_{it} = F(\vec{w}_{it}, \vec{M}_{it}) + u_{it}, \tag{2.1}$$

where $\vec{w}_{it}$ denotes weather, $\vec{M}_{it}$ denotes management, and u_{it} collects other influences on yield. Weather includes growing degree days, exceedance degree days, and precipitation. Management includes irrigation, planting dates, varieties, and fertilizer use. We distinguish the farming system already in place from adjustments within a season:

$$\vec{M}_{it} = \vec{M}_{i0} + \vec{s}(\vec{w}_{it}; \vec{M}_{i0}). \tag{2.2}$$

The farming system $\vec{M}_{i0}$ includes past adaptation to the local climate, such as investments in irrigation and the choice of varieties and crop calendars. The function $\vec{s}$ describes adjustments to a season's weather within that system, such as changes in irrigation or input use. Holding the farming system fixed, the weather response combines the direct crop response with these within-season adjustments. Persistent climate change can also lead farmers to change the farming system itself. That additional response is a different object from the response to annual weather variation.

Season-location fixed effects absorb persistent differences in yield levels, including differences associated with management. They do not remove management's influence on weather sensitivity. Season-year fixed effects absorb shocks common to locations in a given season and year. The empirical weather coefficients are estimated from annual variation within locations and combine the direct crop response with farmers' within-season adjustments. They average over the farming systems observed in 1978–2016, which already reflect past adaptation; management changes during that period can also affect the coefficients when they move with weather. The regression does not separately identify how farming systems would adjust to a persistent change in climate.

For projection, we assume that the short-run response estimated from annual weather applies to long-run climate change. This is the short-run-equals-long-run assumption: the historical response is carried forward without an additional response from future adaptation. The assumption is imposed by the projection, rather than established by the regression. The direction of any resulting bias is unknown. Future adaptation could reduce losses, while short-run buffers such as drawing on stored water may be difficult to sustain under persistent warming. The response can still be nonlinear in temperature and rainfall even though its estimated relationship is held fixed.

2.3.2 Process-based models and empirical responses

A PBM supplies a second representation of the yield response. Using the management variable from Equation 2.1, model m produces simulated log yield as

$$\tilde{y}_{it}^{m} = F_m\left(\vec{w}_{it}, \vec{M}_{it}^{m}\right), \qquad \vec{M}_{it}^{m} = \vec{M}_{0}^{m} + \vec{r}_m(\vec{w}_{it}), \tag{2.3}$$

where F_m is its crop-growth function, $\vec{M}_0^m$ denotes the farming system specified by the simulation protocol, and $\vec{r}_m$ collects any weather-responsive agronomic rules it contains. The simulations used here hold management inputs at 2015 levels and combine simulated irrigated and rainfed yields using fixed harvested-area shares (Section 2.4). Their response to weather includes the model's crop processes and agronomic rules under these settings. Applying that response to future climate without changing the farming system makes the same short-run-equals-long-run assumption as the empirical projection.

The two approaches obtain their weather responses differently. PBM developers specify crop-growth functions and parameterize them using experimental evidence,

prior research, and model calibration. This chapter takes those physiological parameters as given; it does not re-estimate them. The empirical panel estimates the net weather response directly from observed yields. We then statistically calibrate each PBM to observed yields using the same fixed-effect structure as the empirical model. This step estimates a scale factor for the simulated response and absorbs yield-level differences; it does not re-estimate the PBM's physiological parameters or its response shape. A calibrated PBM therefore combines the model's response shape with a scale estimated from observed yields.

Differences between the responses can also arise from management, weather forcing (ERA5 for the empirical regressors and GSWP3-W5E5 for the historical PBM simulations), yield measurement, and spatial aggregation. Past adaptation can shape both observed farming systems and the management settings specified in a PBM. A gap between the responses therefore does not identify adaptation. Table 2.1 compares the historical responses and the assumptions used in projection.

2.3.3 Hybrid prediction and constraints

The empirical model and the three PBMs give different predictions of historical yields. The hybrid first calibrates each PBM prediction to the observed yields, then combines the calibrated predictions with the fitted empirical prediction. Its crop-specific weights are nonnegative and sum to one, so the final prediction is a weighted average of the four fitted components. The restrictions keep large positive and negative weights from offsetting each other and give each weight a direct reading as a component's share of the combination. They do not guarantee better prediction: an unrestricted regression on the same components can fit or predict better, and it can also be used for projection. Section 2.5 gives the estimator and explains how the weights are re-estimated when a scenario lacks one component.

Table 2.1: Weather Responses and Projection Assumptions

	Process-based models	Empirical panel
How the weather response is obtained	Model crop processes and parameters; one response scale estimated from observed yields	Net response estimated from annual weather variation in observed yields
Farming system in the historical response	Specified management settings, including 2015-level inputs and fixed irrigation shares	Existing farming systems during 1978–2016, including past adaptation
Within-season adjustment to weather	Model's agronomic rules, where included	Farmers' adjustments, included in the net response and not isolated
Statistical treatment of yield levels	Season-location and season-year fixed effects in calibration	Season-location and season-year fixed effects in estimation
Long-run response used in projection	Assumed equal to short-run response; farming system unchanged	Assumed equal to short-run response; estimated relationship unchanged

Notes: The PBM column describes the simulations and statistical calibration used in this chapter. The empirical column describes the fixed-effects regression in Section 2.5.1. Neither projection adds future adaptation, technological change, or CO_2 fertilization. The empirical response is averaged over historical farming systems, rather than evaluated at observed 2015 management inputs.

The hybrid is useful if the components make different errors and their combination predicts yields well in locations outside the estimation sample. We test that possibility against each component in held-out regions. A favorable comparison would support the hybrid as a prediction rule; it would not establish which economic or biological process produced its advantage.

2.3.4 What the hybrid can reveal about adaptation

The empirical weight measures the fitted observed-yield component's contribution to the historical prediction, conditional on the calibrated PBMs. The weight is its share in the fitted combination, rather than a ranking of its predictive accuracy. The source of that help is not identified: farmers' adjustments, PBM error, the different weather forcing, and differences in measurement or spatial scale can all contribute. The weights are also estimated on yield levels, whereas the projections concern changes. The fitted empirical value used to estimate the weights includes the fixed effects, while its projection uses only changes in the weather terms. For these reasons, neither the weights nor a gap between empirical and PBM projections measures the share of climate damage avoided through adaptation.

The hybrid does reveal whether information from realized yields changes predictive performance and projected yield losses relative to simulations under prescribed management. The management framework explains the limits of this comparison: the data do not record the management choices needed to separate short-run adjustment from the direct crop response, and the projections add no change in the farming system.

2.4 Data

The analysis uses two distinct panels. Historical subnational yields, weather, and crop simulations form the common sample on which the hybrid is estimated and evaluated. Future climate and crop simulations supply the inputs to its projections; they do not enter estimation. The historical sample contains main-season observations from 1978–2016, whereas the projection panel runs from 2015–2100 and is aggregated at a coarser administrative level.

2.4.1 Observed yields and historical weather

The raw yield archive records annual production and harvested area for maize, rice, and wheat in 33 countries from 1949 to 2024: 27 European countries, Brazil, China, India, Kenya, Nigeria, and the United States. Yield is production divided by harvested area for the same crop category, and each source contributes one observation per administrative unit, crop, season, and year. Statistical agencies often report a crop total alongside its components, so we never combine a total with its own parts. United States maize and wheat come from USDA county survey statistics for all classes and production practices, with maize measured as grain (USDA National Agricultural Statistics Service, 2026); where no all-wheat total is available, production and harvested area are summed over the wheat classes reported for that county-year. United States rice uses archival county records with a single reported production value and harvested-area value; their detailed category labels are not retained. Brazilian yields use harvested rather than planted area. European wheat combines the available common-wheat and durum records, and Indian seasonal records assigned to the same growing season are summed. We harmonize yields to tonnes per hectare and retain each crop's Main season. European green maize is excluded because it measures fresh biomass rather than the grain yield represented by the crop models.

Historical weather comes from ERA5 reanalysis (Hersbach et al., 2020). The processed hourly temperature and precipitation series cover 1978–2019. Temperature enters as growing degree days (GDD) and exceedance degree days (EDD) over crop-specific growth windows. The two precipitation measures are the seasonal total and the sum of squared daily precipitation over the planting, middle-growth, and harvest periods. The latter records how rainfall is concentrated across days; it is not the square of the seasonal total. Weather exposures for irrigated and rainfed land are combined using crop-specific harvested-area shares. Section 2.5 explains the choice of temperature thresholds and growth windows, and Appendix B.1 gives the construction formulas. Historical crop simulations end in 2016, setting the end of the matched estimation period despite the longer weather and yield archives.

2.4.2 Historical crop simulations

Historical simulated yields from EPIC, pDSSAT, and LPJmL come from the ISIMIP3a agricultural archive. The models provide annual irrigated and rainfed outputs on a 0.5° grid under harmonized GSWP3-W5E5 climate forcing and 2015-level socioeconomic conditions. Their archive begins in 1901, but only years shared with observed yields and ERA5 enter this analysis. Available spring- and winter-wheat outputs are matched to the observed wheat category, and available rice outputs to the observed rice category. Simulated yields are aggregated to the administrative units of the observations. Because these simulations are driven by GSWP3-W5E5 rather than ERA5, observed and simulated yields respond to closely related but not identical weather series; Section 2.3 lists this forcing difference among the reasons their responses can differ. Before ensemble weighting, each PBM series is calibrated to observed yields on the common sample; the hybrid does not weight raw simulated yield levels.

2.4.3 Spatial alignment and the common sample

GADM 4.1 boundaries link the gridded inputs to observed administrative units (GADM, 2023). China, Kenya, and Nigeria enter at ADM1; Brazil, India, and the United States enter at ADM2. European NUTS2 observations are matched to GADM ADM2. Grid cells are aggregated using spatial overlap and crop-specific harvested-area weights (Schwarzwald & Geil, 2024). MIRCA2000 harvested-area shares determine how irrigated and rainfed weather and simulated yields are combined. Appendix B.1 documents these matching and weighting steps.

An observation enters the common sample only if observed log yield, GDD, EDD, both precipitation measures, and all three simulated log yields are finite. This leaves 247,824 maize observations in 6 countries, 133,486 rice observations in 11 countries, and 102,669 wheat observations in 29 countries (Table 2.2). Administrative-unit counts mix ADM1 and ADM2, so they do not describe a common geographic resolution. Figure 2.1 maps the crop coverage after matching. The matched observations are concentrated: Brazil supplies 94.6% of the rice sample and 69.2% of the maize sample, while the United States supplies 67.7% of the wheat sample (Appendix Table B1). African coverage is sparse. The geographic validation in Section 2.5.3 therefore tests transfer among the represented regions; it provides limited evidence about places unlike those in the historical sample.

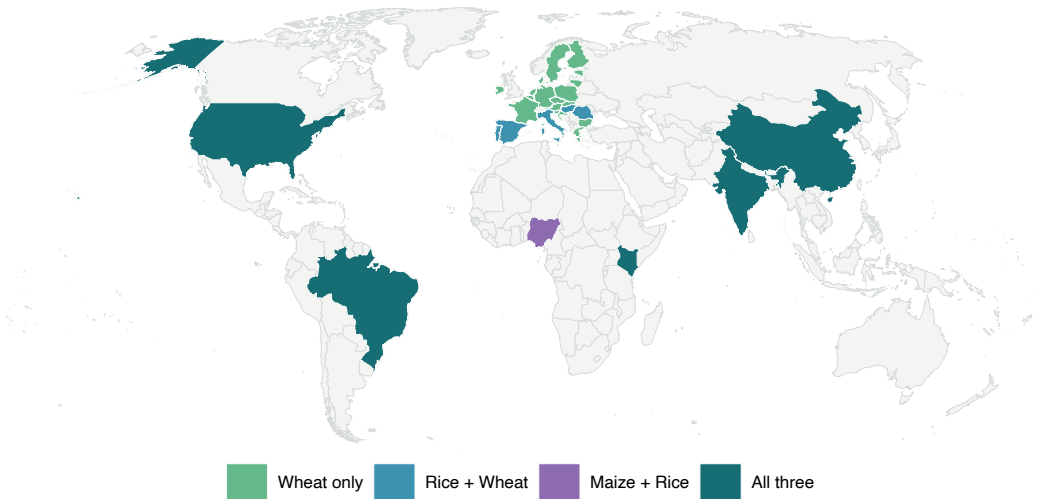


Figure 2.1: Geographic and Crop Coverage of the Historical Estimation Sample. Colors identify the crop combinations retained after matching observed yields, weather variables, and all three historical PBM outputs. Countries in gray do not enter the sample.

Table 2.2: Sample Coverage and Mean Values in the Historical Estimation Sample

Panel A. Sample coverage

Crop	Observations	Countries	Administrative units	First year	Last year
Maize	247,824	6	8,078	1978	2016
Rice	133,486	11	5,222	1978	2016
Wheat	102,669	29	4,177	1978	2016

Panel B. Mean (SD) by crop

Crop	Observed log yield	GDD	EDD	Precipitation
Maize	0.811 (1.124)	2.084 (0.996)	0.012 (0.019)	0.873 (0.451)
Rice	0.424 (0.717)	1.194 (0.259)	0.043 (0.043)	1.160 (0.400)
Wheat	0.838 (0.511)	1.465 (0.534)	0.036 (0.034)	0.739 (0.490)

Notes: Main-season observations with finite observed yield, GDD, EDD, both precipitation measures, and all three PBM yields. Administrative-unit counts combine ADM1 and ADM2 units. Yield is the log of tonnes per hectare. GDD (growing degree days) sum degree-days between each crop's lower and upper temperature thresholds, and EDD (exceedance degree days) sum degree-days above the upper threshold (Appendix Table B2); both are weighted across irrigated and rainfed harvested area and divided by 1,000. Precipitation is the seasonal total in millimeters, weighted the same way and divided by 1,000. Full distributions, including the raw PBM inputs, appear in Appendix B.1.

2.4.4 Future projection inputs

The projection panel covers 2015–2100 under SSP1-2.6 and SSP3-7.0 for three CMIP6 climate models: MPI-ESM1-2-HR, IPSL-CM6A-LR, and MRI-ESM2-0. Daily temperature and precipitation come from the W5E5-bias-corrected ISIMIP3b climate files. We use the historical crop calendars, temperature thresholds, and harvested-area weights to construct future GDD, EDD, and both precipitation measures. GFDL-ESM4 and UKESM1-0-LL are excluded because their saved temperature-threshold panels fail a Celsius-unit consistency check.

Projected PBM yields come from ISIMIP3b for the same scenarios and climate models. Irrigated and rainfed outputs on a 0.5° grid are aggregated to ADM1, which is coarser than much of the historical estimation sample. We use the simulations that hold management inputs and atmospheric CO_2 at 2015 levels (ISIMIP3b `2015soc` and `2015co2`). These PBM runs retain their specified management settings and rules and exclude future CO_2 fertilization. The empirical component instead carries forward the weather response estimated over 1978–2016; its within-season farmer responses may be embedded in that estimate. Neither component adds future adaptation or a technology-driven yield trend, and the empirical equation contains no CO_2 term. pDSSAT is unavailable under SSP3-7.0, so the historical combination is re-estimated using the three components available for that scenario, as described in Section 2.5.4.

The reported end-of-century comparisons use 2096–2100 relative to 2015. A precipitation-processing error affected 2021–2025, 2031–2035, and each corresponding five-year block through 2091–2095: hourly files were passed to crop-calendar calculations that indexed time in days. These years are excluded from the reported projection comparisons. The 2015 baseline and 2096–2100 use daily-derived precipitation inputs in all three climate models and both scenarios. The five-year reporting window is common to every crop and component; it is not an estimate for the excluded years.

2.5 Methodology

In plain terms, the hybrid is a weighted average of four predictions of the same observed yields: one from a weather–yield regression and three from calibrated process-based crop models (PBMs). Stage 1 estimates the regression for each crop. Stage 2 calibrates each PBM's simulated yields to the observations and chooses the weights that best fit observed yields, subject to the weights being nonnegative and summing to one. We then re-estimate everything with each region held out to test prediction elsewhere, and apply the historical weights to projected changes in weather and simulated yields.

2.5.1 Stage 1: Empirical climate-yield model

For each crop $c \in \{\text{maize, rice, wheat}\}$, the empirical component relates observed log yield to temperature and precipitation exposure within locations and seasons:

$$\log(Y_{i,t}^{(c)}) = \beta_1 \text{GDD}_{i,t} + \beta_2 \text{EDD}_{i,t} + \beta_3 R_{i,t} + \beta_4 Q_{i,t} + \alpha_{s(i)} + \lambda_{st} + \varepsilon_{i,t}, \tag{2.4}$$

where $Y_{i,t}^{(c)}$ is observed yield in location i and year t. GDD sums degree-day exposure between the crop-specific lower and upper temperature thresholds; EDD sums degree-days above the upper threshold. Both are calculated over the selected crop window, combined across irrigated and rainfed land using harvested-area shares, and divided by 1,000. R is total precipitation across planting, middle-growth, and harvest periods, similarly weighted and divided by 1,000. Q sums *squared daily precipitation* across those periods before the same weighting and scaling. Thus Q measures rainfall concentration across days; it is not R^2. The coefficient on Q cannot be interpreted as curvature in a quadratic response to seasonal rainfall. Season-location fixed effects,

$\alpha_{s(i)}$, absorb persistent yield differences; season-year fixed effects, λ_{st}, absorb shocks common to locations in a given season and year. Standard errors are clustered by season-location.

We estimate the equation on the crop-specific sample shared with Stage 2, from 1978 to 2016. The temperature thresholds and crop windows were selected by leave-one-region-out cross-validation on a broader observed-yield and weather panel, before requiring complete PBM outputs. Maize uses $T_\ell = 0°\text{C}$ and $T_u = 30°\text{C}$ during harvest; rice uses $T_\ell = 12°\text{C}$ and $T_u = 28°\text{C}$ during harvest; wheat uses $T_\ell = 8°\text{C}$ and $T_u = 28°\text{C}$ during planting and middle growth. The precipitation variables cover all three periods for every crop. Appendix B.1 gives the exposure formulas and selection results.

The historical fitted value $\hat{y}_{i,t}^{\text{emp}}$ includes both sets of fixed effects and enters the ensemble fit. Its weather coefficients describe the net yield association in the observed sample. Future predictions use changes in the weather terms only, as specified below; we do not estimate or carry forward future season-year effects.

2.5.2 Stage 2: Constrained ensemble weighting

We first aggregate each PBM yield series to the corresponding observed administrative units. On the crop-specific common sample, we estimate a separate calibration regression for each PBM m:

$$\log(Y_{i,t}^{(c)}) = \gamma_m \tilde{y}_{i,t}^{m} + \alpha_{s(i)}^{m} + \lambda_{st}^{m} + e_{i,t}^{m}, \tag{2.5}$$

where $\tilde{y}_{i,t}^{m}$ is simulated log yield. The season-location and season-year fixed effects have the same structure as in Stage 1. The calibrated fitted value $\hat{y}_{i,t}^{m}$ includes these fixed effects and the estimated slope $\hat{\gamma}_m$. Stage 2 combines the three calibrated values with the fitted empirical value; it does not weight raw simulated yield levels.

This calibration prevents additive log-yield bias in the fixed-effect span from mechanically lowering a PBM's weight in the historical fit. Adding a location-specific constant or a season-year-specific common shift to simulated log yields can be offset by the calibration fixed effects, leaving the regression's fitted values unchanged on the estimation sample. This invariance does not remove weather-dependent error, identify the source of a bias, or provide a correction for location effects in unseen regions. The slope rescales the simulated variation associated with observed yields. It also multiplies projected PBM changes, as shown below, so calibration affects the projections as well as the historical fit. Table 2.5 reports the slopes, and Appendix Table B12 compares projections with and without this rescaling while retaining the same weights.

The fitted hybrid is

$$\hat{y}_{i,t}^{\text{hyb}} = \omega_{\text{emp}}\hat{y}_{i,t}^{\text{emp}} + \omega_{\text{EPIC}}\hat{y}_{i,t}^{\text{EPIC}} + \omega_{\text{PDSSAT}}\hat{y}_{i,t}^{\text{PDSSAT}} + \omega_{\text{LPJmL}}\hat{y}_{i,t}^{\text{LPJmL}}, \tag{2.6}$$

with crop-specific weights satisfying

$$\omega_m \geq 0 \quad \text{for every component } m, \qquad \sum_m \omega_m = 1. \tag{2.7}$$

We choose the weights to minimize squared differences between observed log yields and $\hat{y}^{\text{hyb}}$ on the common sample. In implementation, $\omega_m = \exp(a_m)/\sum_j \exp(a_j)$, and BFGS minimizes the residual sum of squares over the a_m. The fitted weights are positive and sum to one. They describe contributions to the historical fitted prediction conditional on the included components, not shares of physiological damage or farmer adaptation.

2.5.3 Leave-one-region-out validation

To assess geographic transfer, we hold out Brazil, the United States, Europe, India and China together, or Kenya and Nigeria together. For each eligible fold, we re-estimate the empirical and PBM calibration regressions on the remaining regions. We align the fitted component values with the observations retained after fixed-effect singleton removal, then solve the nonnegative, sum-to-one least-squares problem directly by comparing its feasible component subsets. The full-sample weights use the softmax optimizer described above; the direct fold solver also allows a component to receive exactly zero weight. Fixed effects for an unseen region cannot be predicted from its training sample. Held-out predictions therefore use the estimated slopes only, and prediction errors are computed after subtracting one mean for the held-out region from each series. The resulting root mean squared error (RMSE) includes both variation over time and yield-level differences between locations within that region; it removes only the overall regional mean. Because scoring uses the held-out mean of observed yield, it is not an operational forecast of yield levels. In-sample residual RMSE includes fitted fixed effects, so its magnitude is not directly comparable with the held-out score.

A crop-fold combination is omitted if it has no held-out observations or too few training observations. Maize has four eligible folds because its common sample contains no European observations. We compare the constrained hybrid with its components and record how its empirical weight changes when each region is omitted. The temperature thresholds and windows above were selected using the same geographic grouping on the broader weather–yield panel. Both exercises have four eligible maize folds, but their observation samples differ. Geographic RMSE therefore assesses transfer *conditional on the previously selected specification*; it is not an independent nested cross-validation of the entire model-selection procedure.

2.5.4 Projection framework

For each projected location, year, scenario, and climate model, we express the empirical response as a log change from 2015:

$$\hat{\delta}^{\text{emp}}_{i,t} = \hat{\beta}_1 \Delta \text{GDD}_{i,t} + \hat{\beta}_2 \Delta \text{EDD}_{i,t} + \hat{\beta}_3 \Delta R_{i,t} + \hat{\beta}_4 \Delta Q_{i,t}, \tag{2.8}$$

where Δ is the difference from that location's 2015 exposure under the same scenario and climate model. This calculation uses the historical weather slopes and omits future fixed effects. For PBM m, let $\hat{\gamma}_m$ be the slope in Equation 2.5. Holding its calibration fixed effects at the baseline contribution, the projected change in that fitted component is

$$\hat{\delta}^{m}_{i,t} = \hat{\gamma}_m \left[\log(\hat{Y}^{m}_{i,t,\text{future}}) - \log(\hat{Y}^{m}_{i,2015,\text{baseline}}) \right]. \tag{2.9}$$

The four-component hybrid change is the same weighted combination of component changes:

$$\hat{\delta}^{\text{hyb}}_{i,t} = \omega_{\text{emp}} \hat{\delta}^{\text{emp}}_{i,t} + \omega_{\text{EPIC}} \hat{\delta}^{\text{EPIC}}_{i,t} + \omega_{\text{PDSSAT}} \hat{\delta}^{\text{PDSSAT}}_{i,t} + \omega_{\text{LPJmL}} \hat{\delta}^{\text{LPJmL}}_{i,t}. \tag{2.10}$$

We apply the historical weights to Main-season projections at ADM1, using a unique 2015 baseline for each location, crop, scenario, and climate model. This is a change in resolution from the historical sample, which includes ADM2 observations. The geographic holdouts provide a partial check on transferring the fitted response beyond calibration regions, although the historical sample has limited African coverage.

pDSSAT projections are unavailable under SSP3-7.0. For that scenario, we refit the historical empirical model, PBM calibrations, and constrained weights on the

common sample of the three available components: the empirical model, EPIC, and LPJmL. We calculate its change using Equation 2.10 with these three refitted components and their weights. The refit uses the historical sample common to the reduced library, which can differ from the four-component sample. Its empirical coefficients also generate the SSP3-7.0 empirical-only comparison. Appendix B.5 compares this refitted combination with two alternatives: rescaling the four-component weights after dropping pDSSAT, and keeping the four-component weights while assigning pDSSAT no change.

These are climate-only changes relative to 2015. The PBM simulations hold management inputs and atmospheric CO_2 at 2015 levels. The empirical component instead carries forward the weather response estimated over 1978–2016; it does not condition on observed 2015 management. Both projections assume the short-run weather response applies to persistent climate change, without adding future adaptation, technological change, or CO_2 fertilization. Historical season-year effects are not extrapolated as a secular yield trend. The hybrid–empirical difference reflects the component weather responses, their calibration scales, and their fitted weights. Results use the three retained climate models. Their spread describes variation within this set, not uncertainty across all CMIP6 models.

2.5.5 Model comparison

We compare the empirical model, each PBM, an equal-weight PBM combination, an unrestricted three-PBM regression, an unrestricted four-component regression, and the constrained hybrid on the same crop-specific historical sample. We report in-sample RMSE and fit statistics, and use leave-one-region-out RMSE for the geographic comparison. The unrestricted regressions show the fit available without the adding-up and sign restrictions; their coefficients can be negative and need not

sum to one. The constrained weights instead give a combination rule whose terms are shares, which we apply to the projected component changes. Appendix B.3 reports the unrestricted PBM coefficients and compares the ensemble benchmarks.

End-of-century summaries pool the 2096–2100 projection records and retain 2015 as the baseline. The window excludes the precipitation-processing error described in Section 2.4.

2.6 Results

The results answer three questions in turn: how observed yields respond to weather, how the hybrid weights the four components and how well it predicts in regions left out of estimation, and how the fitted combination changes projected yields under future climate.

2.6.1 Historical climate-yield relationships

Extreme heat is associated with lower observed yields for all three crops (Table 2.3). Growing degree days (GDD) and exceedance degree days (EDD) are measured in thousands of degree-days. Holding the other regressors and fixed effects constant, 10 additional EDD are associated with approximately 4.5% lower maize yield, 6.9% lower rice yield, and 0.8% lower wheat yield. These are net associations in observed fields, where crops and contemporaneous management both respond to weather.

Exposure in the selected GDD band is positively associated with maize and rice yields but negatively associated with wheat yields. For wheat, this variable measures exposure between 8°C and 28°C during planting and middle growth. It should not be interpreted as a beneficial temperature band: for a cool-season crop, its upper portion may already include damaging heat, so GDD and EDD may describe

two levels of heat exposure. The cutoffs were selected for prediction, rather than to identify physiological thresholds; several nearby choices have similar mean validation errors, and wheat's selected upper cutoff is the lowest value searched (Appendix B.1). Seasonal rainfall total has a positive coefficient for maize and rice and a negative coefficient for wheat. Conditional on that total, the sum of squared daily rainfall has a negative coefficient for all three crops. This second term describes the distribution of rainfall across days; it does not establish a concave response to seasonal rainfall. Both rainfall measures cover planting, middle growth, and harvest, including for wheat. Appendix B.2 compares the observed weather coefficients with corresponding PBM associations on the matched sample.

2.6.2 Ensemble weights and predictive performance

The four-component hybrid places very different historical weights on its empirical component: 0.329 for maize, 0.937 for rice, and 0.251 for wheat (Table 2.4). EPIC and LPJmL receive the remaining maize weight; pDSSAT receives most of the remaining wheat weight. These shares show how the components combine to fit observed yield levels. They are the weights used for the SSP1-2.6 projections. Because pDSSAT has no SSP3-7.0 simulations, the SSP3-7.0 projections use weights re-estimated with the empirical model, EPIC, and LPJmL only, on the larger sample available without pDSSAT (Appendix Table B10). Removing pDSSAT shifts weight to the empirical component for wheat, from 0.251 to 0.534, while the maize and rice empirical weights change little. The rice empirical weight remains between 0.86 and 0.91 when each eligible geographic group is omitted in turn. Maize is less stable: omitting Brazil lowers its empirical weight from 0.33 to 0.00, while omitting the United States raises it to 0.57. The location-bootstrap intervals in Table 2.4 give a

Table 2.3: Weather and Observed Log Yield: Fixed-Effects Estimates

	Maize	Rice	Wheat
GDD	0.287***	0.816***	-0.284***
	(0.018)	(0.054)	(0.019)
EDD	-4.621***	-7.118***	-0.825***
	(0.211)	(0.240)	(0.079)
Precipitation	0.335***	0.356***	-0.094***
	(0.013)	(0.021)	(0.009)
Squared daily-rainfall sum (Q)	-0.018***	-0.023***	-0.003**
	(0.003)	(0.005)	(0.002)
Season-year fixed effects	Yes	Yes	Yes
Season-location fixed effects	Yes	Yes	Yes
Observations	247,764	133,430	102,503
R^2	0.846	0.669	0.722
Within R^2	0.020	0.075	0.010
RMSE	0.441	0.413	0.269

Notes: The dependent variable is log yield in tonnes per hectare. GDD (growing degree days) sum degree-days between the crop-specific lower and upper temperature thresholds, and EDD (exceedance degree days) sum degree-days above the upper threshold: 0 and 30°C during harvest for maize, 12 and 28°C during harvest for rice, and 8 and 28°C during planting and middle growth for wheat. Both are measured in thousands of degree-days. Precipitation is the seasonal total (R); Q sums squared daily rainfall. Both are weighted across irrigated and rainfed area and divided by 1,000, and Q is not R^2. Standard errors, clustered by season-location, are in parentheses. Main-season observations only. Observation counts are slightly below those in Table 2.2 because the fixed-effects estimator removes singleton groups. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

separate measure of sampling uncertainty; the rice interval is wide even though its point estimate is high.

Table 2.4: Historical Weights of the Four-Component Constrained Hybrid

Crop	Statistic	Empirical	EPIC	pDSSAT	LPJmL
Maize	Full-sample estimate	0.329	0.505	0.000	0.166
	Bootstrap median	0.321	0.496	0.021	0.161
	Bootstrap 95% interval	[0.280, 0.372]	[0.444, 0.556]	[0.000, 0.049]	[0.071, 0.237]
Rice	Full-sample estimate	0.937	0.026	0.003	0.033
	Bootstrap median	0.957	0.011	0.010	0.019
	Bootstrap 95% interval	[0.263, 1.000]	[0.000, 0.256]	[0.000, 0.267]	[0.000, 0.356]
Wheat	Full-sample estimate	0.251	0.023	0.707	0.019
	Bootstrap median	0.096	0.024	0.817	0.025
	Bootstrap 95% interval	[0.000, 0.325]	[0.000, 0.263]	[0.219, 1.000]	[0.000, 0.302]

Notes: Weights of the empirical model and the calibrated EPIC, pDSSAT, and LPJmL components in the four-component constrained hybrid, estimated on each crop's historical sample. Weights are nonnegative and sum to one within each crop. These full-sample weights are used for the SSP1-2.6 projections. pDSSAT has no SSP3-7.0 simulations, so the SSP3-7.0 projections use three-component weights re-estimated without it (Appendix Table B10). The location bootstrap resamples locations with replacement and re-estimates both stages; it reports the median and the 2.5th and 97.5th percentiles of 100 draws. Because the fit can be flat near the optimum, a full-sample estimate need not lie inside its bootstrap interval.

Table 2.5 reports the PBM calibration slopes for both projection specifications. All are below one, so applying the fitted components to future changes reduces the scale of each raw PBM response. Calibration affects the projected changes as well as the historical component predictions.

A high historical weight does not imply the best prediction elsewhere (Table 2.6). In sample, the lowest root mean squared error (RMSE) among individual components belongs to calibrated EPIC for maize, the empirical component for rice, and calibrated pDSSAT for wheat, and the constrained hybrid fits at least as well as each individual component. In held-out regions, scored after removing each region's overall mean, the ranking changes. The constrained hybrid has RMSEs of 0.815, 0.694, and 0.411 for maize, rice, and wheat. It predicts better than the empirical

Table 2.5: PBM Calibration Slopes by Historical Sample

Sample	Crop	EPIC	pDSSAT	LPJmL	Observations
Four components	Maize	0.592	0.421	0.775	247,764
Four components	Rice	0.195	0.009	0.304	133,430
Four components	Wheat	0.138	0.243	0.136	102,503
Three components	Maize	0.587	–	0.762	260,902
Three components	Rice	0.192	–	0.303	137,398
Three components	Wheat	0.138	–	0.130	104,219

Notes: Each slope comes from a separate regression of observed log yield on simulated log yield with season-location and season-year fixed effects. The four-component sample requires all three PBMs and is used for SSP1-2.6. The larger three-component sample omits pDSSAT and is used for SSP3-7.0. Slopes multiply raw PBM log-yield changes in the calibrated projections.

component alone for every crop (0.850, 0.696, and 0.429) and better than the equal-weight combination of calibrated PBMs for maize and wheat, but not for rice. It is not the most accurate predictor for any crop. Calibrated pDSSAT predicts maize better (0.752), a gap of 0.063 log points; for rice and wheat, pDSSAT and the constrained hybrid differ by less than 0.001. Calibrated EPIC performs best for rice (0.682), and the unconstrained hybrid performs best for wheat (0.409). These are descriptive rankings; the reported errors do not establish statistical equivalence or significant differences.

The validation score pools squared errors across observations, rather than giving each region equal weight. Brazil supplies 95% of rice and 69% of maize test observations, and the United States supplies 68% of wheat test observations. Moreover, weights are fitted to component predictions including fixed effects, whereas held-out predictions use only slopes and are centered once within each region. Differences between locations in that region remain in the score. Better fit in the training sample therefore need not improve this different prediction task. Appendix B.4 reports fold-specific errors and weight changes. The constrained hybrid supplies a

transparent nonnegative combination rule for projections; these validation results do not establish its predictive superiority.

Table 2.6: Prediction Errors of Empirical, Calibrated Process-Based, and Hybrid Models

	In-sample RMSE			Held-out-region RMSE		
Model	Maize	Rice	Wheat	Maize	Rice	Wheat
Empirical model	0.441	0.413	0.269	0.850	0.696	0.429
Calibrated EPIC	0.439	0.428	0.270	0.844	0.682	0.434
Calibrated pDSSAT	0.442	0.429	0.268	0.752	0.694	0.412
Calibrated LPJmL	0.441	0.426	0.270	0.922	0.683	0.426
Equal-weight calibrated PBMs	0.439	0.427	0.269	0.830	0.686	0.418
Unconstrained hybrid	0.434	0.412	0.267	0.798	0.697	0.409
Constrained hybrid	0.437	0.413	0.268	0.815	0.694	0.411

Notes: RMSE in log-yield units; lower is better. All predictors are evaluated against observed yields on each crop's common historical sample. PBM rows evaluate statistical calibrations of simulated yields, not raw crop-model outputs. For each held-out region, the empirical slopes, PBM calibration slopes, and hybrid weights are fitted using the remaining regions. The PBM simulations themselves are not retrained, and these scores do not establish independent out-of-sample performance relative to the crop models' original development data. Held-out predictions omit unavailable fixed effects; both observed and predicted log yields are centered within each fold before scoring. Centering removes one overall regional mean; differences between locations within the held-out region remain in the score. These are not operational forecasts of yield levels. In-sample errors include fitted fixed effects, so the two columns assess different prediction tasks. Maize has four eligible folds; rice and wheat have five. Temperature windows were selected on a broader observed-yield and weather panel before this common-sample validation.

2.6.3 Projected yield changes

Under SSP3-7.0, the hybrid projects median yield changes over 2096–2100, relative to 2015, of −16.6% for maize, −47.8% for rice, and −10.7% for wheat (Table 2.7). The corresponding changes under SSP1-2.6 are −2.4%, −8.7%, and −1.3%. These medians pool finite subnational-year-climate-model records; they are neither changes in global production nor harvested-area-weighted world-yield estimates.

The empirical climate component alone projects SSP3-7.0 median losses of −30.1% for maize, −50.6% for rice, and −16.8% for wheat. The hybrid projects

smaller maize and wheat losses than the empirical component, while the rice projection changes less. These differences reflect both the ensemble weights and the calibration slopes applied to PBM changes. Using the same weights without PBM rescaling changes the SSP3-7.0 median from −16.6% to −19.4% for maize and from −10.7% to −18.0% for wheat (Appendix Table B12). For wheat, the unscaled hybrid loss is therefore slightly larger than the empirical loss: the smaller hybrid loss depends on the calibration slopes rather than on the combination alone. This comparison holds the weights fixed and does not identify why the calibration slopes are below one. SSP1-2.6 uses the four-component hybrid. SSP3-7.0 uses the three-component hybrid refitted without pDSSAT, which has no projection for that scenario. Appendix B.5 reports sensitivity to that missing member and to PBM calibration.

Table 2.7: Median Yield Changes in 2096–2100 Relative to 2015

Crop	Scenario	Empirical model (%)	Hybrid (%)
Maize	SSP1-2.6	−2.51	−2.38
Maize	SSP3-7.0	−30.12	−16.61
Rice	SSP1-2.6	−9.12	−8.74
Rice	SSP3-7.0	−50.62	−47.79
Wheat	SSP1-2.6	−2.99	−1.32
Wheat	SSP3-7.0	−16.77	−10.74

Notes: Entries are medians of the percentage change in yield relative to 2015 across finite subnational projection records for 2096–2100, pooling three climate models (MPI-ESM1-2-HR, IPSL-CM6A-LR, and MRI-ESM2-0). They are not weighted by harvested area and are not changes in production. The empirical model applies the historical weather coefficients alone. The SSP1-2.6 hybrid uses the four-component weights in Table 2.4; the SSP3-7.0 hybrid uses three-component weights re-estimated without pDSSAT, which has no SSP3-7.0 simulations (Appendix Table B10). Projection assumptions are stated in Section 2.5.4.

The end-of-century maps show why the pooled medians cannot describe every location (Figure 2.2). Among locations with positive harvested-area weights and

complete three-model projections, the figure's area-weighted medians are −11.9% for maize, −9.4% for rice, and −0.2% for wheat. These changes are closer to zero than those in Table 2.7 because the two summaries weight differently: the map weights locations by harvested area and keeps only locations with complete three-model projections, whereas the table gives every projection record equal weight. Appendix Tables B13 and B14 give regional and country summaries on their stated aggregation scales.

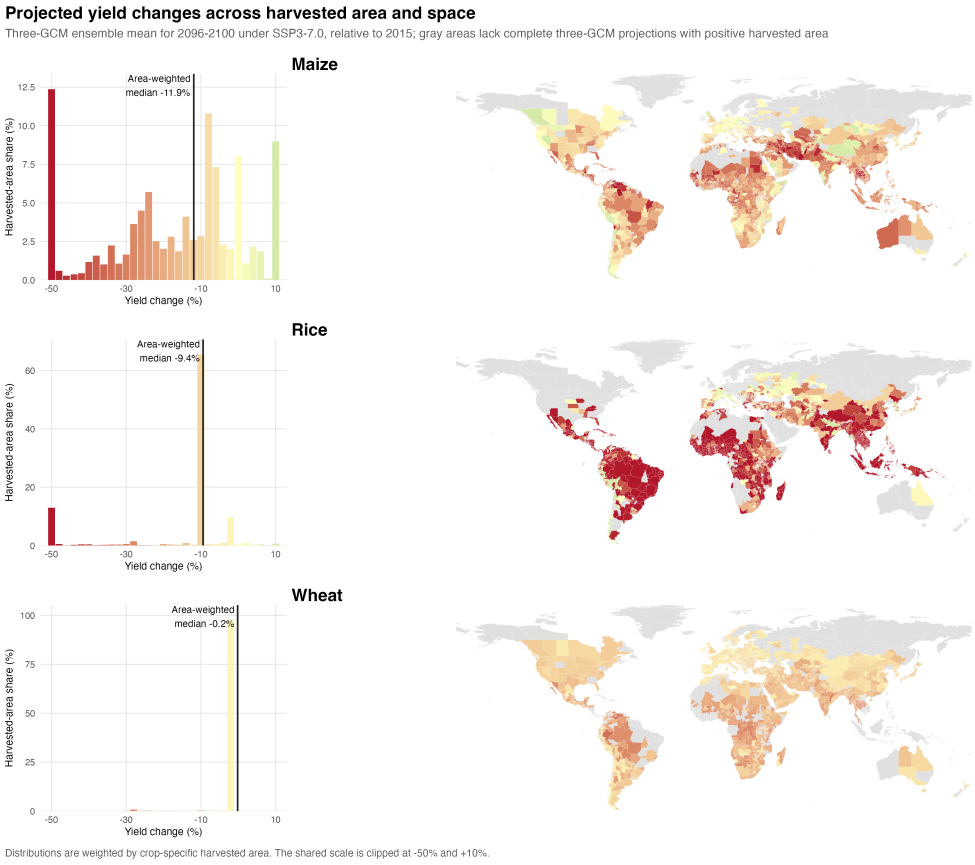


Figure 2.2: Subnational Hybrid Yield Changes under SSP3-7.0, 2096–2100. For each mapped location, the change relative to 2015 is averaged over 2096–2100 and three climate models. Histograms weight locations by crop-specific harvested area, and black lines mark area-weighted medians. Gray areas lack a complete three-model projection with positive harvested area. The color scale is clipped at −50% and +10%. Projection assumptions are stated in Section 2.5.4.

2.7 Discussion

2.7.1 What the hybrid contributes

Empirical weather models and process-based crop models (PBMs) are often treated as competing ways to project agricultural climate damage. This chapter instead matches both to the same subnational yields for maize, rice, and wheat and estimates crop-specific weights for one fitted empirical component and three calibrated PBMs. It extends the comparison of statistical and process-based maize models by Roberts et al. (2017) to three crops and several continents, adds explicit weights to the coordinated PBM ensembles of Jägermeyr et al. (2021) and Müller et al. (2021), and complements predictors that feed crop simulations into flexible learners (Shahhosseini et al., 2021; von Bloh et al., 2024) by keeping each component's contribution visible.

Three findings follow. First, the fitted balance differs by crop: the empirical response dominates the historical fit for rice, shares it with EPIC and LPJmL for maize, and receives a quarter of the weight for wheat, where pDSSAT dominates. Second, historical weight is not the same as transferability. The rice empirical weight coexists with better held-out prediction from the calibrated EPIC component, and the maize weight depends heavily on whether Brazil is in the estimation sample. A fitted weight should therefore be read as a property of the historical sample. Third, the choice of combination changes the size of projected losses, most for maize and wheat. These geographic comparisons are conditional on the weather specification selected earlier on a broader observed-yield and weather panel.

The constrained hybrid predicts held-out regions better than the empirical component alone for every crop but is not the most accurate predictor for any crop. Calibrated pDSSAT has lower held-out RMSE for maize, and for rice and wheat the two differ by less than 0.001. The weights are estimated from historical fitted values, so their fit within the estimation sample need not carry over to excluded regions. The

constrained hybrid supplies one weighted average for comparing climate projections; its interpretable shares do not establish predictive superiority. Historical prediction and future response also answer different questions: a comparison of held-out yield levels does not validate extrapolation to persistent warming.

2.7.2 What the comparison says about adaptation

The framework in Section 2.3 explains why the hybrid cannot measure adaptation directly. The empirical response reflects the weather sensitivity of existing farming systems, including the sensitivity that past adaptation produced; the PBM responses reflect prescribed management and each model's crop processes. Their difference can also come from the different weather forcing behind the two kinds of yield, from measurement error and spatial aggregation, and from differences in crop-model structure and parameterization. Both the empirical model and PBM calibration include fixed effects, so their presence does not distinguish the two approaches.

The matched heat responses show why a simple adaptation interpretation fails. Observed maize and rice yields have more negative extreme-heat coefficients than the three PBMs, while for wheat the PBM coefficients are more negative than the observed one (Appendix B.2). The simulations therefore do not uniformly understate heat damage, and neither the sign nor the size of an empirical–PBM gap identifies what farmers did. The comparisons point to specific questions: whether irrigation, planting dates, varieties, or input use change after heat exposure, and whether persistent warming produces responses different from those to annual shocks. Answering them requires management data or a design that separates those decisions from direct crop responses.

2.7.3 Interpreting the climate projections

The projections apply the empirical weather response estimated over 1978–2016 and PBM responses simulated with management inputs and atmospheric CO_2 fixed at 2015 levels. Applying the short-run response over the long run is a projection assumption, not an estimated equality between the two responses. We project neither additional adaptation nor future technology or yield trends. The effects of relaxing that assumption need not have a known sign: future adjustment could reduce losses, while short-run responses need not remain feasible under persistent warming.

The smaller hybrid losses for maize and wheat depend on both the estimated weights and the calibration slopes. Each slope rescales an entire PBM response, including its projected change. This statistical scale adjustment is especially consequential for wheat, so the gap between hybrid and empirical projections cannot be attributed to combining unaltered crop-model responses. With the same weights and projection records, removing the scale adjustment changes the SSP3-7.0 wheat median from -10.7% to -18.0%, slightly below the empirical model's -16.8% (Appendix Table B12). This sensitivity comparison isolates the effect of rescaling at the fitted weights; it is not a re-estimated hybrid.

The headline record-level medians and the harvested-area-weighted maps answer different descriptive questions. The former give each subnational-year-climate-model record equal weight; the latter describe changes across mapped harvested area with complete climate-model coverage. Neither is a change in global production, which would also depend on acreage, crop choice, prices, and trade. The local disagreement among component projections (Appendix Figure B2) shows that one median cannot describe every growing region.

The projections also rest on three climate models, on an ADM1 grid coarser than much of the historical sample, and on weights estimated where African observations are sparse. A broader climate-model set would test sensitivity to the selected climate inputs. Allowing CO_2 and management to change would extend the projection beyond the conditions studied here.

2.8 Conclusion

Projected climate damage to agriculture depends on how observed field evidence and crop simulations are combined, and the fitted combination differs by crop. The hybrid model proposed here gives the empirical weather response nearly all of the historical weight for rice, about a third for maize, and a quarter for wheat, with the rest going to calibrated process-based components. Those weights summarize historical fit rather than predictive rank: in regions held out of estimation, the constrained hybrid improves on the empirical model alone but is not the most accurate predictor for any crop, and the maize weight shifts sharply when Brazil is excluded.

The balance between the two kinds of evidence changes projected losses. Under SSP3-7.0, using the historical empirical response and PBM simulations with management inputs and atmospheric CO_2 fixed at 2015 levels, the hybrid's median subnational yield changes for 2096–2100 are -16.6% for maize, -47.8% for rice, and -10.7% for wheat, compared with -30.1%, -50.6%, and -16.8% from the empirical model alone. These differences depend on both the fitted weights and the calibration slopes that rescale projected PBM changes. Without that rescaling, the smaller wheat loss disappears.

The projections assume that short-run weather responses apply over the long run. They include no additional adaptation or future technology or yield trend. Linking yields and crop-model responses to consistent records of irrigation, planting

dates, varieties, and inputs would help distinguish farmers' adjustments from direct crop responses. The present results show that historical fit and held-out prediction do not consistently favor the constrained hybrid, even though the choice of weights and calibration materially changes projected climate losses.

Chapter 3

ECONOMIC VALUE OF RECREATIONAL CRABBING AND CLAMMING IN THE MID-ATLANTIC REGION

Muhammad Taky Tahmid, George R. Parsons

Abstract

Recreational access decisions require values for the activities people undertake. We estimate the value of blue-crab and clam recreation using a 2026 survey of Delaware fishing-license holders about their last activity day in 2025. Respondents considered whether they would still have gone at a randomly assigned additional cost. Nonparametric lower-bound mean willingness to pay is $65.91 per crabbing day and $58.38 per clamming day, based on 3,531 and 1,227 responses. Restricting responses to certainty of 10 lowers these values to $53.80 and $45.12; continuously mapping certainty into the response raises the valuation indices by about $7. Parametric values depend materially on the bid's functional form. These sensitivity exercises describe measurement choices rather than corrections for hypothetical bias. The estimates provide local activity-day values; applying them to a policy also requires comparable participants and evidence on changes in participation.

JEL codes: Q22, Q26, Q51, C83

Keywords: clamming, contingent behavior, crabbing, Delaware, recreational shellfishing, travel cost, Turnbull estimator, willingness to pay

3.1 Introduction

The benefits of recreational access depend on what people do at a site. A day spent harvesting blue crabs or clams may differ in value from a day of finfish angling, even in nearby coastal waters. The activities use different gear, require different effort, and face different access conditions. A value borrowed from another fishery can therefore be a poor guide to the benefits of preserving a particular activity. This problem matters when evaluating closures, habitat improvements, and changes in access to coastal recreation.

We ask how much participants value a completed crabbing or clamming day, using Delaware fishing-license holders as the setting. We estimate a value for each activity with a question tied to a trip the participant actually took. The preferred estimates imply lower-bound mean willingness to pay of about $66 for a crabbing day and $58 for a clamming day. These values provide activity-specific evidence for assessing the recreational benefits of access to coastal resources.

The survey, conducted in 2026, asked respondents about trips taken in 2025. Each participant described a recent trip and considered whether they would still have taken it at a randomly assigned additional out-of-pocket cost. This contingent-behavior travel-cost question ties the hypothetical choice to a trip the respondent actually took (Englin & Cameron, 1996). We estimate the distribution of stated willingness to pay over the offered cost range using 3,531 crabbing and 1,227 clamming responses. Our main nonparametric estimates measure a lower bound on mean trip-day value. Binary parametric models provide conditional medians, while certainty-mapped models supply a separate sensitivity index.

The survey also lets us examine how the measured values depend on respondents' certainty and the questions they have already answered. Restricting the sample

to the most certain respondents lowers estimated values; mapping certainty continuously into the response raises the resulting indices. The randomized-order check among dual-activity respondents gives lower clamming values when that module comes second. Parametric values also depend materially on whether the bid enters in levels or logs. The checks show which survey and modeling choices matter for valuation, without treating a particular adjustment as a measured correction for hypothetical bias.

Earlier studies value clamming in Massachusetts (Smith, Conrad, & Storey, 1976), mixed shellfishing on Cape Cod (Damery & Allen, 2004), and clam and oyster harvest under closures in Puget Sound (Anderson & Plummer, 2016, 2017). Their species, valued choices, and units differ. We add day values for blue-crab and clam recreation measured directly in this setting. The common question and unit provide a consistent approach to valuing both activities. The design avoids converting a mixed annual permit value into an activity-day value or borrowing an estimate from another shellfishery.

The estimates apply to completed trips reported by survey respondents. Using them in a benefit calculation requires comparable activity days and a separate prediction of the change in participation. The data section describes the sampling frame and response rate relevant to extending the values to all license holders. After presenting the setting and estimators, we report the activity values, examine survey-design sensitivity, and compare the estimates with the choices valued in earlier studies.

3.2 Background and Literature

Blue-crab and clam recreation illustrate why the activity being valued matters for benefit transfer. In the Delaware survey area, crabbing takes place in bays, rivers, and connected waters, while clamming is concentrated in the Inland Bays. Differences

in gear, trip patterns, and exposure to water-quality closures shape the recreational opportunities available to participants. Figure 3.1 shows the survey location zones.

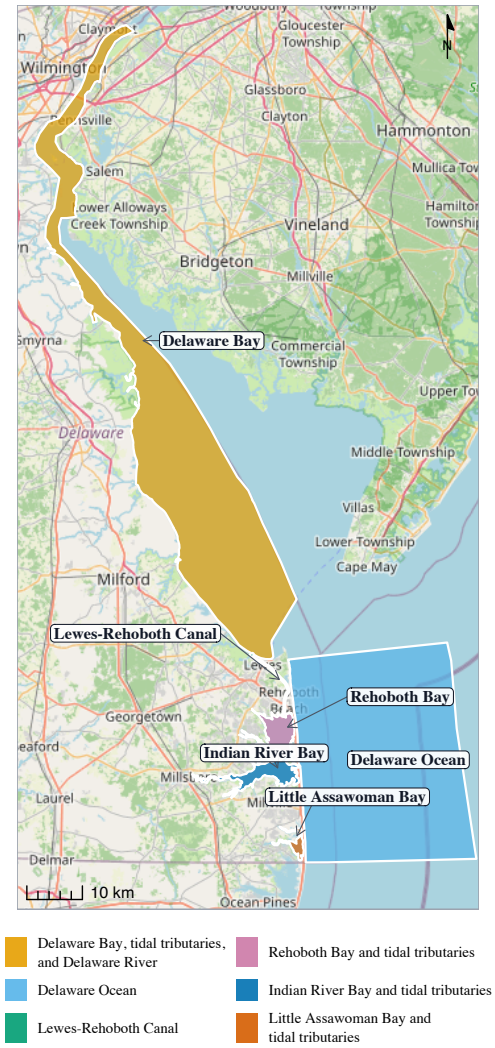


Figure 3.1: Shellfishing Locations Used in the Survey

Note: The broad zones match the map interface respondents used to report their crabbing and clamming locations.

The recreation-demand literature offers established methods for measuring the benefit from a fishing day. Travel-cost and stated-preference studies estimate consumer surplus for many finfish fisheries (Phaneuf & Smith, 2005; Haab & Mcconnell,

2002). Shellfishing studies use related methods but value several different choices. Smith et al. (1976) estimates the value of recreational clamming in Massachusetts. Damery and Allen (2004) values an annual permit for mixed shellfishing on Cape Cod; expressing that estimate per day requires dividing the annual value by reported trips. A NOAA technical report (Anderson & Plummer, 2016) and a subsequent peer-reviewed study (Anderson & Plummer, 2017) examine expected clam and oyster trips under pollution and biotoxin closures in Puget Sound. Their policy scenarios allow participation to change with access conditions. The Recreation Use Values Database converts results from the Cape Cod and Puget Sound studies to common per-day units (Rosenberger, 2025). We use those benchmarks to interpret the scale of our estimates after discussing the differences in what each study values.

Regional surveys supply evidence on participants and harvest. Studies of crabbers in South Carolina and North Carolina document effort and catch (Low, 1998; Vogelsong & Nobles, 2004), while a Chesapeake-region marine recreation survey describes catch and participants (Ashford, Jones, & Fegley, 2010). The studies reviewed here provide context for shellfishing participation. Their effort and catch measures answer different questions from the benefit participants receive from an outing. Our valuation evidence concerns blue-crab and clam day values in a Delaware-centered setting.

We follow the contingent-behavior travel-cost approach of Englin and Cameron (1996). Respondents consider an additional cost for an outing they have already taken. The reported outing supplies a concrete reference for the choice, and randomized cost increments reveal acceptance at prices beyond those observed in travel data. Discrete-choice and travel-cost models provide the demand framework (Cameron, 1992; Whitehead, Pattanayak, Houtven, & Gelso, 2008). Within that framework, the Turnbull estimator gives a lower bound on mean stated willingness to pay (Haab

& Mcconnell, 2002); binary and fractional logits summarize binary acceptance and certainty-adjusted responses (Papke & Wooldridge, 1996). The certainty follow-up allows us to examine the role of reported confidence in the hypothetical answer (Blumenschein, Johannesson, Blomquist, Liljas, & O'Conor, 1998; Samnaliev & Stevens, 2006). The next sections describe the survey choices and the assumptions behind these summaries.

3.3 Data and Survey Design

The data come from a 2026 University of Delaware internet survey of Delaware Division of Fish and Wildlife license holders about the 2025 fishing season. The license registry included an email address for about 76 percent of holders. Beginning in late March 2026, the university sent invitations and subsequent reminders to 78,189 holders. The survey received 9,481 completed responses: 12.1 percent of all invitations and 12.4 percent of the 76,387 invitations recorded as delivered. The incomplete email frame and low completion rate limit how far respondent values can be generalized to the licensed population.

The questionnaire screened respondents for crabbing and clamming participation. Activity modules asked about trip frequency, harvest, fishing locations, and the cost of the most recent trip. Respondents active in both activities completed both modules in randomized order. Each module ended with the additional-cost valuation question and a certainty follow-up. Those who had not fished for either species proceeded to demographics; participants active in the previous five years but not in 2025 received a shorter questionnaire. The study followed Institutional Review Board protocols, and respondents were assured of confidentiality.

The final valuation samples are smaller than the participant groups described in the survey. Among completers, 4,717 reported crabbing and 1,958 reported clamming in the previous five years; 3,595 and 1,254, respectively, reported participation in 2025. The cleaned activity files exclude missing or "unsure" activity-day categories, leaving 3,549 crabbing and 1,234 clamming records. Requiring valid assigned bids and yes/no valuation answers then gives 3,531 crabbing and 1,227 clamming responses. The two samples overlap: 984 respondents contribute to both. These are the samples used for the preferred Turnbull estimates. Full-covariate models require additional non-missing items. We predict household income for 55 clamming and 177 crabbing records without reported income (4.5 and 5.0 percent of the respective valuation samples) and include an imputation indicator (Appendix Table C11).

Table 3.1 describes all completers and the broader five-year participant groups. About half are ages 50–69, roughly four in five are male, and about seven in ten report household income above $75,000. Sixty-four percent of completers live in Delaware; most others live in Pennsylvania, Maryland, New Jersey, or Virginia. In the final valuation samples, the Delaware resident shares are 48 percent for clamming and 54 percent for crabbing. Appendix Table C12 reports the characteristics of those samples and compares single- and dual-activity respondents. The license-holder frame thus reaches participants from neighboring states as well as Delaware residents.

Figure 3.1 shows the zones used in the survey's location questions. Crabbing is more common than clamming among respondents. Crabbers also report more activity days and longer outings, and the lifetime-experience distributions differ (Figure 3.2). These descriptions identify the participation and experience of the people whose reported outings we value. The resulting estimates pertain to those outings and participants.

Table 3.1: Demographics of Survey Completers and Five-Year Participants

	All completers	Clammers	Crabbers
Age			
Under 30	7%	4%	6%
30–49	23%	23%	25%
50–69	49%	54%	50%
70 and over	22%	19%	19%
Gender			
Male	82%	83%	81%
Education			
Less than high school	1%	1%	1%
High school or equivalent	22%	23%	23%
Some college	28%	28%	29%
Bachelor's degree	27%	28%	26%
Graduate or professional degree	21%	21%	20%
Employment			
Full time	52%	57%	55%
Part time	6%	5%	5%
Retired	38%	36%	36%
Student	1%	0%	1%
Other	2%	1%	2%
Household income			
Less than $50,000	12%	10%	11%
$50,000–$75,000	16%	16%	16%
$75,001–$150,000	42%	43%	43%
More than $150,000	30%	31%	30%
State of residence			
Delaware	64%	58%	64%
Pennsylvania	22%	33%	26%
Maryland	9%	5%	7%
New Jersey	3%	1%	2%
Virginia	2%	2%	2%
Boat owner			
Yes	40%	61%	51%
Environmental concern			
Strong environmentalist	27%	27%	26%
Somewhat environmentalist	33%	33%	33%
Concerned, not environmentalist	39%	39%	39%
Not a top concern	1%	1%	1%
N	9481	1958	4717

Note: Percentages are within-group shares among respondents with non-missing answers for that item; rows may not sum to 100% due to rounding. Education uses the five survey categories (including Some college). The $75,001–$150,000 income band combines the original $75,001–$100,000 and $100,001–$150,000 categories. Boat owner is the share reporting ownership. Sample: all survey completers (All completers) and those who crabbed or clammed in the last five years (Crabbers / Clammers; respondents may appear in both activity columns). These shares differ from the contingent-behavior estimation samples in the valuation results (non-missing bid and WTP response).

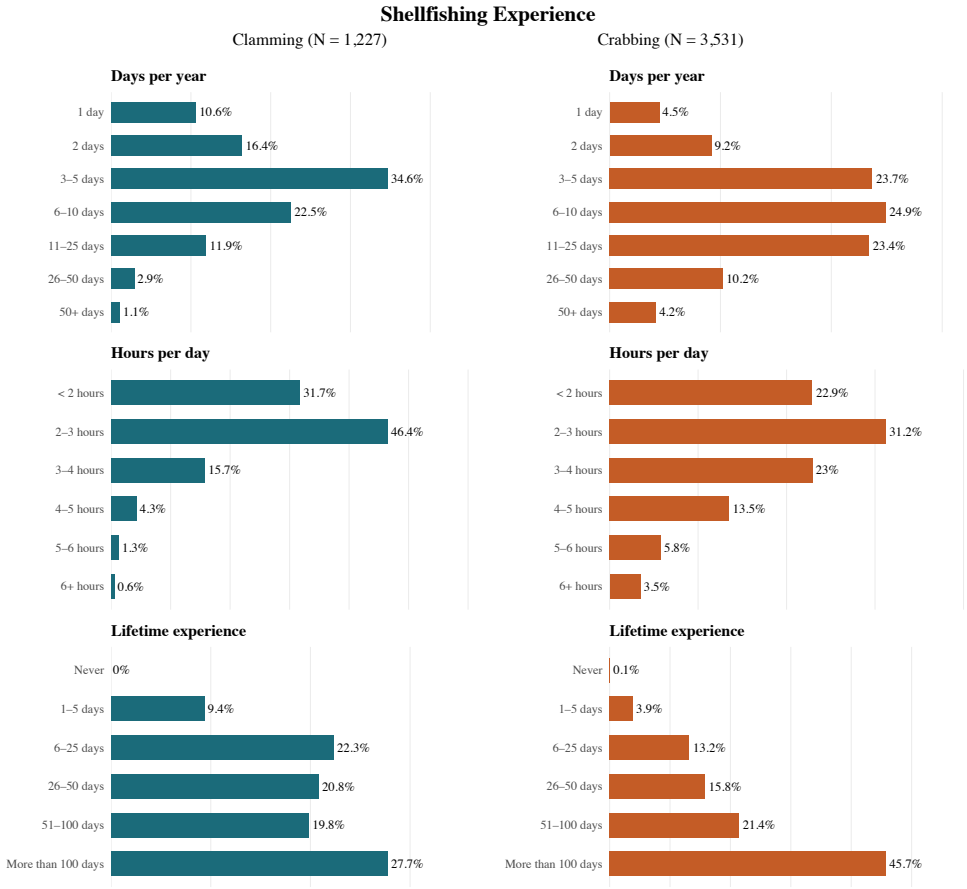


Figure 3.2: Participation and Experience in the Valuation Samples

Note: Bars show percentages within each reported response category for annual activity days, hours per outing, and lifetime activity days. The samples contain 1,227 clamming and 3,531 crabbing responses with a valid valuation bid and answer; each panel excludes missing answers to its item. Categories have open endpoints, so the figure reports distributions rather than means inferred from assigned category values.

3.4 Empirical Framework

We value the participant's most recent crabbing or clamming day. The questionnaire asks, "Suppose the cost of making your last day clamming possible had been $X more than what you actually spent. Would you still have gone clamming?" The crabbing question uses the same wording for that activity. A reminder asks respondents to treat the increment as a real expense that leaves less money for other

purchases. The preceding expense question includes fuel, parking, food, and bait and excludes reusable equipment.

Separate survey-flow randomizers assign each activity one of seven increments, {\$5, \$15, \$25, \$50, \$75, \$150, \$300}, using the survey platform's balanced-presentation option. The valued unit is therefore a reported activity day. Willingness to pay is additional to that day's existing expenses, rather than total trip spending. We report nominal dollars from the 2026 valuation question about 2025 activity, without an inflation adjustment. Interpreting acceptance as willingness to pay assumes respondents understand the cost increase, take the budget reminder seriously, and report the choice they would make at that cost (Englin & Cameron, 1996; Whitehead et al., 2008). The estimate concerns completed days and does not identify a participation response to closures or changes in resource quality.

The main estimate is the nonparametric Turnbull lower bound on mean stated willingness to pay (Haab & Mcconnell, 2002). Let $t_1 < \cdots < t_K$ denote the $K = 7$ bids and $\tilde{p}_k$ the observed acceptance share at t_k. Acceptance should be non-increasing in cost. If the sample shares violate that restriction, we pool adjacent violating cells using their respondent counts as weights to obtain constrained shares $\hat{p}_k$. For binary answers, a pooled share equals the combined number of yes answers divided by the combined cell count. With $t_0 = 0$, the lower-bound mean is

$$\widehat{\mathrm{WTP}}_{\mathrm{NP}} = \sum_{k=1}^{K} (t_k - t_{k-1}) \hat{p}_k. \tag{3.1}$$

Each bid interval contributes its width times acceptance at the upper endpoint. The first contributes $t_1 \hat{p}_1$, and the unobserved tail above the highest bid contributes zero. Those choices make the estimate a lower bound over nonnegative willingness to pay. Appendix Table C10 examines an alternative treatment of the first interval.

Confidence intervals use 1,000 respondent bootstrap resamples. Respondents active in both activities completed both modules in random order, so their answers appear in both valuation samples. For activity differences and certainty-screen comparisons, shared respondent draws preserve that overlap. The randomized module order also provides variation for the survey-order checks described below.

After each valuation answer, respondents rated their certainty from 1 to 10. Certainty follow-ups have been used to examine hypothetical responses (Blumenschein et al., 1998; Samnaliev & Stevens, 2006). As a sensitivity exercise, we combine the binary answer and certainty score c into a mapped response:

$$p_{adj} = \begin{cases} 0.5 + \dfrac{c}{10} \times 0.5 & \text{if yes,} \\ 0.5 - \dfrac{c}{10} \times 0.5 & \text{if no.} \end{cases} \tag{3.2}$$

At certainty 10, a yes remains 1 and a no remains 0; at certainty 1, they become 0.55 and 0.45. The mapping moves less-certain answers toward one half. We apply the same count-weighted monotone calculation to these mapped responses and, separately, to samples meeting progressively higher certainty thresholds. For mapped responses the weights minimize respondent-level squared error; they do not establish that the mapping is a calibrated probability. The first approach changes the response measure, while the second changes which respondents are retained. Both are descriptive sensitivity exercises. Because certainty is reported after the bid and varies with the answer, screening does not automatically identify the WTP distribution of a fixed group independent of the assigned cost.

Parametric models describe how acceptance varies with the bid and participant characteristics:

$$\Pr(\text{accept}_i = 1) = \Lambda(\beta_0 + \beta_1 b_i + \mathbf{x}_i'\boldsymbol{\gamma}), \tag{3.3}$$

where Λ is the logistic function, b_i is the assigned cost increment, and $\mathbf{x}_i$ contains observed covariates (Englin & Cameron, 1996). Median willingness to pay is the bid at which predicted acceptance equals one half. In a bid-only model it is $-\hat{\beta}_0/\hat{\beta}_1$; in a full model we evaluate the corresponding ratio at estimation-sample mean covariates. The binary-logit crossing is a conditional median under the fitted model, not the population median averaged over participants. With a log bid, we exponentiate the corresponding index. These statistics differ from the Turnbull lower-bound mean and use smaller complete-case samples.

We estimate binary logits for the yes/no answer and fractional logits for p_{adj} (Papke & Wooldridge, 1996). For the latter, the 0.5 crossing is a threshold of the expected mapped response. Calling it a WTP median would require the additional assumption that the certainty mapping is a calibrated acceptance probability; we retain it as a sensitivity index. Controls include household income in thousands of dollars, lifetime activity-specific experience, age, gender, boat ownership, hours per day, fishing location, and an income-imputation indicator. Crabbing models also include fishing-method indicators. Experience and hours use values assigned to response categories; for example, lifetime experience above 100 days is coded as 150 days. These controls summarize categories rather than exact counts.

The full-covariate models include an indicator for whether the other module came first. We also estimate the models without prior exposure, retaining records whose module came first or whose order record indicates no other module. This restriction changes sample composition as well as exposure. Appendix Table C4 provides a separate order check among the 984 respondents with valid valuation answers for both activities. It compares randomized first and second modules and models binary acceptance using current-bid categories, module order, and the absolute preceding bid. Using absolute bids avoids forming groups whose definition changes with

the current cost. Standard errors are HC3 robust; parametric intervals use 1,000 Krinsky–Robb draws.

3.5 Results

3.5.1 Empirical Patterns in Responses

Acceptance falls steadily with the randomized additional trip cost (Figure 3.3). At $5, 87.9 percent of clammers and 93.6 percent of crabbers say they would still have taken the outing. At $300, the shares are 5.1 and 8.1 percent. The raw rates decline at every bid in both activities, so the pattern is present before imposing the Turnbull monotonicity restriction. Appendix Table C1 gives the underlying cell counts.

Acceptance remains substantial at intermediate costs in each activity. At $50, 47.0 percent of clammers and 45.5 percent of crabbers would still take the outing; at $75, the shares are 23.9 and 30.4 percent, respectively. These responses contribute to each activity's estimated value over the offered cost range.

Respondents' confidence changes with the choice they face. Among yes answers, the share reporting certainty of 9 or 10 generally falls as the bid rises; among no answers, it generally rises (Appendix Figure C1). The valuation checks below examine how using that reported confidence changes the dollar estimates. Appendix Table C2 reports additional certainty thresholds by bid and answer.

3.5.2 Willingness to Pay Estimates

The preferred Turnbull lower-bound means are $58.38 per clamming trip-day and $65.91 per crabbing trip-day. Their respondent-bootstrap 95 percent confidence intervals are $51.62–$65.52 and $61.68–$70.71, respectively. These estimates measure

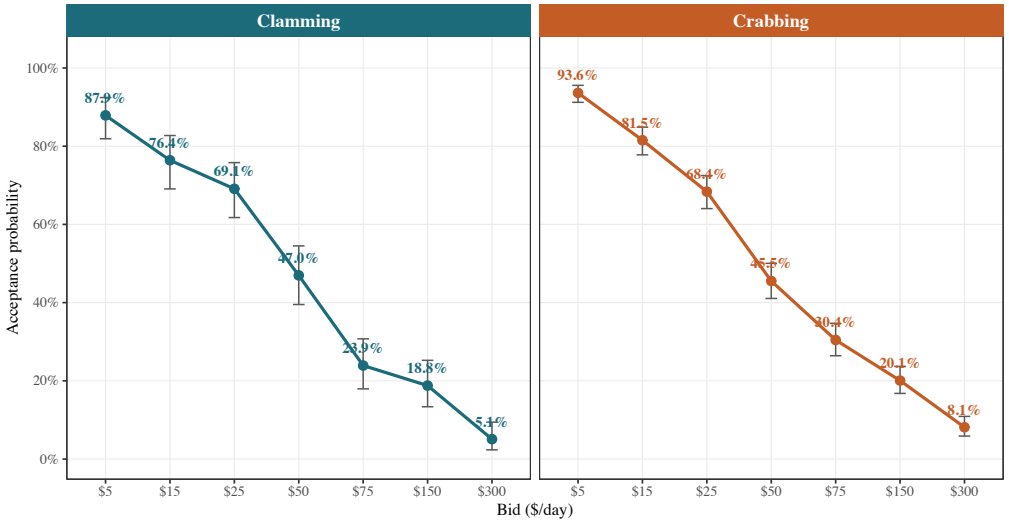


Figure 3.3: Acceptance of Randomized Additional Trip Costs

Note: Points show the share accepting each assigned additional cost among 1,227 clamming and 3,531 crabbing responses. Whiskers are exact binomial 95 percent confidence intervals. Bid categories are equally spaced on the horizontal axis; Appendix Table C1 gives the cell counts.

the recreational benefit respondents attach to their reported outings within each activity. Table 3.2 reports the preferred day values alongside the parametric summaries discussed below.

Continuous certainty mapping raises the valuation index by about $7 per day in each activity (Figure 3.4; Appendix Table C3). This calculation replaces each binary answer with the probability defined in Equation 3.2. Without a calibrated mapping, the resulting index need not be a lower bound on mean underlying WTP. It also excludes the 28 clammers and 61 crabbers who do not report certainty, so the difference from the preferred estimate includes a small sample change. We retain the binary answers for the main estimates because the continuous mapping assumes how stated confidence should translate into acceptance.

Certainty screening moves the values in the opposite direction. Estimates remain fairly stable through a minimum score of 5, then decline as the cutoff rises

Table 3.2: Day Values from Binary and Certainty-Mapped Responses

Statistic	N	Value ($/day)	95% CI
Clamming			
Turnbull lower-bound mean	1,227	$58.38	[51.62, 65.52]
Binary median: linear bid	1,178	$60.25	[52.70, 68.37]
Binary median: log bid	1,178	$38.31	[33.85, 42.76]
Mapped threshold: linear bid	1,156	$61.76	[54.80, 68.73]
Crabbing			
Turnbull lower-bound mean	3,531	$65.91	[61.68, 70.71]
Binary median: linear bid	3,379	$73.23	[68.22, 78.76]
Binary median: log bid	3,379	$45.48	[42.61, 48.55]
Mapped threshold: linear bid	3,324	$74.92	[70.22, 79.85]
Crabbing minus clamming: means	—	$7.54	[-0.92, 15.00]

Turnbull estimates are lower-bound means with percentile intervals from 1,000 respondent bootstrap resamples. Binary-logit medians and fractional-logit 50-percent mapped-response thresholds use estimation-sample mean covariates and 1,000 Krinsky–Robb draws with HC3 covariance. The mapped-response threshold is a sensitivity index; interpreting it as a WTP median additionally requires a calibrated certainty mapping.
The activity difference uses shared respondent draws ($p = 0.084$). The models use the activity-order controls in Table C5. Values are additional to reported trip expenses.

(Figure 3.5). Among respondents reporting certainty of 10, the values are $45.12 for clamming and $53.80 for crabbing, about 23 and 18 percent below the preferred estimates. Both declines have respondent-bootstrap intervals excluding zero.

The strictest screen retains 537 clammers and 1,612 crabbers, compared with 1,227 and 3,531 in the full samples. It describes responses retained by a stricter confidence rule. Because certainty depends on the assigned bid and answer, the retained groups can differ across bid cells as well as across cutoffs. Continuous adjustment instead changes the answers of respondents who report certainty. Because the two exercises change the response measure and sample differently, their opposite movements cannot tell us the size or direction of hypothetical bias in the preferred estimates.

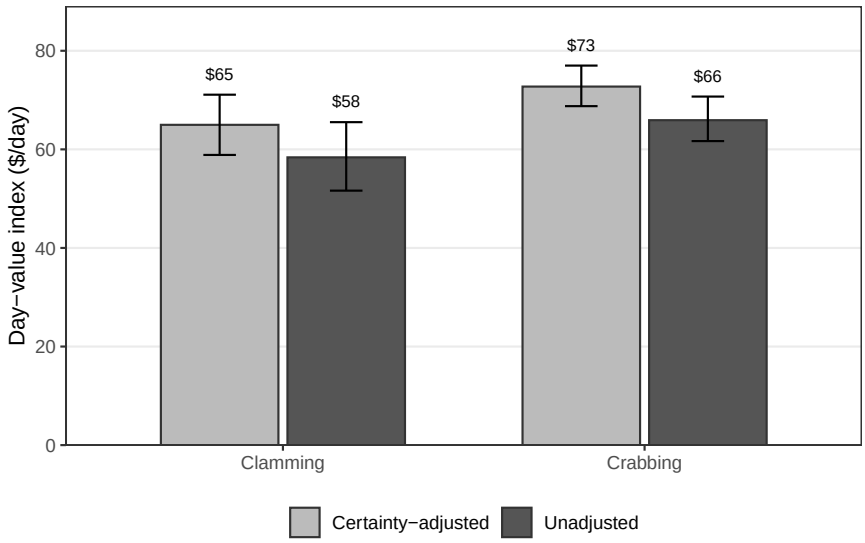


Figure 3.4: Trip-Day Values with Continuous Certainty Adjustment

Note: Bars show binary-response lower-bound means and corresponding mapped-response indices in dollars per trip-day. Whiskers are 95 percent respondent-bootstrap intervals. Unadjusted samples contain 1,227 clammers and 3,531 crabbers; adjusted samples contain 1,199 and 3,470 respondents who also report certainty. Adjustment uses the probability mapping in Equation 3.2. The comparison includes both the mapping and the change in sample.

As a secondary comparison, the full-sample crabbing estimate exceeds the clamming estimate by \$7.54 (95 percent CI: −\$0.92 to \$15.00; $p = 0.084$). Among respondents reporting certainty of 10, the difference is \$8.69 (95 percent CI: \$0.54–\$16.34). The full-sample interval leaves the ordering uncertain; the restricted-sample contrast concerns a different group of participants.

3.5.3 Parametric Estimates

Parametric values depend on how the bid enters the acceptance model (Table 3.2). At sample-mean covariates, binary logits with a linear bid give conditional medians of \$60.25 for clamming and \$73.23 for crabbing. Using a log bid reduces them to \$38.31 and \$45.48. On the same samples and with the same controls, log-bid

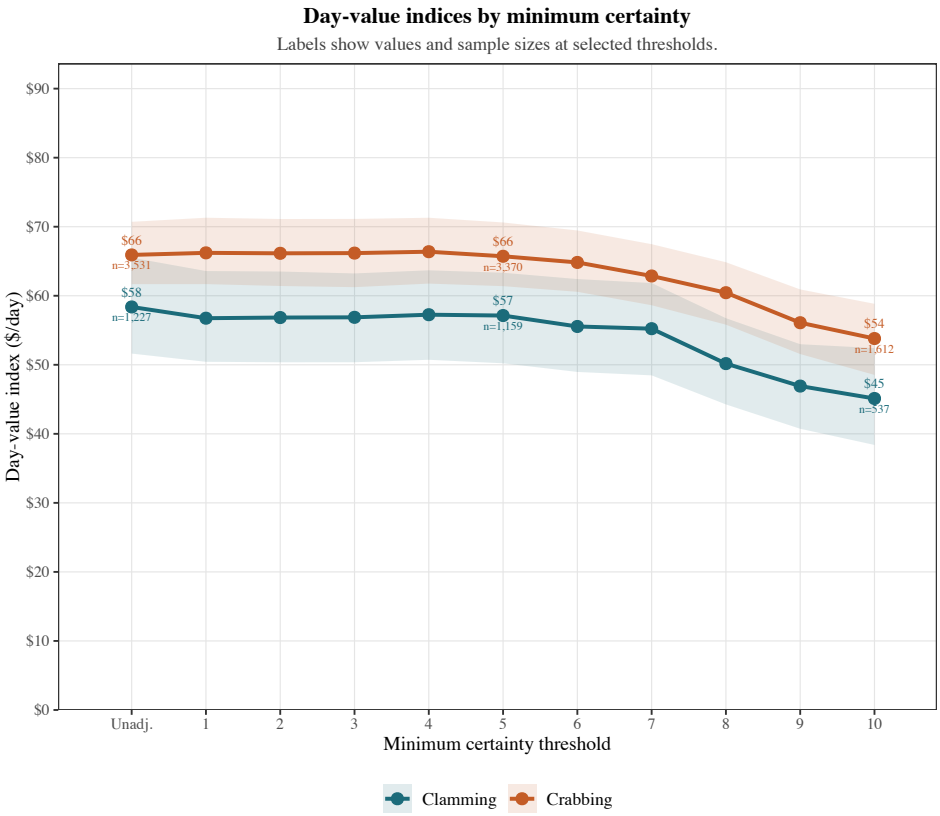


Figure 3.5: Trip-Day Values by Minimum Reported Certainty

Note: Each point is a Turnbull lower-bound mean for respondents whose certainty score meets the indicated threshold. Shaded bands are pointwise 95 percent percentile intervals from 1,000 respondent bootstrap resamples. The full-sample point includes responses with missing certainty; thresholds 1 to 10 exclude them. Sample sizes are shown at selected thresholds. At certainty 10, 537 clammers and 1,612 crabbers remain. Changes along the paths reflect post-bid selection and changes in sample composition; the curves are descriptive indices rather than distributions of fixed participant groups.

models also have lower AIC: 1,182.6 versus 1,237.9 for clamming and 3,196.5 versus 3,497.2 for crabbing (Appendix Table C7). Thus the data favor the log-bid form by this in-sample criterion. We report both forms to show the size of the specification dependence; neither replaces the preferred nonparametric lower-bound mean. Logging income alone changes the medians little (Appendix Table C6).

For comparison, fractional logits with a linear bid cross a mapped response of 0.5 at \$61.76 for clamming and \$74.92 for crabbing. These are thresholds of the certainty-mapped response rather than independently identified WTP medians. Adding certainty as a binary-logit covariate changes its conditional median little (Appendix Table C9).

Higher bids lower acceptance in every reported specification. Income and hours per activity day are positively associated with acceptance in both activities; the association with lifetime experience is more consistent for clamming. Appendix Table C5 reports the full linear-bid coefficients and HC3 standard errors. Participant characteristics are observational controls, so their coefficients describe conditional associations.

The order check holds the participant group fixed at 984 respondents with valid answers for both activities (Appendix Table C4). Clamming's lower-bound mean is \$63.56 when its module comes first and \$49.17 when it comes second; crabbing's corresponding values are \$62.24 and \$68.11. All seven current bids occur in each order group. In binary logits controlling for current-bid categories and the absolute preceding bid, the second-module coefficients are -0.344 (HC3 SE 0.164) for clamming and 0.308 (SE 0.163) for crabbing, evaluated at the mean offered preceding bid. A \$100 increase in the preceding bid has coefficients of 0.239 (SE 0.114) and 0.199 (SE 0.132), respectively. These estimates concern completed dual-activity responses; they do not separate anchoring from other features of answering a second module.

Excluding prior exposure in the broader model samples raises clamming's linear-bid binary median to \$68.41 and its mapped-response threshold to \$69.91. Crabbing's corresponding values are \$71.55 and \$73.52. Appendix Table C8 reports their sample sizes and intervals. This comparison changes the participants as well as removing a preceding bid, so the differences cannot be attributed entirely to order.

3.5.4 Comparison with Previous Literature

Our day values are similar in scale to the Cape Cod shellfishing benchmark and below the Puget Sound benchmarks. For context, in the database's reported 2025-dollar units, the Recreation Use Values Database reports about $70 per day based on Damery and Allen (2004) and roughly $131–$177 based on Anderson and Plummer (2017) (Rosenberger, 2025). The database records are 470001 for Cape Cod and 602001–602003 for Puget Sound.[1] The comparisons provide context for the scale of the estimates, while the underlying choices explain why the figures are not interchangeable. Our valuation bids were stated in the 2026 survey without inflation adjustment, so this is a comparison of scale rather than a harmonized price-year transfer.

Cape Cod's per-day figure divides an annual mixed-shellfishing permit value by reported trips. Our question instead places an additional cost on one completed outing for a particular activity. The Puget Sound studies model expected trip counts under pollution and biotoxin closures. Species, trip lengths, substitute sites, and policy changes also differ. The dollar ranking therefore reflects several differences in the valued opportunity, rather than geography alone.

For blue-crab and clam recreation, the local survey supplies values tied to completed outings in this setting. Earlier studies remain useful benchmarks for their scale and for assessing whether a transfer fits a proposed policy. A transfer to a particular project should consider whether its activities and changes in access match those valued in the source study.

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[1] The 2025 RUVD workbook, accessed 19 October 2026, reports $69.83 for record 470001 and $131.25–$176.50 across records 602001–602003. Their source value years are 2002 and 2013, respectively. The Puget Sound records identify the published Anderson and Plummer study.

3.6 Conclusion

This chapter measures the recreational benefit of blue-crab and clam outings for participants in a Delaware-centered survey. The values are tied to trips participants actually took and expressed per activity day, providing local evidence for economic appraisal of recreational access. The preferred lower-bound means are about $66 per crabbing day and $58 per clamming day. The contribution is a value for each activity and an account of the survey and modeling choices that affect its measurement.

The checks clarify which choices matter when using those values. Treatment of reported certainty changes the valuation indices, and the order check among dual-activity respondents gives lower clamming values when that module comes second. Binary parametric medians are lower under a log bid, which also improves in-sample fit. This dependence on specification supports reporting the nonparametric lower bound alongside clearly labeled parametric summaries. A common questionnaire improves comparability, but does not remove the need to examine how respondents form their answers.

For a closure, habitat project, or access improvement, the day values are one input to an economic benefit calculation. The other is the number of comparable activity days the project changes. The survey measures the first for reported outings; predicting the second requires evidence on participation under the policy. Extending the values to all license holders also requires addressing email coverage and survey completion. A broader sampling frame and repeated trip observations would help connect the local estimates to population benefits and participation responses. Until then, the chapter provides activity-specific evidence that can inform a carefully matched benefit transfer.

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Appendix A

IN-KIND PAYMENT AND AGRICULTURAL CASH WAGES: SEASONAL AND YIELD VARIATION IN BANGLADESH

A.1 Guide to this appendix

Table A1 lists the specifications used in the chapter, what each compares, and where it is reported. The appendix follows that order: the main IV design first, then the companion fitted-yield estimates, the harvest comparison, prices and assumed meal values, contextual comparisons, and the data. Throughout, parentheses report district-clustered standard errors. In tables of two-step estimates, brackets report a joint district bootstrap (999 draws) that re-estimates the weather–yield regression in each draw, and significance stars follow that bootstrap. The intervals in the leave-one-year-out figure for the all-category fitted-yield meal gap and in the assumed-meal-value sensitivity curve use district-clustered standard errors and treat fitted yield as known; other figures state their own construction.

Table A1: Specifications in Chapter 1 and Where They Are Reported

Comparison	Outcome and regressor	Sample	Fixed effects	Inference	Reported in
Main IV (pooled slope and meal gap)	Log crop-season cash wage on observed log yield; 9 weather instruments (pooled), 18 with meal interactions (meal gap)	No-meal and three-meal wage cells, 2017–2025 ($N = 4{,}140$)	Wage cell; year–meal or season–year–meal	Clustered; AR sets for pooled slope only	Table 1.2; Section A.2
Two-step fitted yield	Same outcome on fitted log yield, production, area, or lagged yield	No-meal and three-meal cells; matching, lags, gender change N	Wage cell; year–meal	Clustered; joint bootstrap	Section A.3
All-category fitted yield	Same outcome on fitted log yield by meal category	All four meal categories ($N = 7{,}708$)	Year; district–season	Clustered; joint bootstrap	Section A.4
Harvest calendar	Log (or taka) monthly cash wage on a harvest-month indicator	No-meal and three-meal monthly cells ($N = 16{,}605$)	District–year–gender–meal	District-clustered	Table 1.3; Section A.5
Prices and assumed meal values	Log retail price on fitted yield; log cash wage with price controls; log of cash plus assumed meal value	Price-matched or all-category samples	Year; district–season	Clustered; joint bootstrap	Section A.6
Context	Meal gap by district proxies; household characteristics	All-category sample; BIHS 2018–19 households	As in the all-category model; none for households	Proxies: clustered, joint bootstrap; households: t-tests, OLS	Section A.7

Notes. The N values identify each comparison's reference sample; alternative groupings, omissions, balanced or matched samples and gender splits change N, as reported in the relevant tables. Only the main IV design produces the pooled yield slope of -0.200 discussed in the chapter. The other yield specifications are alternative estimands or companion estimates that use the same weather variation; they are not independent confirmation, and the harvest comparison does not identify the effect of yield.

A.2 Main IV design: first stage, inference, and year sensitivity

This section examines the pooled IV estimate of −0.200 in Table 1.2 using exactly that model's sample, instruments, and fixed effects.

First stage and reduced form

Appendix Table A2 reports the first stage and reduced form instrument by instrument. In the baseline first stage, the Aus and Aman extreme-degree-day coefficients and the Boro precipitation coefficient are positive and statistically detectable; these associations weaken with season–year–meal controls, under which neither the first stage nor the reduced form is jointly significant. The individual reduced-form coefficients are small and mostly imprecise. The table describes the variation the instruments use; the signs of individual coefficients neither validate nor invalidate the instruments.

Anderson–Rubin confidence sets

Appendix Table A3 reports Anderson–Rubin sets obtained by numerical inversion, with refined boundaries and separately evaluated tail limits. With year–meal effects the set is a single interval, [−0.30, −0.10]. With season–year–meal effects the first stage is not significant, so the set is unbounded; it consists of two rays and excludes values between 0.53 and 1.28. These sets are robust to weak instruments only if the instruments are valid.

Omitting one year at a time

Appendix Table A4 re-estimates the pooled model, omitting one year at a time. Six of the nine omissions leave the slope between −0.186 and −0.243. Omitting 2017 or 2021 gives −0.143 or −0.092, and omitting 2024 gives −0.008 (standard error 0.104), with the first-stage F falling from 6.6 to 2.8. The negative baseline estimate

Table A2: First Stage and Reduced Form of the Main IV Design, by Instrument

	Year–meal effects (col. 4)		Season–year–meal effects (col. 5)	
Instrument	Log yield	Log wage	Log yield	Log wage
Boro: GDD	0.0030	−0.0070	0.0092*	−0.0033
	(0.0048)	(0.0044)	(0.0049)	(0.0103)
Boro: EDD	−0.0058	0.0028	0.0015	0.0051
	(0.0036)	(0.0029)	(0.0037)	(0.0052)
Boro: Precipitation	0.0090**	−0.0034	0.0012	0.0006
	(0.0038)	(0.0020)	(0.0028)	(0.0027)
Aus: GDD	−0.0070	−0.0111	−0.0095	−0.0113
	(0.0086)	(0.0089)	(0.0131)	(0.0100)
Aus: EDD	0.0074***	0.0001	0.0020	−0.0016
	(0.0020)	(0.0015)	(0.0033)	(0.0029)
Aus: Precipitation	0.0001	−0.0033**	0.0011	−0.0047**
	(0.0013)	(0.0015)	(0.0021)	(0.0021)
Aman: GDD	−0.0166*	0.0103*	−0.0243**	0.0031
	(0.0098)	(0.0058)	(0.0112)	(0.0079)
Aman: EDD	0.0086**	0.0003	0.0052	0.0025
	(0.0040)	(0.0037)	(0.0049)	(0.0043)
Aman: Precipitation	0.0034*	−0.0030	−0.0023	−0.0025
	(0.0021)	(0.0020)	(0.0028)	(0.0028)
Joint F	6.63	2.76	1.26	1.28
Joint p	<0.001	0.009	0.279	0.268
N	4,140	4,140	4,140	4,140

Notes. Same sample, instruments and fixed effects as the pooled models in Table 1.2; GDD: growing-degree days between 10 and 30°C, per 100; EDD: extreme-degree days above 30°C, per 10; precipitation: seasonal total, per 100 mm. Each weather variable is set to zero outside its own season. "Log yield" columns are the first stage; "Log wage" columns are the reduced form for the log real daily cash wage. District-clustered standard errors; joint tests are cluster-robust Wald F tests of the nine instruments with (9, 63) degrees of freedom. These regressions differ from the 2013–2025 weather–yield prediction regressions in Appendix Table A6. Reduced-form coefficients combine any effect of weather through yield with any direct effect of weather on farm work.

* Significant at 10%; ** at 5%; *** at 1%.

Table A3: Anderson–Rubin Confidence Sets for the Pooled Yield Slope

	95% set	First-stage F	Tail p	Minimum AR F
Year–meal effects (column 4)	$[-0.30, -0.10]$	6.63	<0.001	1.83
Season–year–meal effects (column 5)	$(-\infty, 0.53] \cup [1.28, \infty)$	1.26	0.279	0.99

Notes. Pooled (common-slope) 2SLS specifications of Table 1.2, N = 4,140. For each value b of the yield slope, the Anderson–Rubin test regresses $\log w - b\,y$ on the nine season-specific weather instruments after the fixed effects and tests them jointly with a district-clustered Wald F test with (9, 63) degrees of freedom; the set collects the values not rejected at 5% (critical F = 2.03). The sets are obtained by numerical inversion: the statistic has a closed form in b, checked against direct re-estimation at nine values of b per specification and at each reported boundary and points 0.001 on either side; a finite scan over $b = \tan\theta$ (400,001 points) locates acceptance changes, whose boundaries are then refined numerically. As $|b| \to \infty$ the statistic tends to the cluster-robust first-stage F, evaluated separately; "Tail p" is that limit's p-value, so a tail p above 0.05 means the set is unbounded. "Minimum AR F" is the minimum on the numerical scan. Both sets are non-empty; an empty set would reject the model's joint restrictions under the maintained approximation, but non-emptiness does not establish that the instruments are valid. The sets are robust to weak instruments only if the instruments are valid.

is therefore sensitive to the inclusion of 2024, and the sample without 2024 provides weaker instrument relevance; this does not show that the 2024 data are in error or that the true relationship is zero.

A restricted weather–wage comparison

Appendix Table A5 relates cash wages directly to temperature, without yield. It keeps only Boro growing-degree days and Aus extreme-degree days, so it is an ancillary comparison rather than the reduced form of the main design (column 3 of Table 1.2 and Appendix Table A2). Neither the association with average cash wages nor the three-meal minus no-meal difference is statistically significant at the 5 percent level. A non-significant restricted regression does not test the exclusion restriction.

Table A4: Pooled IV Yield Slope, Omitting One Year at a Time

Year omitted	Slope	SE	p	First-stage F	N	Singleton rows dropped
None (full sample)	−0.200	(0.084)	0.019	6.63	4,140	34
2017	−0.143	(0.103)	0.167	10.83	3,796	35
2018	−0.234	(0.086)	0.009	7.76	3,715	32
2019	−0.186	(0.085)	0.033	6.93	3,651	38
2020	−0.208	(0.091)	0.026	11.66	3,635	40
2021	−0.092	(0.078)	0.244	7.91	3,654	38
2022	−0.192	(0.078)	0.016	7.48	3,641	35
2023	−0.243	(0.082)	0.004	5.79	3,657	36
2024	−0.008	(0.104)	0.942	2.81	3,662	43
2025	−0.207	(0.079)	0.011	11.29	3,651	33

Notes. Each row re-estimates the pooled (common-slope) 2SLS model of Table 1.2, column (4): log real daily cash wage on observed log rice yield instrumented by the nine season-specific weather variables, with year–meal and wage-cell fixed effects, after dropping all observations from the year shown. District-clustered standard errors; all 64 districts remain in every sample. The first-stage F is the cluster-robust joint test of the nine instruments in the corresponding sample. These are ordinary clustered results, not weak-instrument-robust intervals; the nine omissions are not independent replications.

Table A5: Restricted Temperature and Cash-Wage Associations

Dependent variable: log(real daily wage) Coefficients scaled per 100 degree-days		
Panel A: Average cash wage and temperature		
	(1) $EDD_{>30}$	(2) $GDD_{[10,30]}$
Temperature coefficient	0.005	−0.007
	(0.015)	(0.004)
Temperature p-value	0.750	0.092
Observations	7,518	7,518
Year effects	Yes	Yes
District–season effects	Yes	Yes
Panel B: Three-meal minus no-meal difference		
	Boro (GDD)	Aus (EDD)
Temperature × three meals	−0.001	0.009
	(0.002)	(0.014)
Interaction p-value	0.542	0.534
Observations	2,592	2,378
Year effects	Yes	Yes
District–season effects	Yes	Yes

Notes. These regressions relate log cash wages directly to temperature, without fitted yield. These regressions do not separate direct weather effects from any effect that operates through rice yield. District-clustered SEs in parentheses. Coefficients ×100 (per 100 degree-days).
Panel A. Log cash wages on one temperature measure at a time, with gender and meal-category indicators and year and district–season fixed effects.
Panel B. Within one season, whether the temperature association differs between three-meal and no-meal jobs. These restricted comparisons retain only Boro growing-degree days and Aus extreme-degree days. They are not the weather specification used in the current yield analysis, which includes both temperature measures and precipitation in every season. The joint reduced form using all nine weather terms is in Table 1.2.
Neither panel is significant at 5 percent. These tests do not establish whether direct climate effects on wages contribute to the yield–wage association.

A.3 Two-step fitted-yield estimates with the same wage cells

The estimates in this section replace observed yield with weather-predicted yield in the wage-cell specification. They use the same weather variation as the IV design, so they are companion estimates rather than independent confirmation.

Weather–yield prediction

Appendix Table A6 regresses log rice yield on growing-degree days, extreme-degree days, and precipitation in every season, using 2013–2025 data. These regressions produce fitted yield. Their F statistics describe prediction in that longer sample and differ from the first-stage statistics of the main IV design in Appendix Table A2. Weather predicts yield weakly after district and year effects. The temperature-only and pooled comparisons show how prediction depends on the controls; a stronger restricted prediction does not resolve identification. Degree-day thresholds follow Schlenker and Roberts (2009a).

Two-step estimates and sample reconciliation

Appendix Table A7 estimates equation (1.4) by ordinary least squares using weather-predicted yield. This is a generated-regressor comparison, not two-stage least squares; its bootstrap repeats the weather–yield and wage regressions. The common slope is negative but not distinguishable from zero with the bootstrap standard error, and the meal gap is imprecise. Appendix Table A8 separates differences in samples from differences in controls: on records containing both meal categories, the wage interaction and a regression of their log-wage difference give the same estimate. Matching changes which wage records enter; it does not identify employment contracts or correct the instruments.

Table A6: Growing-Season Weather and Rice Yield

Dependent variable: log(rice yield)						
Panel A: Weather used for fitted yield and as instruments						
Season	$GDD_{[10,30]}$	$EDD_{>30}$	Precip.	Joint F	p	N
Boro	0.008*	0.021	−0.004	2.26	0.090	832
	(0.004)	(0.020)	(0.018)			
Aus	−0.011	0.008	0.014	0.58	0.631	823
	(0.012)	(0.032)	(0.019)			
Aman	−0.014*	−0.010	−0.040**	3.02	0.036	832
	(0.007)	(0.030)	(0.019)			
Panel B: Temperature-only models						
Season	Weather			Joint F	p	N
Boro	$GDD_{[10,30]}$			5.58	0.021	832
Aus	$EDD_{>30}$			0.08	0.772	823
Aman	GDD + EDD			1.16	0.321	832
Panel C: Pooled across seasons						
Sample	Weather			Joint F	p	N
All seasons	GDD + EDD + Pr + Pr^2			7.57	$< .001$	2487
All seasons	GDD + EDD			12.23	$< .001$	2487

Notes. Regressions of log rice yield on weather, with year and district fixed effects, estimated separately for each season on 2013–2025 (Panel C: year and district×season fixed effects). District-clustered standard errors in parentheses. Degree-day coefficients are per 100 degree-days and precipitation per 1,000 mm. Joint F tests all weather terms in the row. Panel A generates the fitted log yield used in the two-step estimates; the same three variables, by season, are the instruments in Table 1.2.

* Significant at 10%; ** at 5%; *** at 1%.

Table A7: Fitted Rice Yield and Cash Wages by Meal Category (Bangladesh, 2017–2025)

Dependent variable: log(real daily wage); no-meal and three-meal cells only				
	(1) Pooled	(2) Full sample	(3) Female	(4) Male
Meal gap (three minus no meals)		0.117	−0.052	0.239*
		(0.120) [0.138]	(0.274) [0.296]	(0.125) [0.133]
No-meal slope		−0.103**		
		(0.037) [0.047]		
Three-meal slope		0.014		
		(0.126) [0.146]		
Common slope	−0.082			
	(0.043) [0.053]			
Year–meal effects	Yes	Yes	Yes	Yes
Wage-cell effects	Yes	Yes	Yes	Yes
N	4,140	4,140	1,726	2,414

Notes. Unit: district×season×gender×meal×year. Only no-meal and three-meal cells enter columns (2)–(4). The meal gap is the three-meal slope minus the no-meal slope on fitted log yield. These are ordinary least-squares regressions on weather-predicted yield, not the two-stage least-squares estimates of Table 1.2. Fixed effects: one for each district–season–gender–meal wage cell, and year effects that differ by meal category. District-clustered standard errors in parentheses. Brackets: district bootstrap SE (999 draws) that re-estimates the weather–yield regression in each draw; clustered SEs treat fitted yield as known. Stars and reported p-values use the bootstrap SE with a t_{63} reference distribution. Gender subsamples drop four district×season×gender×meal fixed-effect singletons. Table A8 compares these estimates with a matched-sample version and with a specification that uses all four meal categories.

* Significant at 10%; ** at 5%; *** at 1% (bootstrap-SE t_{63} tests).

Table A8: Fitted-Yield Estimates under Alternative Samples and Controls

Specification	$\hat{\beta}$	SE		p	N
All meal categories; district–season effects	−0.002	(0.012)	[0.012]	0.900	7,708
No-meal and three-meal jobs; wage-cell effects	0.117	(0.120)	[0.138]	0.399	4,140
Same regression; matched wage records	0.250*	(0.134)	[0.145]	0.091	2,080
Wage-gap regression; matched records	0.250*	(0.134)	[0.145]	0.091	1,040

Coverage statistic	Value
No-meal and three-meal wage records	4,174
District–season–gender–year cells with both categories	1,069
Wage records in matched cells	2,138
Matched cells retained in estimation	1,040
Absolute difference between matched estimates	1.33e-15

Notes. Rows 1–3 regress cash wages on fitted log yield and its interaction with the three-meal category. Row 4 instead regresses the three-meal minus no-meal log wage on fitted log yield. Wage-cell effects control for persistent differences within district, season, gender and meal category; year–meal effects allow national wage changes to differ by category. Row 2 is the two-step wage-cell specification of Table A7. Rows 3–4 use the matched sample in which both no-meal and three-meal wages are observed in the same district×season×gender×year cell; on that sample the wage interaction and wage-gap regression are algebraically equivalent (difference $< 10^{-8}$). There are 1,069 matched cells before estimation (2,138 wage observations); fixed-effect singleton removal leaves 1,040 gap cells (2,080 wage observations). Row 1 is the all-meal specification with district×season fixed effects and all meal categories; it is retained for comparison only. Matching selects on availability of meal-category records. District-clustered standard errors. Brackets: district bootstrap SE (999 draws) that re-estimates the weather–yield regression in each draw; clustered SEs treat fitted yield as known. Stars and reported p-values use the bootstrap SE with a t_{63} reference distribution.
* Significant at 10%; ** at 5%; *** at 1%.

Alternative rice measures and timing

Replacing fitted yield with fitted log rice production gives a meal gap of −0.110 (0.091) (Appendix Table A9); fitted log rice area gives −0.092 (0.077) (Appendix Table A10); and the previous year's fitted yield gives 0.114 (0.155) (Appendix Table A11). Changing the regressor changes the question, so these are alternatives rather than estimates of one yield effect. None of the gaps is distinguishable from zero.

Table A9: Fitted Rice Production and Cash Wages by Meal Category (Bangladesh, 2017–2025)

Dependent variable: log(real daily wage); no-meal and three-meal cells only				
	(1) Pooled	(2) Full sample	(3) Female	(4) Male
Meal gap (three minus no meals)		−0.110	−0.088	−0.109
		(0.091) [0.088]	(0.156) [0.144]	(0.074) [0.078]
No-meal slope		−0.022		
		(0.018) [0.023]		
Three-meal slope		−0.132		
		(0.094) [0.091]		
Common slope	−0.032			
	(0.020) [0.026]			
Year–meal effects	Yes	Yes	Yes	Yes
Wage-cell effects	Yes	Yes	Yes	Yes
N	4,140	4,140	1,726	2,414

Notes. Unit: district×season×gender×meal×year. Only no-meal and three-meal cells enter columns (2)–(4). The regressor is fitted log rice production; the meal gap is the three-meal slope minus the no-meal slope. Fixed effects, clustering, and sample restrictions match Table A7. Brackets: district bootstrap SE (999 draws) that re-estimates the weather–yield regression in each draw; clustered SEs treat fitted yield as known. Stars and reported p-values use the bootstrap SE with a t_{63} reference distribution. Fitted production comes from the season-specific weather regressions of Table A6, Panel A, with log production as the outcome. Gender subsamples drop four district×season×gender×meal fixed-effect singletons.
* Significant at 10%; ** at 5%; *** at 1%.

Table A10: Fitted Rice Area and Cash Wages by Meal Category (Bangladesh, 2017–2025)

Dependent variable: log(real daily wage); no-meal and three-meal cells only				
	(1) Pooled	(2) Full sample	(3) Female	(4) Male
Meal gap (three minus no meals)		−0.092	−0.061	−0.101
		(0.077) [0.078]	(0.143) [0.139]	(0.063) [0.068]
No-meal slope		−0.017		
		(0.020) [0.025]		
Three-meal slope		−0.109		
		(0.082) [0.082]		
Common slope	−0.028			
	(0.022) [0.027]			
Year–meal effects	Yes	Yes	Yes	Yes
Wage-cell effects	Yes	Yes	Yes	Yes
N	4,140	4,140	1,726	2,414

Notes. Unit: district×season×gender×meal×year. Only no-meal and three-meal cells enter columns (2)–(4). The regressor is fitted log rice area for the season; the meal gap is the three-meal slope minus the no-meal slope. Fixed effects, clustering, and sample restrictions match Table A7. Brackets: district bootstrap SE (999 draws) that re-estimates the weather–yield regression in each draw; clustered SEs treat fitted yield as known. Stars and reported p-values use the bootstrap SE with a t_{63} reference distribution. Fitted area comes from the season-specific weather regressions of Table A6, Panel A, with log area as the outcome. Gender subsamples drop four district×season×gender×meal fixed-effect singletons.
* Significant at 10%; ** at 5%; *** at 1%.

Table A11: Previous-Year Fitted Yield and Cash Wages by Meal Category (Bangladesh, 2017–2025)

Dependent variable: log(real daily wage); no-meal and three-meal cells only				
	(1) Pooled	(2) Full sample	(3) Female	(4) Male
Meal gap (three minus no meals)		0.114	−0.081	0.220
		(0.155) [0.151]	(0.284) [0.286]	(0.163) [0.172]
No-meal slope		0.051		
		(0.061) [0.063]		
Three-meal slope		0.165		
		(0.156) [0.164]		
Common slope	0.072			
	(0.061) [0.069]			
Year–meal effects	Yes	Yes	Yes	Yes
Wage-cell effects	Yes	Yes	Yes	Yes
N	3,796	3,796	1,574	2,222

Notes. Unit: district×season×gender×meal×year. Only no-meal and three-meal cells enter columns (2)–(4). The regressor is fitted log yield in the previous district×season year $(t-1)$; wages are measured in year t (2018–2025 after the initial lag). The meal gap is the three-meal slope minus the no-meal slope on lagged fitted log yield. Fixed effects, clustering, and sample restrictions otherwise match Table A7. Brackets: district bootstrap SE (999 draws) that re-estimates the weather–yield regression in each draw; clustered SEs treat fitted yield as known. Stars and reported p-values use the bootstrap SE with a t_{63} reference distribution. Contemporaneous yield results are in Table A7. Gender subsamples drop four district×season×gender×meal fixed-effect singletons.
* Significant at 10%; ** at 5%; *** at 1%.

Summary of alternative meal-gap estimates

Appendix Table A12 collects the meal gaps from this section and the next. The rows differ in sample, controls, or regressor, so they are not estimates of one common effect. The large first-difference gap shrinks when national year effects differ by meal category. These exercises leave the meal-gap question open and concern differences between meal categories, not the pooled slope.

Table A12: Alternative Fitted-Yield Comparisons: A Summary

Specification	Meal gap	Standard errors
All meal categories; year controls	−0.002	(0.012) [0.012]
All meal categories; season–year controls	−0.006	(0.013) [0.013]
Yearly changes; common year controls	−0.742***	(0.227) [0.252]
Yearly changes; year–meal controls	−0.061	(0.181) [0.235]
Fitted rice production; wage-cell controls	−0.110	(0.091) [0.088]
Fitted rice area; wage-cell controls	−0.092	(0.077) [0.078]
Previous-year fitted yield; wage-cell controls	0.114	(0.155) [0.151]

The meal gap is the higher-meal minus lower-meal fitted-yield slope in log cash wages. The all-meal comparisons use the three-meal interaction relative to no meals and district–season effects. The first-difference rows compare changes across consecutive years; the second lets national year changes differ by meal category. The final three rows use no-meal and three-meal records with district–season–gender–meal wage-cell and year–meal effects, as in Table A7. The production and area rows use those fitted rice measures in place of yield; the last uses fitted yield from the previous year. Samples differ across these comparisons.
Parentheses contain district-clustered standard errors; brackets contain joint district-bootstrap standard errors from re-estimating both stages (999 draws). Stars use the bootstrap standard error with a t_{63} reference distribution. These comparisons do not identify a causal yield effect or test harvest wage smoothing. The rows summarize different samples and controls; they should not be read as estimates of the same causal effect.
* Significant at 10%; ** at 5%; *** at 1%.

A.4 Meal-category differences in a simpler all-category specification

The tables in this section use all four meal categories with year and district–season fixed effects, a simpler specification than the wage-cell models. They address whether the fitted-yield relationship differs across meal categories. They are not checks of the pooled IV slope or of the harvest comparison.

Appendix Table A13 reports the fitted-yield slope for each meal category; all four are small and imprecise. Other tables change the meal groupings, add the local share of three-meal records, and change how years are controlled for (Appendix Table A14); estimate the comparison season by season (Appendix Table A15); and drop one year at a time (Appendix Figure A1). The meal gap under this specification is −0.002 (0.012) and stays within about 0.01 of zero across the regroupings, the

composition control, year effects that differ by meal category, and each dropped year. The season-specific gaps are imprecise. Estimated in first differences, the gap is −0.742 (0.227), but it falls to −0.061 (0.181) once year effects differ by meal category; national wage movements that differ across meal categories, rather than local yield changes, appear to drive the first-difference estimate.

Table A13: Fitted-Yield Associations for Each Meal Category

Dependent variable: log(real daily wage)			
	(1) Full sample	(2) Female	(3) Male
0 meals	−0.032 (0.040) [0.050]	−0.041 (0.076) [0.087]	−0.033 (0.042) [0.049]
1 meal	−0.029 (0.043) [0.052]	−0.045 (0.079) [0.090]	−0.023 (0.044) [0.051]
2 meals	−0.018 (0.045) [0.052]	−0.037 (0.083) [0.091]	−0.014 (0.048) [0.053]
3 meals	−0.033 (0.044) [0.052]	−0.062 (0.081) [0.089]	−0.025 (0.048) [0.052]
Common slope (all meals)	−0.032 (0.041) [0.050]		
Year effects	Yes	Yes	Yes
District–season effects	Yes	Yes	Yes
N	7,708	3,174	4,530

Notes. Each column regresses log cash wages on fitted log yield interacted with the four meal categories, with year and district–season fixed effects. Each row is the fitted-yield slope for one meal category. Table A7 instead uses only no-meal and three-meal records, with a fixed effect for each wage cell. Brackets: district bootstrap SE (999 draws) that re-estimates the weather–yield regression in each draw; clustered SEs treat fitted yield as known. Stars and reported p-values use the bootstrap SE with a t_{63} reference distribution.
* Significant at 10%; ** at 5%; *** at 1%.

Table A14: Fitted-Yield Wage Differences under Alternative Meal Groupings

Dependent variable: log(real daily wage) unless noted				
Panel A: Alternative meal-category groupings				
	(1) 0 vs. 3	(2) 0–1 vs. 3	(3) 0 vs. 2–3	(4) 0–1 vs. 2–3
Meal gap	−0.002	−0.001	0.007	0.010
	(0.012) [0.012]	(0.012) [0.012]	(0.009) [0.009]	(0.008) [0.008]
Panel B: Composition control				
	(1) Baseline	(2) Share three-meal		
Meal gap	−0.002	0.003		
	(0.012) [0.012]	(0.012) [0.012]		
Panel C: Alternative controls and changes across years				
	(1) Yield minus year mean	(2) Year–meal controls	(3) Yearly changes	(4) Changes; year–meal controls
Meal gap	0.004	0.004	−0.742***	−0.061
	(0.012) [0.012]	(0.012) [0.012]	(0.227) [0.252]	(0.181) [0.235]

Notes. Each cell reports a meal gap: the fitted-yield slope of the higher-meal group minus that of the lower-meal group, with year and district–season fixed effects. Table A7 instead uses only no-meal and three-meal records with wage-cell fixed effects. District-clustered standard errors in parentheses. Brackets: district bootstrap SE (999 draws) that re-estimates the weather–yield regression in each draw; clustered SEs treat fitted yield as known. Stars and reported p-values use the bootstrap SE with a t_{63} reference distribution.

Panel A. Column (1) is the three-meal minus no-meal contrast estimated within the four-category sample of Table A13. Column (2) compares $\{0, 1\}$ with three meals; column (3) compares zero with $\{2, 3\}$; column (4) compares $\{0, 1\}$ with $\{2, 3\}$.

Panel B. Column (1) repeats Panel A, column (1). Column (2) adds the district×season×year share of three-meal wage cells.

Panel C. Column (1) subtracts the calendar-year mean from fitted log yield. Column (2) lets the year effects differ by meal category. Column (3) estimates the wage regression in first differences within each wage cell (Δ log wage on Δ fitted log yield, consecutive years); column (4) also lets the year effects differ by meal category.

*** Significant at 1%; ** at 5%; * at 10%.

Table A15: Fitted-Yield Wage Differences by Rice Season and Meal Grouping

Dependent variable: log(real daily wage)			
	Boro	Aus	Aman
Panel A: Zero meals vs. three meals			
Meal gap	0.115	−0.053	−0.007
	(0.126) [0.133]	(0.059) [0.061]	(0.041) [0.041]
Observations	2,656	2,438	2,614
Panel B: $\{0, 1\}$ *meals vs. three meals*			
Meal gap	0.139	−0.073	0.016
	(0.124) [0.132]	(0.053) [0.056]	(0.042) [0.043]
Observations	2,173	2,004	2,137
Panel C: Zero meals vs. $\{2, 3\}$ *meals*			
Meal gap	−0.032	0.026	−0.000
	(0.085) [0.089]	(0.056) [0.056]	(0.033) [0.034]
Observations	1,901	1,793	1,874
Panel D: $\{0, 1\}$ *meals vs.* $\{2, 3\}$ *meals*			
Meal gap	−0.024	0.015	0.024
	(0.081) [0.083]	(0.045) [0.045]	(0.030) [0.031]
Observations	2,656	2,438	2,614

Notes. Each cell is a meal gap (the fitted-yield slope of the higher-meal group minus that of the lower-meal group) from a regression estimated separately for each season, with year and district–season fixed effects; district-clustered standard errors in parentheses. Brackets: district bootstrap SE (999 draws) that re-estimates the weather–yield regression in each draw; clustered SEs treat fitted yield as known. Stars and reported p-values use the bootstrap SE with a t_{63} reference distribution. Panel A reports the three-meal minus no-meal contrast estimated within the four-category sample (all meal categories enter the regression, as the sample sizes show), with these fixed effects; Table A7 instead uses only no-meal and three-meal records with wage-cell fixed effects. Panels B–D re-estimate the same season splits under the alternative meal groupings in Table A14, Panel A. Sample sizes vary when intermediate meal categories are excluded.
*** Significant at 1%; ** at 5%; * at 10%.

Appendix Table A16 replaces the year effects with season–year effects, which absorb shocks common to all districts in a season and year, much like the richer seasonal controls in column (5) of Table 1.2. This leaves much less variation in fitted yield. The meal gap is −0.006 (0.013), and the separate meal slopes become very imprecise.

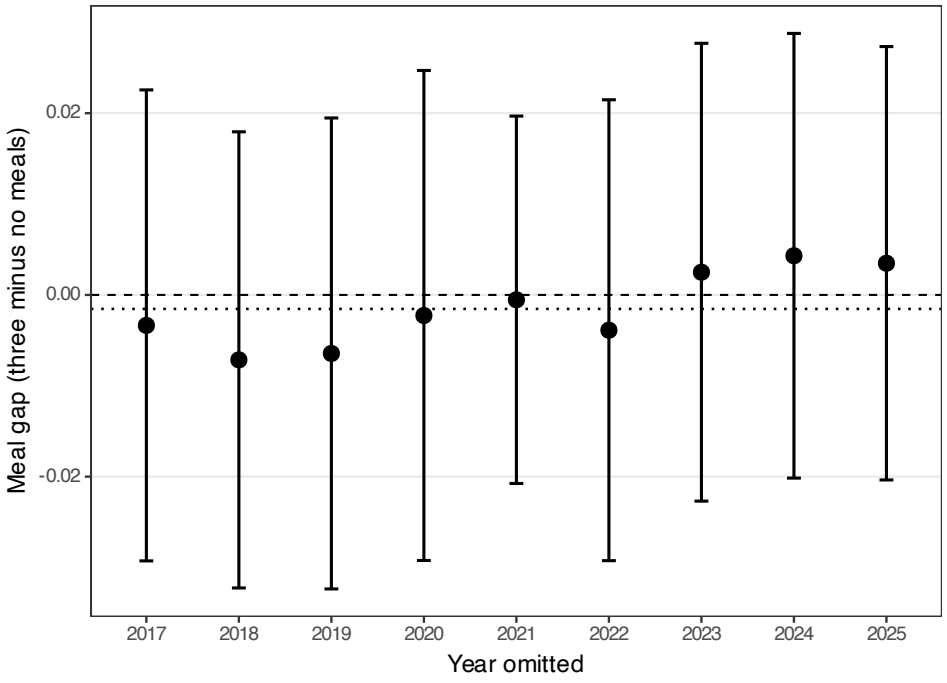


Figure A1: Fitted-Yield Meal Gap in the All-Category Specification, Dropping One Year at a Time. Each marker drops one year from 2017–2025 and re-estimates the three-meal minus no-meal gap with all four meal categories and year and district–season fixed effects. The dotted line is the full-sample estimate (−0.002). This figure concerns the meal gap in the simpler all-category model, not the pooled IV slope; Appendix Table A4 omits years from that model. Error bars are approximate 95% intervals using district-clustered standard errors, treating fitted yield as known; they are not the joint-bootstrap intervals used in the tables.

Table A16: Fitted-Yield Wage Associations with Different Controls for Common Shocks

Dependent variable: log(real daily wage)		
	(1) Year controls	(2) Season–year controls
No-meal slope	−0.032 (0.040) [0.050]	0.253 (0.373) [0.395]
One-meal slope	−0.029 (0.043) [0.052]	0.251 (0.374) [0.397]
Two-meal slope	−0.018 (0.045) [0.052]	0.263 (0.375) [0.398]
Three-meal slope	−0.033 (0.044) [0.052]	0.247 (0.375) [0.397]
Meal gap (three minus no meals)	−0.002 (0.012) [0.012]	−0.006 (0.013) [0.013]
Meal gap for one SD of fitted yield	−0.00004	−0.00003
Year effects	Yes	No
Season–year effects	No	Yes
District–season effects	Yes	Yes
N	7,708	7,708

Notes. Column (1) repeats the all-meal specification of Table A13 (year and district–season fixed effects). Column (2) replaces the year effects with season–year effects, which absorb shocks common to all districts in a season and year. Meal rows are fitted-yield slopes for each meal category; the meal gap is the three-meal slope minus the no-meal slope. Residual SD of fitted log yield is 0.0246 in column (1) and 0.0051 in column (2). The last row multiplies the meal gap by that residual SD, giving the implied log-wage difference for a one-standard-deviation change in fitted yield after the fixed effects. District-clustered standard errors in parentheses. Brackets: district bootstrap SE (999 draws) that re-estimates the weather–yield regression in each draw; clustered SEs treat fitted yield as known. Stars and reported p-values use the bootstrap SE with a t_{63} reference distribution. The separate meal slopes in column (2) are imprecise.
*** Significant at 1%; ** at 5%; * at 10%.

Appendix Table A17 asks whether the mix of published wage records changes with fitted yield, which would affect how the meal-category comparisons are read. Each district–season–year contributes shares of its wage records by meal category; these are shares of records, not of workers or jobs. The three-meal share changes little with the change in fitted yield (−0.018, with a clustered standard error of 0.043

and a bootstrap standard error of 0.046), and neither the no-meal share nor the ratio of the two changes significantly. This does not establish that the jobs or workers behind each category are unchanged.

Table A17: Meal-Category Shares in Published Wage Cells

Dependent variable: share or ratio of published wage cells					
	Coef.		SE	p	N
Panel A: Pooled sample					
Share three-meal cells	−0.018	(0.043)	[0.046]	0.697	1,512
Share no-meal cells	0.019	(0.069)	[0.088]	0.829	1,512
Three-meal / no-meal ratio	−0.067	(0.185)	[0.182]	0.713	1,512
Panel B: Share three-meal, by season					
Boro	0.338	(0.737)	[0.892]	0.706	504
Aus	0.486	(0.471)	[0.509]	0.343	504
Aman	−0.228	(0.291)	[0.304]	0.455	504

Notes. These regressions ask whether changes in fitted yield change the *mix* of published meal-category wage records in a district–season–year, which would affect how Table A7 is read.
Unit: district×season×year. Share three-meal (no-meal) is the fraction of wage cells in that unit recorded as three (zero) meals; denominator = all meal categories 0–3. The ratio is three-meal counts divided by no-meal counts (dropped when the no-meal count is zero).
Regressor: within-district–season change in fitted log yield, $\Delta \log \widehat{y}$. Panel A includes year and district×season FEs; Panel B includes year and district FEs within each season. District-clustered SEs in parentheses. Brackets: district bootstrap SE (999 draws) that re-estimates the weather–yield regression in each draw; clustered SEs treat fitted yield as known. Stars and reported p-values use the bootstrap SE with a t_{63} reference distribution.
A nonzero coefficient would mean that the mix of published records moves with fitted yield. The shares count wage records, not workers or jobs.
*** Significant at 1%; ** at 5%; * at 10%.

A.5 Harvest comparison

This section checks the descriptive harvest comparison of Table 1.3.

Definitions, samples, trends, and placebo months

Appendix Table A18 asks whether the similar harvest increases depend on how the harvest is defined, who is in the sample, or how the seasonal wage pattern

is modeled. Each row re-estimates column (1) of Table 1.3, reporting the no-meal harvest increase and the meal gap. The gap stays small for women and for men, when any one year is dropped, and when the Aus harvest months are added. Three-meal wages drift down relative to no-meal wages over the calendar year (Figure 1.3), which can help account for why the Boro and Aman harvests taken separately give gaps of opposite sign, and why the placebo harvests at the two ends of the year do the same. Adding a common linear month trend and its interaction with three meals gives a gap of 0.001 (0.007) in logs and −0.5 (2.9) taka per day. Restricting the sample to wage cells reported in all twelve months, or to district–year–gender cells in which both job types are reported all year, also leaves the gap small. These checks describe the sensitivity of the observed contrast; they do not identify why the two series move together.

Appendix Table A19 reports the harvest increase for each of the four meal categories, which ranges from 4.7 to 5.7 percent, and shows that within a year the monthly cash wage is slightly more variable in meal jobs than in no-meal jobs.

Rice-intensive districts

Appendix Table A20 examines the rice-intensity pattern in column (2) of Table 1.3. Rice intensity is standardized, so each coefficient is the change in the meal gap for a district one standard deviation more rice-intensive than average. The coefficient stays negative under other measures of rice intensity, although it is not significant when intensity is measured by Boro rice area alone. It is driven by women's wages and loses significance for men and when the five largest rice districts are excluded. We therefore treat the pattern as exploratory. It does not show that meal provision marks a particular employment arrangement in those districts.

Table A18: Harvest Wage Differences: Alternative Definitions, Samples, and Placebo Months

	No-meal harvest increase	Meal gap	95% CI	N
Baseline (Boro + Aman)	0.057***	−0.0061	[−0.019, 0.006]	16,605
	(0.005)	(0.0064)		
Women	0.059***	−0.0160*	[−0.033, 0.001]	6,718
	(0.006)	(0.0089)		
Men	0.057***	−0.0007	[−0.013, 0.012]	9,887
	(0.005)	(0.0063)		
Taka per day	25.6***	−3.6	[−8.6, 1.4]	16,605
	(2.3)	(2.5)		
Month trends	0.052***	0.0013	[−0.013, 0.015]	16,605
	(0.006)	(0.0071)		
Month trends, taka	23.9***	−0.5	[−6.2, 5.2]	16,605
	(2.7)	(2.9)		
Cells with all 12 months	0.058***	−0.0052	[−0.019, 0.009]	14,412
	(0.005)	(0.0073)		
Both types, all 12 months	0.059***	−0.0080	[−0.018, 0.001]	6,432
	(0.007)	(0.0048)		
Leave one year out (range)		−0.0089 to −0.0016		
Add Aus harvest (Jul–Aug)	0.052***	−0.0047	[−0.015, 0.006]	16,605
	(0.004)	(0.0053)		
Boro harvest only (Apr–May)	0.056***	0.0132	[−0.006, 0.032]	16,605
	(0.008)	(0.0098)		
Aman harvest only (Nov–Dec)	0.036***	−0.0216***	[−0.037, −0.006]	16,605
	(0.004)	(0.0078)		
Placebo harvests in adjacent non-harvest months				
Jan–Feb	−0.035***	0.0140** (0.0055)		10,933
Feb–Mar	−0.030***	0.0051 (0.0063)		10,933
Jun–Jul	0.025***	−0.0030 (0.0063)		10,933
Jul–Aug	0.027***	−0.0022 (0.0066)		10,933
Aug–Sep	0.019***	−0.0067 (0.0056)		10,933
Sep–Oct	0.009***	−0.0155*** (0.0055)		10,933

Notes. Each row re-estimates column (1) of Table 1.3: the no-meal harvest increase and the meal gap (the three-meal minus the no-meal harvest increase); in the placebo rows the meal gap's standard error is shown beside it. The taka rows use the real daily wage in taka instead of its log. Placebo rows use non-harvest months only. Three-meal wages drift down relative to no-meal wages over the calendar year (Figure 1.3), so the single-harvest rows differ in sign; the trend rows add a linear month trend and its interaction with three meals. Three-meal rows are reported in 8.8 months per district–year–gender cell on average (11.4 for zero meals); the coverage rows keep cells observed in all twelve months. District-clustered standard errors in parentheses. * Significant at 10%; ** at 5%; *** at 1%.

Table A19: Harvest Increases and Within-Year Wage Variability by Number of Meals

	No meals	One meal	Two meals	Three meals
Harvest increase in log cash wage	0.057***	0.047***	0.050***	0.051***
	(0.005)	(0.005)	(0.005)	(0.007)
Within-year SD of log wage, relative to no-meal jobs	—	0.002	0.010***	0.019***
		(0.001)	(0.003)	(0.003)

Notes. Row 1: harvest-month increase in the log real daily cash wage for each meal category, estimated with all four categories and the fixed effects of Table 1.3, column (1). Row 2: difference in the within-year standard deviation of the log wage relative to no-meal jobs, using district×year×gender×meal cells with at least ten months. District-clustered standard errors in parentheses.
* Significant at 10%; ** at 5%; *** at 1%.

Table A20: Harvest Wage Differences by Rice Intensity

	Change in meal gap	SE	Districts	N
Baseline: log rice area, 2013–2016	−0.0139**	(0.0066)	64	16,605
Boro rice area	−0.0106	(0.0085)	64	16,605
Aman rice area	−0.0094	(0.0058)	64	16,605
Rice production	−0.0139**	(0.0067)	64	16,605
Rank of rice area	−0.0173***	(0.0061)	64	16,605
Women	−0.0221***	(0.0079)	62	6,718
Men	−0.0094	(0.0066)	64	9,887
Boro harvest only	−0.0086	(0.0083)	64	16,605
Aman harvest only	−0.0131*	(0.0072)	64	16,605
Drop five largest rice districts	−0.0058	(0.0073)	59	15,033
Leave one year out (range)	−0.0166 to −0.0111			
Permutation p (999 shuffles across districts)	0.036			

Notes. The coefficient is the change in the three-meal minus no-meal harvest wage gap for a one-standard-deviation increase in rice intensity. A negative value means that meal-job wages rise less relative to no-meal wages where rice is more important. These are exploratory comparisons; the result is sensitive to gender and to excluding the largest rice districts. Specification as in column (2) of Table 1.3. The five largest rice districts (2013–2016 rice area) are Mymensingh, Naogaon, Dinajpur, Bogura, Cumilla. District-clustered standard errors.
* Significant at 10%; ** at 5%; *** at 1%.

Matched monthly rice-price check

We match WFP retail coarse-rice prices to wage records containing both no-meal and three-meal wages in the same district, year, month, and gender. Prices

are available for 1,872 of the 4,517 paired records (41 percent), covering 47 districts; coverage is sparse before 2020. After dropping singleton fixed-effect groups, the regressions use 1,848 observations. Rice prices and wages use the same annual CPI adjustment. Each wage-gap regression includes district–year–gender fixed effects and a linear calendar-month trend, with standard errors clustered by district.

The percentage harvest difference remains close to zero in the paired and price-matched samples (Appendix Table A21). The absolute taka difference is more sensitive to matching: in the full paired sample, three-meal cash wages rise by 4.7 taka less than no-meal wages, with a 95 percent confidence interval from 1.0 to 8.5 taka less. Thus, equal taka increases are not a consistent explanation across sample definitions. In the price-matched sample, adding rice prices changes the estimated harvest gap only from -3.48 to -3.39 taka. Higher rice prices are associated with a more negative cash-wage gap, which is consistent with more valuable meals substituting for cash, but does not establish that explanation. After allowing for the month trend, rice prices show no clear harvest increase. Rice prices cover only part of meal costs, and actual meal values and changes in the workers or jobs recorded are not observed. This exploratory check does not identify why the percentage harvest increases are similar.

Table A21: Exploratory Harvest and Rice-Price Comparisons

Comparison	Estimate	SE	N
Full paired sample: log gap	−0.0008	0.0045	4,474
Price-matched sample: log gap	0.0016	0.0074	1,848
Full paired sample: taka gap	−4.71	1.87	4,474
Price-matched sample: taka gap	−3.48	3.03	1,848
Price-matched taka gap, price control	−3.39	2.99	1,848
Rice-price association with taka gap	−0.99	0.41	1,848
Harvest change in rice price	0.17	0.13	1,257

Notes: The wage gap is three-meal minus no-meal daily cash pay. Log-gap estimates are log points; taka-gap estimates are real taka per day. All rows include a linear calendar-month trend. Wage-gap regressions use district–year–gender fixed effects. The rice-price association controls for harvest months and is per real taka/kg higher retail coarse-rice price. The final row uses unique district-month observations, district–year fixed effects, and real taka/kg as the outcome. SEs are district-clustered. Matched wage regressions cover 47 districts. Price matching retains 41% of paired wage records before singleton removal. These are descriptive associations; rice prices do not measure complete meal values.

A.6 Rice prices and assumed meal values

The comparisons in this section describe associations and sensitivity to assumptions. They are not tests of mediation or of a mechanism, and they do not explain the similar harvest increases.

Rice prices

Appendix Table A22 reports the association between weather-predicted yield and district retail rice prices discussed in Section 1.7.1. The estimate is negative but imprecise. It uses the same fitted yield as Appendix Table A7 and describes how prices move with predicted yield; it does not test whether prices account for the wage estimates, and retail prices may differ from the prices farmers receive.

Appendix Table A23 adds the retail rice price to the all-category fitted-yield regression on the price-matched sample. The meal gap stays at about −0.013, with or without allowing the price coefficient to differ by meal category. This concerns the

Table A22: Weather-Predicted Rice Yield and District Retail Rice Prices

Dependent variable: log(retail coarse-rice price)						
	Coef.		SE	p	N	Districts
Fitted log rice yield	−0.090	(0.073)	[0.093]	.336	840	62

Notes. Unit: district×season×year. Prices are WFP/HDX district retail coarse-rice prices (World Food Programme, 2026), in nominal BDT per kg, averaged over harvest windows (Boro April–June; Aus July–September; Aman November–January). Fitted log yield is the weather-predicted yield of Appendix Table A6, Panel A. Year and district×season fixed effects; the year effects absorb common annual price movements, so subtracting a common annual log CPI would leave the yield slope unchanged. The sample has 62 districts: the price file lacks Patuakhali, and Cox's Bazar does not match the wage records. Parentheses: district-clustered standard error, treating fitted yield as known. Brackets: district bootstrap standard error (999 draws) that re-estimates the weather–yield regression in each draw; the reported p-value uses the bootstrap standard error with a t_{63} reference distribution. The estimate is an association between predicted yield and retail prices; it does not test whether prices mediate any wage response, and retail prices may differ from the prices farmers receive.
* Significant at 10%; ** at 5%; *** at 1%.

meal gap, not the pooled IV slope, and does not test the price account. Appendix Figure A2 shows the national retail price series.

Assumed meal values

Appendix Table A24 and Appendix Figure A3 add an assumed value of meals to the cash wage before taking logs and re-estimate the fitted-yield meal gap. Under the public-basket and BIHS valuations the gap turns slightly positive, about 0.02, and it reaches the 10 percent significance level under the public-basket and upper BIHS valuations. Because the meal values are assumed, these rows do not measure total daily pay, and they do not test whether meal jobs have smaller seasonal changes in pay.

Table A23: Retail Rice Prices and the Fitted-Yield Meal Gap

	Coef.	SE	p	N	Districts
Dependent variable: log(*real daily cash wage*)					
Coefficient reported: fitted-yield meal gap (three minus no meals)					
Price sample, no price control	−0.013	(0.018) [0.018]	.499	4,201	62
Adding log rice price	−0.013	(0.018) [0.018]	.470	4,201	62
Adding price by meal category	−0.012	(0.018) [0.018]	.493	4,201	62

Notes. WFP/HDX retail coarse-rice prices, harvest-window averages (Boro Apr–Jun; Aus Jul–Sep; Aman Nov–Jan), nominal BDT/kg; year FEs absorb common annual price movements. Year and district×season FEs; SEs clustered by district. Brackets: district bootstrap SE (999 draws) that re-estimates the weather–yield regression in each draw; clustered SEs treat fitted yield as known. Stars and reported p-values use the bootstrap SE with a t_{63} reference distribution. Each row reports the fitted-yield meal gap (three minus no meals) in log real daily cash wages, using all four meal categories on the price-matched sample; rows 2–3 add the log retail price, alone and interacted with meal category. The price–yield association itself is in Appendix Table A22. These are associations; they do not test whether prices mediate any wage response, and they concern the meal gap, not the pooled IV slope. Wholesale/all-rice series omitted (coverage ≤2 districts).
*** Significant at 1%; ** at 5%; * at 10%.

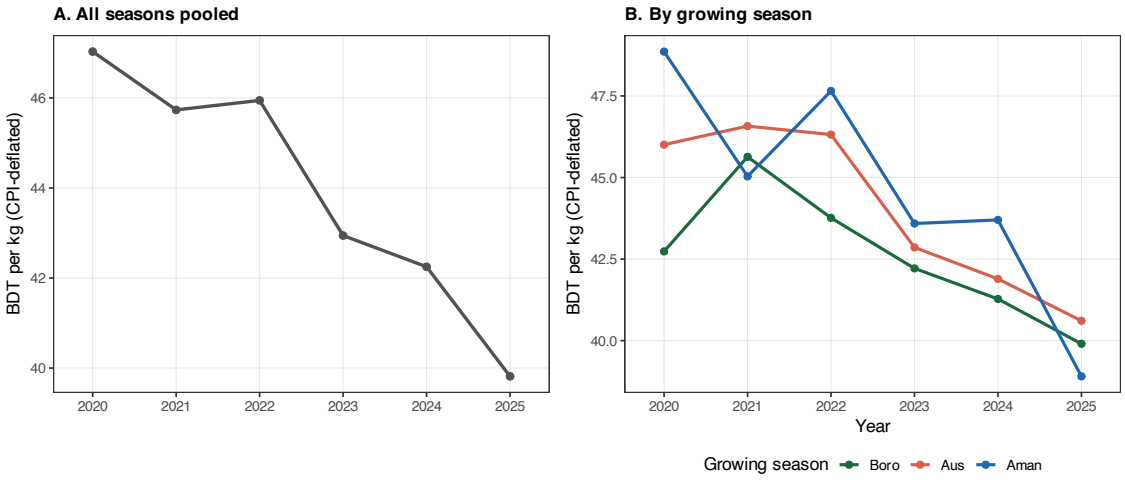


Figure A2: National Mean Retail Coarse-Rice Prices, 2020–2025. Panel A pools growing seasons; Panel B splits them. Values are district harvest-window averages from WFP/HDX prices, deflated with the BBS CPI. Patuakhali is missing throughout, and earlier years, which cover only six districts, are omitted. This figure is descriptive, not a regression estimate; the price regressions use nominal prices with year effects.

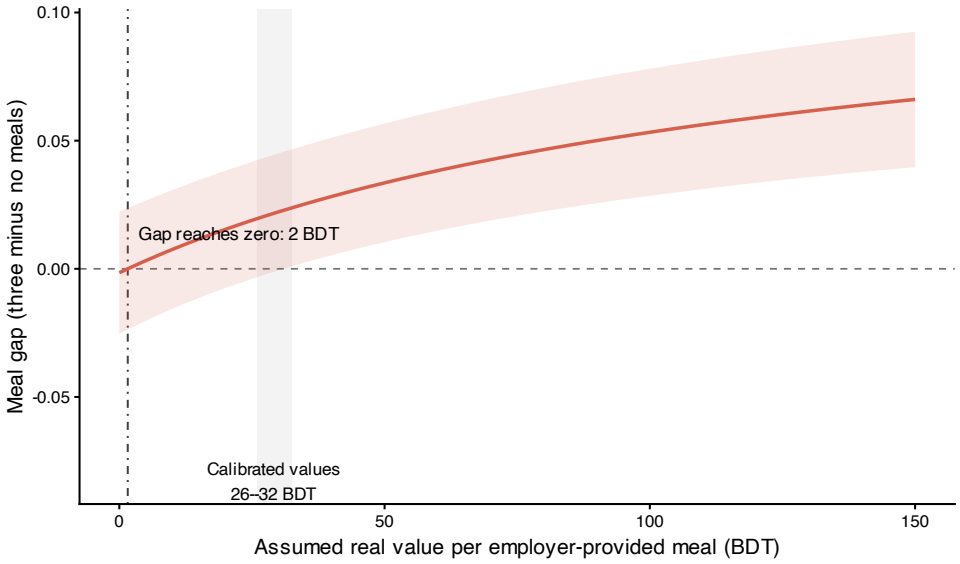


Figure A3: Fitted-Yield Meal Gap under Alternative Assumed Meal Values. The curve re-estimates the meal gap after adding a constant assumed value per meal to daily cash wages (Appendix Table A24). The shaded band is a district-clustered 95% confidence interval that treats fitted yield as known; the gray band marks the BIHS, public-basket, and upper BIHS values. The labelled vertical line marks where the point estimate crosses zero; it is not a confidence statement about a threshold. Meal values are assumed, not observed.

Table A24: Fitted-Yield Wage Differences with Assumed Meal Values

Compensation measure	Coef.	SE	p	N	Districts
Dependent variable: log(*compensation*)*; coefficient: fitted-yield meal gap*					
Cash wage only	−0.002	(0.012) [0.012]	.900	7,708	64
Rice-only floor (cash + meals)	0.008	(0.012) [0.012]	.497	7,708	64
Public-basket valuation	0.019*	(0.011) [0.012]	.096	7,708	64
BIHS employer-food valuation	0.018	(0.012) [0.012]	.125	7,708	64
Upper BIHS valuation (×1.25)	0.022*	(0.012) [0.012]	.063	7,708	64

Notes. Each row uses log(cash wage + imputed meal value); row 1 is cash only. Public basket mean ≈ 31.0 real BDT/meal; BIHS basket ≈ 25.9 (283 employer-meal slots, 207 households). The meal values are assumed, not observed; the rows describe fitted-yield associations and do not measure total pay or test harvest smoothing. Year and district×season fixed effects; standard errors clustered by district. Brackets: district bootstrap SE (999 draws) that re-estimates the weather–yield regression in each draw; clustered SEs treat fitted yield as known. Stars and reported p-values use the bootstrap SE with a t_{63} reference distribution.
*** Significant at 1%; ** at 5%; * at 10%.

A.7 Context: local conditions and household survey

The comparisons in this section are contextual. They do not identify labor demand, outside options, contracts, or the effect of meals.

Local agricultural and economic conditions

Appendix Table A25 and Appendix Table A26 let the meal gap in the all-category fitted-yield specification vary with standardized district proxies. Each coefficient in the "Change per SD" column is the change in the meal gap for a district one standard deviation above the mean of that proxy. The proxies describe district conditions, such as rice area before 2017, the 2019 irrigation share, poverty, or mobile-phone ownership; they are not measures of labor demand or of the workers in the wage records. None of the agricultural proxies changes the gap by much relative to its standard error. Among the nine economic proxies, the gap is larger where poverty is higher (0.036, with clustered and bootstrap standard errors of 0.012 and

0.013; bootstrap $p = .005$) and where mobile-phone ownership is lower (0.021, standard errors of 0.010; $p = .045$). With nine proxies tested, these two results carry little weight on their own.

Table A25: Fitted-Yield Wage Differences across Agricultural Conditions

Dependent variable: log(real daily wage)				
Proxy	Meal gap	Change per SD	p	Years; districts
Panel A: Agricultural conditions (reference years in row labels)				
2019 irrigation share	0.004	0.008	.718	2017–2025; 49
	(0.012) [0.013]	(0.020) [0.022]		
Pre-sample cropping intensity	−0.001	0.016	.430	2017–2025; 59
	(0.013) [0.013]	(0.020) [0.020]		
Pre-sample gross cropped area	−0.009	−0.005	.664	2017–2025; 59
	(0.012) [0.013]	(0.010) [0.011]		
Pre-sample rice area	−0.010	−0.008	.420	2017–2025; 62
	(0.011) [0.011]	(0.008) [0.010]		
Pre-sample rice production	−0.010	−0.006	.590	2017–2025; 62
	(0.011) [0.011]	(0.009) [0.012]		
Panel B: Constructed seasonal rice-work index				
Seasonal rice-work index	0.016	0.019	.278	2017–2023; 63
	(0.013) [0.013]	(0.015) [0.017]		

Notes. Each row adds one standardized proxy (mean zero, standard deviation one) and its interactions with fitted log yield and the three-meal indicator. Meal gap is the three-meal minus no-meal fitted-yield slope at the proxy mean; Change per SD is the change in that gap for a one-standard-deviation higher proxy. The proxies describe district conditions; they do not measure labor demand. The seasonal rice-work index is $\log(1 + \text{rice area} \times \text{season labor weight})$ (weights 90/100/110 for Aus/Aman/Boro); that row uses 2017–2023 only. Meal main effects, gender, year and district×season FEs; district-clustered SEs. Outside-option checks are in Appendix Table A26. Brackets: district bootstrap SE (999 draws) that re-estimates the weather–yield regression in each draw; clustered SEs treat fitted yield as known. Stars and reported p-values use the bootstrap SE with a t_{63} reference distribution.
* Significant at 10%; ** at 5%; *** at 1%.

Table A26: Fitted-Yield Wage Differences across Local Economic Conditions

Proxy	Meal gap	Change per SD	p	Years; districts
Composite market-constraint score	0.004	0.018	.125	2017–2025; 63
	(0.013) [0.013]	(0.010) [0.011]		
Distance to Dhaka	0.003	0.006	.703	2017–2025; 64
	(0.020) [0.019]	(0.016) [0.017]		
Incomplete-primary share	0.003	0.001	.947	2017–2025; 64
	(0.013) [0.013]	(0.011) [0.011]		
Low firm density	−0.004	0.001	.973	2017–2025; 64
	(0.013) [0.013]	(0.003) [0.017]		
Low mobile ownership	0.005	0.021**	.045	2017–2025; 64
	(0.014) [0.014]	(0.010) [0.010]		
Low nonfarm employment	0.001	0.013	.241	2017–2025; 64
	(0.013) [0.013]	(0.010) [0.011]		
Poverty rate	0.014	0.036***	.005	2017–2025; 64
	(0.013) [0.013]	(0.012) [0.013]		
Low night-light radiance	−0.001	0.007	.580	2017–2025; 64
	(0.013) [0.014]	(0.005) [0.013]		
District remittance inflow (USD m)	−0.001	−0.017	.278	2018–2025; 63
	(0.014) [0.014]	(0.013) [0.015]		

Notes. Each row adds one standardized proxy (mean zero, standard deviation one) and its interactions with fitted log yield and the three-meal indicator. Meal gap is the three-meal minus no-meal fitted-yield slope at the proxy mean; Change per SD is the change in that gap for a one-standard-deviation higher proxy. The proxies describe district economic conditions, not the workers in the wage records. All regressions include year and district–season fixed effects; standard errors are clustered by district. Nonfarm employment, mobile ownership, firm density, and night lights are multiplied by −1 so larger values mean weaker outside options. Remittance inflows are log(1 + USD million) from Bangladesh Bank fiscal-year data, mapped to calendar year = fiscal-year end and standardized within year. The corresponding agricultural-condition interactions are in Appendix Table A25. Brackets: district bootstrap SE (999 draws) that re-estimates the weather–yield regression in each draw; clustered SEs treat fitted yield as known. Stars and reported p-values use the bootstrap SE with a t_{63} reference distribution.
* Significant at 10%; ** at 5%; *** at 1%.

Household survey

The chapter's discussion of meals and food needs (Section 1.7.3) raises the question of who eats at an employer's place. In the BIHS 2018–19 survey, households in which someone reports eating at an employer's place differ from other rural households in diet, savings, and food insecurity (Appendix Table A27). This indicator is

not the BBS meal category, and the two surveys are not linked, so the comparison describes households. It is not evidence on the wage patterns in the chapter.

Table A27: Rural Household Characteristics by Reported Employer-Provided Food (BIHS 2018–19)

Outcome	Employer-food	Other	Diff. (p)
Panel A: Dietary diversity			
Mean distinct food items	13.07	10.29	+2.78 ($p < .001$)
Mean food groups	5.15	4.93	+0.22 ($p = .005$)
N households	207	5,396	5,603
Panel B: Savings and food insecurity			
Log total savings (means)	9.06	9.45	−0.389 ($p = .007$)
FIES score (0–8; higher = worse)	2.24	1.50	+0.73 ($p < .001$)
N (savings / FIES)	131 / 207	3,447 / 5,396	

Sources. BIHS 2018–19 (International Food Policy Research Institute, 2020).
Purpose. Descriptive household context only. The survey is not linked to the BBS wage records or to the meal gap in Table A7. BIHS flags households where anyone reports food eaten at an employer's place; it does *not* record BBS meal counts (0–3).
Notes. Columns report group means; Diff. is employer-food minus other (two-sample t-tests for diet and FIES; OLS coefficient on an employer-food indicator for log savings, positive balances only). Cross-section only; selection into employer-food jobs is not random.

A.8 Data construction, sources, and coverage

Cash wages and sample coverage

The BBS agricultural yearbooks report daily cash wages by district, month, gender and number of employer-provided meals (Bangladesh Bureau of Statistics, 2025). The main comparison retains no-meal and three-meal records for 2017–2025 and deflates nominal wages with the national annual CPI. The observations are published wage averages, not a panel of workers or employment contracts. The main data section reports the sample counts and the monthly coverage checks.

Three-meal records are available for fewer months than no-meal records: 8.8 versus 11.4 months per district–year–gender cell on average. The shares observed

in all twelve months are 53 and 87 percent, respectively. Some consecutive monthly values are unchanged. The balanced-sample estimates in Appendix Table A18 address whether uneven reporting changes the harvest comparison; they do not reveal why records are missing or wages repeat. Appendix Figure A4 shows the distribution of reported three-meal wage records. Its shares count records, not workers or employment.

Rice yield, weather, and rice intensity

BBS yearbooks supply seasonal rice production and area (Bangladesh Bureau of Statistics, 2024). Their ratio gives district rice yield. ERA5 supplies hourly temperature and precipitation (Hersbach et al., 2020). Growing-degree days accumulate temperatures between 10 and 30°C; extreme-degree days accumulate heat above 30°C. Temperature indices are area-weighted to districts and precipitation is totaled within the rice growing season. The same three weather measures enter every season's yield regression. BBS seasonal rice area and production also supply the alternative regressors in Appendix Table A9 and Appendix Table A10. Rice intensity is based on district rice area averaged over 2013–2016, before the wage sample. The harvest heterogeneity checks standardize log rice area and examine alternative rice measures. These are district characteristics, not observed labor demand.

Prices, food values, and household data

We obtain district market prices from the WFP/HDX Bangladesh Food Prices file (World Food Programme, 2026). Its sources are FAO GIEWS, FPMA, DAM, and WFP collection. We express district-month prices in BDT per kg-equivalent and average them over harvest windows: April–June for Boro, July–September for Aus, and November–January for Aman. The file has retail coarse-rice prices for 63 of the

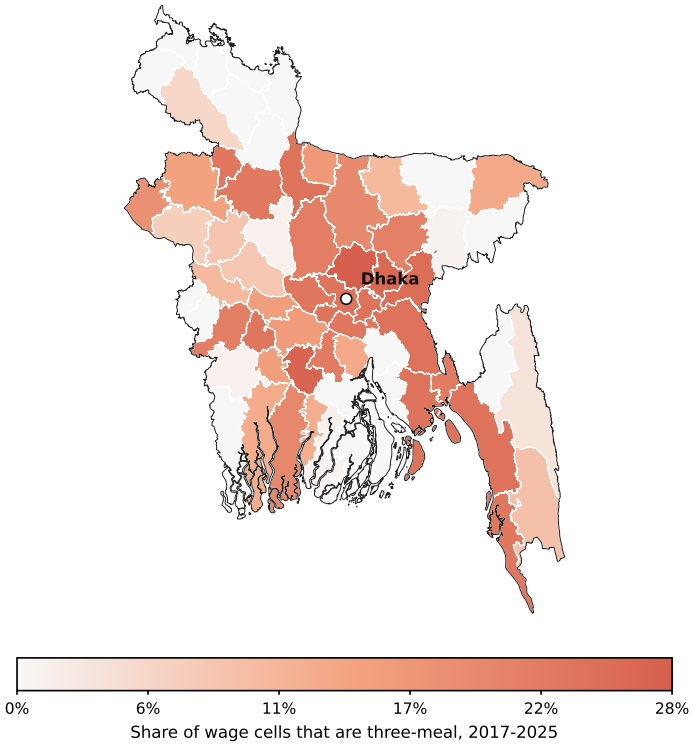


Figure A4: District Share of Three-Meal Wage Cells, 2017–2025. Each district's share counts three-meal records among all four meal categories in the seasonal wage file. The white marker is Dhaka; Satkhira and Chandpur have no three-meal cells.

64 wage-panel districts; Patuakhali is missing. The price comparisons in Appendix Table A22 and Appendix Table A23 use 62 districts, because Cox's Bazar does not match in the merge with the wage records. A fixed ingredient list valued with the same data gives the public-basket meal value in Appendix Table A24, averaging 31.0 real BDT per meal.

The Bangladesh Integrated Household Survey 2018–19 (International Food Policy Research Institute, 2020) supplies the household comparison (Appendix Table A27) and ingredient weights for a second meal valuation (Appendix Table A24), averaging 25.9 real BDT per meal. Its employer-food flag records whether someone in the household ate at an employer's place. It does not correspond to a BBS zero-to-three-meal category, and we do not merge the surveys. BIHS Module C reports about 61 BDT of in-kind food per food-receiving workday. This is a daily benchmark rather than a value for a single BBS meal.

Local economic and agricultural conditions

The World Bank Bangladesh Spatial Database (World Bank, 2022) supplies district poverty, agricultural and nonfarm employment, incomplete primary schooling, distance to Dhaka, mobile ownership, and firm density. The constraint index in Appendix Table A26 counts six indicators: poverty, agricultural employment, incomplete primary schooling, and distance to Dhaka above their medians; mobile ownership and firm density below their medians. The index ranges from zero to six. We also examine its inputs and the nonfarm employment share separately. District-year VIIRS night-light radiance comes from EOG VNL rasters (Elvidge, Zhizhin, Ghosh, Hsu, & Taneja, 2021). District worker remittance inflows come from Bangladesh Bank's fiscal-year Annexure IV tables (Bangladesh Bank, 2025). We assign each fiscal year to

the calendar year in which it ends and standardize log inflows within year. Both variables are proxies for outside options in Appendix Table A26. Pre-2017 BBS land and rice tables supply cropping intensity, gross cropped area, rice area, and rice production. Irrigation share comes from the 2019 agricultural census (Bangladesh Bureau of Statistics, 2019). The constructed seasonal rice-work index in Appendix Table A25 is $\log(1 + \text{rice area} \times \text{season labor weight})$, with weights 90, 100, and 110 for Aus, Aman, and Boro. It is a proxy based on rice area and assumed labor weights, not an observed measure of labor demand or supply. Appendix Table A28 gives the units, coverage, and uses of these sources, and Appendix Table A29 reports their descriptive statistics.

Table A28: Data Sources for the Additional Checks

Block	Contents	Districts	Used in
World Bank index	Index (0–6) *and* all six inputs: poverty, agri. employment, incomplete primary, distance to Dhaka, mobile ownership, firm density	64	A26, A29
Outside options	Nonfarm employment; VIIRS night-light radiance; district remittance inflows (Bangladesh Bank FY)	63–64	A26, A29
Agriculture / land	Irrigation share; cropping intensity; gross cropped area; rice area; rice production; seasonal rice-work index	49–64	A25, A29
Prices / meals	Retail coarse rice price; imputed public and BIHS meal values	62–64	A23–A24, A29
BIHS households	Dietary diversity (food items, food groups); FIES; log savings (employer-food vs other)	—	A27, A29

Sources. World Bank (World Bank, 2022); EOG VIIRS; Bangladesh Bank remittance; WFP/HDX (World Food Programme, 2026); BIHS (International Food Policy Research Institute, 2020).

Notes. The table lists the external data used in the appendix; Appendix Table A29 reports means, SDs, and N for each variable (including all six constraint-index inputs).

Table A29: Summary Statistics for the Additional-Check Data

Variable	*N*	Districts	Mean	SD
A. World Bank composite constraint index and its six inputs (Appendix Table A26)				
Composite constraint index (0–6)	64	64	3.02	1.49
Poverty rate (%)	64	64	27.4	15.3
Agri. employment share (%)	64	64	43.3	15.3
Incomplete primary share (%)	64	64	8.3	3.2
Distance to Dhaka (km)	64	64	149.0	75.9
Mobile ownership (%)	64	64	69.1	9.7
Firm density (per km^2)	64	64	64.3	59.2
B. Other outside-option / friction proxies (Appendix Table A26)				
Nonfarm employment share (%)	64	64	56.7	15.3
VIIRS night-light radiance	576	64	0.53	0.92
District remittance inflow (USD m)	503	63	339.8	889.5
C. Agricultural and land proxies (Appendix Table A25)				
2019 irrigation share	49	49	0.67	0.19
Pre-sample cropping intensity	59	59	1.91	0.35
Pre-sample gross cropped area (000 ha)	59	59	570.9	301.8
Pre-sample rice area (000 ha)	62	62	147.9	102.9
Pre-sample rice production (000 t)	62	62	181.6	140.8
Log seasonal rice-work index	1,335	64	15.80	1.56
D. Rice prices and imputed meal values (Appendix Table A23 and Appendix Table A24)				
Retail coarse rice price (real BDT/kg)	849	62	43.9	4.0
Imputed meal value: public food basket (real BDT/meal)	1,728	64	31.0	2.3
Imputed meal value: BIHS employer basket (real BDT/meal)	1,728	64	25.9	1.4

E. BIHS 2018–19 household context (Appendix Table A27; not merged to wage panel)

Outcome	Employer-food	Other	*N* emp.	*N* other
Distinct food items	13.07	10.29	207	5,396
Food groups	5.15	4.93	207	5,396
FIES score (0–8; higher = worse)	2.24	1.50	207	5,396
Log total savings (positive balances)	9.06	9.45	131	3,447

Sources. A–B: World Bank Spatial Database (World Bank, 2022); EOG VIIRS night lights; Bangladesh Bank district remittance Annexure IV. C: BBS land/rice tables and 2019 agricultural census. D: WFP/HDX (World Food Programme, 2026); BIHS ingredient weights for meal values. E: BIHS 2018–19 (International Food Policy Research Institute, 2020) (IFPRI).

Construction. The composite constraint index (Panel A) is the sum of six binary indicators for above-median poverty, agri. employment, incomplete primary, and distance to Dhaka, and below-median mobile ownership and firm density (range 0–6). Panel A therefore reports the index *and* every input used to build it. The seasonal rice-work index is $\log(1 + \text{rice area} \times \text{season labor weight})$ (weights 90/100/110 for Aus/Aman/Boro). Imputed meal values enter only the cash-plus-meals sensitivity in Appendix Table A24.

Notes. Means and SDs are across the listed unit (district, district-year, or district-season-year). These variables enter only the supplementary appendix comparisons; they are not instruments and do not enter the main wage equations.

District trends

Appendix Figure A5 plots each district's changes in mean rice yield and real cash wages over 2017–2025. Yields rise and cash wages fall in 57 of the 64 districts. These unadjusted changes describe district trends. They do not identify the harvest or yield comparisons in the chapter, which use variation within districts and years.

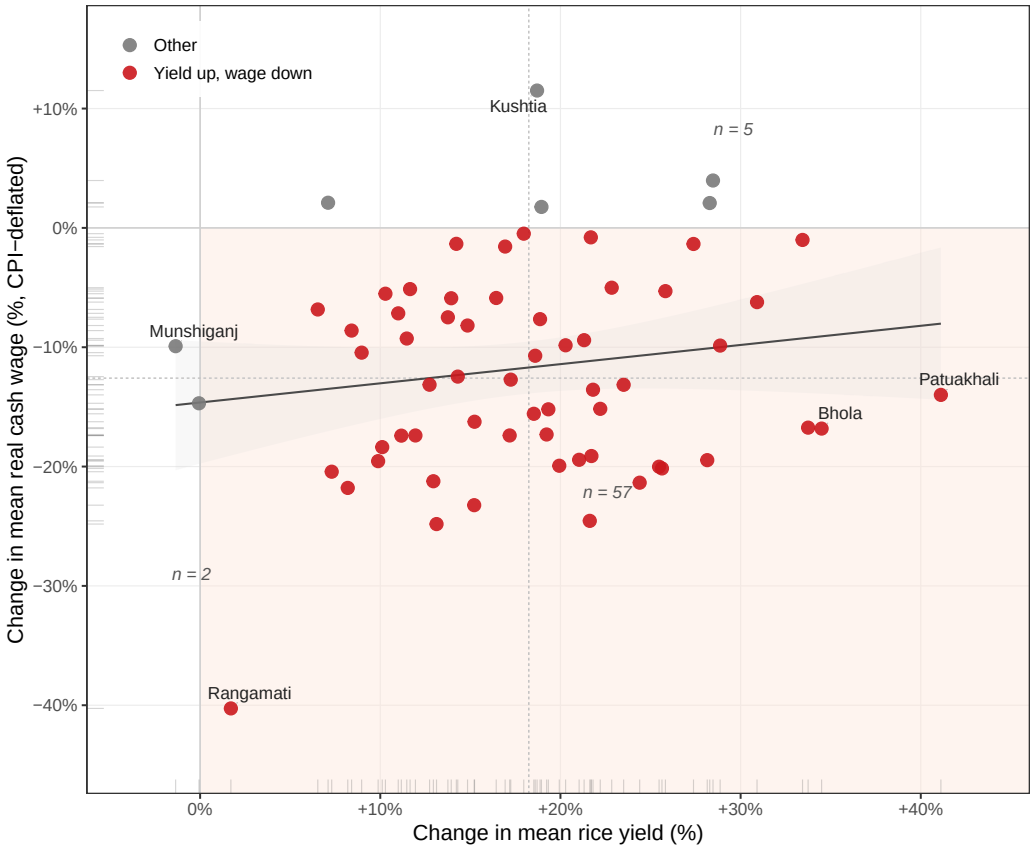


Figure A5: District Yield and Cash-Wage Changes, 2017–2025. Each point is one district. The shaded quadrant marks yield up and cash wage down (57 of 64 districts). The cross-district correlation is 0.16; dashed lines are sample medians (+18% yield, −13% real cash wage). The solid line is an unadjusted linear fit with a 95% confidence band. These district changes are descriptive and do not identify the within-district yield–wage relationship.

Appendix B

HYBRID MODELS FOR CROP YIELD PROJECTIONS: CROP-SPECIFIC WEIGHTING OF EMPIRICAL AND PROCESS-BASED MODELS

The appendix follows the chapter's argument. Sections B.1 and B.2 document data construction and compare the weather responses of observed and simulated yields. Sections B.3 and B.4 provide prediction benchmarks, geographic weight checks, and bootstrap details. Section B.5 examines sensitivity to the missing pDSSAT simulations and PBM calibration, then compares the direction of the component projections. Section B.6 reports the regional and country patterns behind the pooled main-text summaries.

B.1 Data and Climate Construction

Main-text link: Section 2.4, Data.

B.1.1 Observed yields and spatial identifiers

The raw archive contains annual production, harvested area, and yield records for maize, rice, and wheat in 33 countries. Country files differ in units, crop names, seasons, and administrative resolution. We harmonize crop names and seasons, convert yields to tonnes per hectare, and match each observation to a GADM 4.1 identifier at the available administrative level. The matched estimation panel uses China, Kenya, and Nigeria at ADM1, and Brazil, India, and the United States at

ADM2; European NUTS2 observations are matched to GADM ADM2. The European green-maize series is excluded because it measures fresh biomass rather than grain yield.

US maize and wheat use the labeled USDA extract cited in Section 2.4. When an all-wheat total is unavailable, the fallback covers the reported classes; a missing class is not treated as a reported zero. The archival US rice extract retains one production and one harvested-area value per accepted county-year but lacks the detailed category labels recovered for maize and wheat. Its category identity cannot therefore be checked to the same level. Two high source-reported rice yields in Meru, Kenya, are retained: 49.5 tonnes per hectare in 2012 and 42.57 in 2013. Excluding these two records in a diagnostic changes the four-component-sample rice EDD coefficient from -7.1177 to -7.1190; the implied yield change for 10 additional EDD moves from -6.8703% to -6.8715%. This checks the Stage 1 coefficient's sensitivity to those records, not their source accuracy or the sensitivity of every hybrid output.

The harmonized keys are country, year, season, crop, and `poly_id`. They link observed yields to weather variables, irrigation shares, and simulated yields. Table 2.2 in the main text reports coverage and selected variables for the crop-specific common sample. Table B3 reports the full input distributions, including raw simulated PBM yields. These are in the same units as observed yield, so comparing their means with the observed mean measures each model's level bias (Table B9); the calibration regressions absorb mean level differences before ensemble weighting. Table B1 shows how unevenly observations are distributed across the geographic validation folds.

B.1.2 Process-based yields and irrigation weighting

Historical yields from EPIC, pDSSAT, and LPJmL come from ISIMIP3a. Each model reports irrigated and rainfed yields on a 0.5° grid through 2016. Grid cells are

Table B1: Geographic Composition of the Historical Estimation Sample

Geographic fold	Maize	Rice	Wheat
Brazil	171,478 (69.2%)	126,287 (94.6%)	30,432 (29.6%)
United States	71,382 (28.8%)	2,088 (1.6%)	69,492 (67.7%)
Europe	0 (0.0%)	128 (0.1%)	2,155 (2.1%)
India and China	4,396 (1.8%)	4,568 (3.4%)	544 (0.5%)
Kenya and Nigeria	568 (0.2%)	415 (0.3%)	46 (0.0%)
Total	247,824	133,486	102,669

Notes: Cells give observations and within-crop shares. Europe groups European countries; India and China, and Kenya and Nigeria, are combined validation folds. The sample requires finite observed yield, climate variables, and all three PBM yields. Fold sizes differ substantially; the reported held-out RMSE pools squared errors over observations in eligible folds.

aggregated to the administrative unit of each observed series using spatial overlap weights based on harvested area.

MIRCA2000 harvested-area rasters provide crop-specific irrigated and rainfed shares. For a climate variable or simulated yield X, the combined value is

$$X_{it} = \frac{X_{it}^{\mathrm{IRC}} A_i^{\mathrm{IRC}} + X_{it}^{\mathrm{RFC}} A_i^{\mathrm{RFC}}}{A_i^{\mathrm{IRC}} + A_i^{\mathrm{RFC}}}, \tag{B.1}$$

where A_i^{IRC} and A_i^{RFC} are irrigated and rainfed harvested areas for the relevant crop. Observed yields, weighted climate variables, and weighted PBM yields are then merged by year, crop, season, and `poly_id`.

B.1.3 Temperature and precipitation variables

Hourly ERA5 temperature and precipitation are available from 1978 to 2019. For threshold T, we calculate hourly exceedance and divide its daily sum by 24, expressing heat exposure in degree-days. GDD and EDD are summed over crop-specific phenological windows and combined across irrigated and rainfed areas using Equation B.1. Specifically,

$$H_{id}(T) = \frac{1}{24}\sum_{h=1}^{24}\max(0, T_{idh} - T), \tag{B.2}$$

$$\text{GDD}_{it} = \sum_{d\in P}\left[H_{id}(T_\ell) - H_{id}(T_u)\right], \tag{B.3}$$

$$\text{EDD}_{it} = \sum_{d\in P} H_{id}(T_u), \tag{B.4}$$

where P is the selected phenological window. Both variables are divided by 1,000 before estimation. For each irrigation regime, R sums daily precipitation and Q sums squared daily precipitation over the planting, middle-growth, and harvest periods. We combine irrigated and rainfed values using Equation B.1 and divide each measure by 1,000. Thus Q records rainfall concentration across days; it is not the square of the seasonal total R.

B.1.4 Threshold and phenology selection

Candidate lower thresholds are 0°C, 4°C, 8°C, and 12°C; candidate upper thresholds are 28°C, 30°C, and 32°C. The threshold search used observed yields and weather before the PBM complete-case restriction. Each candidate fixed-effects model was estimated with each eligible geographic group held out in turn: four groups for maize and five for rice and wheat. Candidates were ranked by mean held-out RMSE. The selected definitions were then held fixed for the historical estimates and the projections. The hybrid validation in Section 2.5.3 uses the narrower common sample and has four eligible folds for maize.

Table B2: Temperature Thresholds and Growth Windows for the Weather Variables

Crop	T_ℓ (°C)	T_u (°C)	Growth window
Maize	0	30	harvest
Rice	12	28	harvest
Wheat	8	28	plant + middle

Notes: T_ℓ and T_u are the lower and upper temperature thresholds. Growing degree days (GDD) sum degree-days between them; exceedance degree days (EDD) sum degree-days above T_u. Both are summed over the listed growth window. The thresholds and windows were selected by cross-validation over eligible geographic regions (four for maize and five for rice and wheat) on a broader observed-yield and weather sample, before requiring complete PBM output, and are held fixed in the historical estimates and the projections. The four-component estimation sample has no European maize observations, so its geographic validation uses four regions for maize.

The selected configurations differ across crops. Maize uses the harvest window with thresholds of 0°C and 30°C, rice uses the harvest window with 12°C and 28°C, and wheat uses the combined planting and middle-growth window with 8°C and 28°C. These choices are fixed before estimating the historical models and are carried unchanged into the projection climate variables. For wheat, the four best candidate bands, all using planting and middle growth, are 8–28°C, 4–28°C, 8–30°C, and 4–30°C, with mean fold RMSEs of 0.4779, 0.4783, 0.4783, and 0.4788. For maize, the two best harvest-window bands, 0–30°C and 0–28°C, have mean fold RMSEs of 0.6757 and 0.6768. The small spread describes the saved search results; it is not a test of statistical equivalence. Because the search considers upper cutoffs only between 28°C and 32°C, and wheat's selected cutoff is the lowest of these, it cannot locate a lower physiological damage threshold for wheat.

Table B3: Distributions of Yield, Weather, and Simulated Variables in the Historical Estimation Sample

Variable	Observations	Mean	SD	P10	Median	P90
Maize						
Observed log yield	247,824	0.811	1.124	-0.694	1.019	2.061
GDD (°C-days/1,000)	247,824	2.084	0.996	0.722	1.706	3.364
EDD (°C-days/1,000)	247,824	0.012	0.019	0.000	0.004	0.036
Seasonal precipitation (mm/1,000)	247,824	0.873	0.451	0.289	0.839	1.461
Daily rain squares (Q)	247,824	2.157	1.685	0.387	2.046	3.842
EPIC log yield	247,824	1.565	0.484	0.954	1.552	2.243
pDSSAT log yield	247,824	1.449	0.684	0.345	1.626	2.239
LPJmL log yield	247,824	1.582	0.400	1.115	1.525	2.137
Rice						
Observed log yield	133,486	0.424	0.717	-0.357	0.405	1.319
GDD (°C-days/1,000)	133,486	1.194	0.259	0.880	1.251	1.461
EDD (°C-days/1,000)	133,486	0.043	0.043	0.003	0.029	0.108
Seasonal precipitation (mm/1,000)	133,486	1.160	0.400	0.640	1.164	1.626
Daily rain squares (Q)	133,486	2.911	2.012	1.285	2.700	4.490
EPIC log yield	133,486	0.958	0.529	0.266	1.062	1.498
pDSSAT log yield	133,486	0.924	0.589	0.158	0.993	1.574
LPJmL log yield	133,486	1.078	0.356	0.747	1.096	1.419
Wheat						
Observed log yield	102,669	0.838	0.511	0.182	0.878	1.439
GDD (°C-days/1,000)	102,669	1.465	0.534	0.831	1.371	2.241
EDD (°C-days/1,000)	102,669	0.036	0.034	0.004	0.027	0.079
Seasonal precipitation (mm/1,000)	102,669	0.739	0.490	0.288	0.570	1.535
Daily rain squares (Q)	102,669	2.020	2.120	0.482	1.480	4.217
EPIC log yield	102,669	1.098	0.418	0.541	1.192	1.549
pDSSAT log yield	102,669	0.965	0.803	0.127	1.066	1.759
LPJmL log yield	102,669	1.266	0.515	0.595	1.386	1.816

Notes: All rows use the crop-specific main-season estimation sample. GDD (growing degree days) and EDD (exceedance degree days) are defined in Table 2.2; both are weighted across irrigated and rainfed area and divided by 1,000. Seasonal precipitation (R) and the sum of squared daily precipitation (Q) are each weighted the same way and divided by 1,000; Q is not the square of R. Observed and simulated yields are log tonnes per hectare. The PBM rows show raw simulated yields, in the same units as observed yield, before the calibration used for ensemble weighting. Comparing their means with the observed mean measures each model's level bias (Table B9). Calibration removes such level differences, so this raw comparison is distinct from the calibrated predictions evaluated in Table 2.6.),

B.2 Weather Responses of Observed and Simulated Yields

Main-text link: Section 2.6.1, Historical climate-yield relationships.

We estimate the climate sensitivity of each PBM output on the crop-specific common sample used in Stage 1. The dependent variable is simulated log yield from EPIC, pDSSAT, or LPJmL. Every column uses the same GDD, EDD, precipitation variables, scaling, season-year fixed effects, and season-location fixed effects as the observed-yield regression. Because every column shares the regressors, sample, and fixed effects, the coefficients can be compared within each crop. The comparison is not a controlled experiment, however: the regressors are built from ERA5, whereas the simulations were driven by GSWP3-W5E5 weather, and the simulated yields follow the prescribed management described in Section 2.3.

For maize and rice, the observed EDD coefficients are more negative than the corresponding PBM coefficients. The wheat pattern differs: all three PBMs have more negative EDD coefficients than the observed-yield regression. Other coefficients also vary in sign and magnitude across crops and models. These regressions document differences between observed and simulated responses within the estimation sample. Within the framework of Section 2.3, such differences can arise from crop-model structure, the management specification, the weather forcing, or the measurement and aggregation of yields; the regressions do not apportion them among these sources, and they are not used to estimate the ensemble weights.

Table B4: Weather Coefficients for Observed and Simulated Log Yields on the Matched Sample

Crop	Variable	Empirical	EPIC	pDSSAT	LPJmL
Maize	GDD	0.2871^{***}	0.0245^{***}	0.0987^{***}	0.0358^{***}
		(0.0184)	(0.0060)	(0.0067)	(0.0042)
	EDD	-4.6213^{***}	-1.5378^{***}	-1.5828^{***}	-0.9483^{***}
		(0.2110)	(0.0522)	(0.0672)	(0.0373)
	Precipitation	0.3353^{***}	0.0665^{***}	0.0406^{***}	-0.0064^{**}
		(0.0129)	(0.0036)	(0.0033)	(0.0029)
	Daily rain squares (Q)	-0.0184^{***}	-0.0032^{***}	-0.0017^{***}	0.0013^{**}
		(0.0029)	(0.0007)	(0.0005)	(0.0006)
Rice	GDD	0.8155^{***}	-0.1299^{***}	0.1134^{***}	0.1158^{***}
		(0.0536)	(0.0150)	(0.0296)	(0.0166)
	EDD	-7.1177^{***}	-1.2303^{***}	-0.3431^{**}	-1.6344^{***}
		(0.2405)	(0.0665)	(0.1438)	(0.0889)
	Precipitation	0.3561^{***}	0.1253^{***}	-0.0433^{***}	0.1031^{***}
		(0.0212)	(0.0105)	(0.0126)	(0.0075)
	Daily rain squares (Q)	-0.0233^{***}	-0.0109^{***}	-0.0075^{***}	-0.0062^{***}
		(0.0048)	(0.0025)	(0.0025)	(0.0015)
Wheat	GDD	-0.2844^{***}	0.1063^{***}	-0.0087	0.0543^{***}
		(0.0191)	(0.0146)	(0.0142)	(0.0182)
	EDD	-0.8254^{***}	-2.9349^{***}	-2.4038^{***}	-3.7249^{***}
		(0.0795)	(0.1064)	(0.0996)	(0.1653)
	Precipitation	-0.0937^{***}	-0.0689^{***}	-0.1651^{***}	-0.0437^{***}
		(0.0091)	(0.0046)	(0.0064)	(0.0057)
	Daily rain squares (Q)	-0.0033^{**}	-0.0003	0.0018^{**}	-0.0029^{**}
		(0.0016)	(0.0006)	(0.0007)	(0.0012)

Notes: Each column regresses one log-yield series on the same weather variables and fixed effects, using the same crop-specific estimation sample: observed yield (Empirical) or the raw simulated yield of EPIC, pDSSAT, or LPJmL. GDD (growing degree days) and EDD (exceedance degree days) are in thousands of degree-days, with the crop-specific thresholds listed in Table 2.3. R is seasonal precipitation and Q the sum of squared daily precipitation across planting, middle growth, and harvest; each is weighted across irrigated and rainfed area and divided by 1,000. All models include season-year and season-location fixed effects. Standard errors, clustered by season-location, are in parentheses. The regressors use ERA5 weather, whereas the historical simulations are driven by GSWP3-W5E5. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.3 Alternative Specifications and Predictive Performance

Main-text link: Section 2.5.5, Model comparison.

B.3.1 Unconstrained hybrid

The unconstrained hybrid includes the three PBM log yields and four climate variables in one fixed-effects regression:

$$\log(Y_{it}) = \theta_1 \log(\hat{Y}_{it}^{\mathrm{EPIC}}) + \theta_2 \log(\hat{Y}_{it}^{\mathrm{pDSSAT}}) + \theta_3 \log(\hat{Y}_{it}^{\mathrm{LPJmL}}) + \beta_1 \mathrm{GDD}_{it} + \beta_2 \mathrm{EDD}_{it} + \beta_3 R_{it} + \beta_4 Q_{it} + \alpha_{s(i)} + \lambda_{st} + \varepsilon_{it}. \quad \text{(B.5)}$$

The θ coefficients may be negative and need not sum to one. This specification provides a flexible predictive benchmark. It could also be used to project, but its coefficients are not shares of the prediction. Table B5 reports the three PBM coefficients. Their sum is approximately 0.88 for maize, 0.19 for rice, and 0.23 for wheat; the EPIC coefficient for wheat is slightly negative and not statistically different from zero at the 5% level. The unrestricted regression therefore does not produce a common set of nonnegative shares, even when it predicts well.

B.3.2 Ensemble benchmark comparison

Table B6 compares four ways of combining the components introduced in main-text Table 2.6. The unconstrained hybrid has the lowest held-out-region RMSE for maize and wheat. For rice, the equal-weight combination of calibrated PBMs performs best among the four combinations, although main-text Table 2.6 shows that the calibrated EPIC component alone performs slightly better. The constrained hybrid is retained because its weights are nonnegative shares that sum to one and can be

Table B5: Process-Based Model Coefficients in the Unconstrained Hybrid Regression

Crop	EPIC	pDSSAT	LPJmL
Maize	0.3282^{***}	0.1675^{***}	0.3820^{***}
	(0.0145)	(0.0097)	(0.0307)
Rice	0.0237^{***}	0.0017	0.1599^{***}
	(0.0089)	(0.0075)	(0.0202)
Wheat	-0.0191^{*}	0.2201^{***}	0.0283^{***}
	(0.0106)	(0.0096)	(0.0077)

Notes: Each row reports one regression of observed log yield on the three simulated log yields, GDD, EDD, seasonal precipitation (R), the sum of squared daily precipitation (Q), season-year fixed effects, and season-location fixed effects (Equation B.5). Standard errors, clustered by season-location, are in parentheses. The coefficients are unrestricted: they can be negative and need not sum to one, so they are not ensemble shares. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

applied directly to projected component changes, not because it dominates every benchmark.

Table B6: Prediction Errors of Alternative Ensemble Combinations

	In-sample RMSE			Held-out-region RMSE		
Model	Maize	Rice	Wheat	Maize	Rice	Wheat
Equal-weight calibrated PBMs	0.4386	0.4270	0.2686	0.8297	0.6863	0.4182
Equal-weight four components	0.4380	0.4205	0.2684	0.8315	0.6864	0.4191
Unconstrained hybrid	0.4343	0.4117	0.2670	0.7979	0.6966	0.4089
Constrained hybrid	0.4375	0.4125	0.2676	0.8150	0.6938	0.4112

Notes: Entries are root mean squared errors in log-yield units on each crop's historical estimation sample; lower values indicate better prediction. The equal-weight rows average the calibrated fitted components (three PBMs, or three PBMs and the empirical model). Held-out-region scores refit each specification without the held-out region, omit its unavailable fixed effects, and center observed and predicted log yields within the region. In-sample errors include fitted fixed effects, so the two sets of columns assess different prediction tasks. Maize has four eligible held-out regions because its sample contains no European observations; rice and wheat have five. The constrained hybrid imposes nonnegative weights that sum to one.

B.4 Geographic Validation and Weight Stability

Main-text link: Section 2.6.2, Ensemble weights and predictive performance.

B.4.1 Leave-one-region-out estimates

The candidate folds are Brazil, the United States, Europe, India and China, and a final group containing Kenya and Nigeria. Stage 1 and Stage 2 are re-estimated after excluding each eligible fold. Maize has no European observations in the common sample, so that crop has four rather than five eligible folds. Table B7 reports the empirical weight from each training sample, and Figure B1 plots the same estimates against the full-sample weight.

Maize is the most geographically sensitive: its empirical weight falls from 0.33 in the full sample to zero when Brazil is excluded and rises to 0.57 when the United States is excluded, while the other eligible folds leave it at 0.33. Rice retains a high empirical weight, ranging from 0.86 to 0.91, while wheat ranges from 0.11 to 0.32. The exercise therefore supports the importance of the empirical component for rice but does not support treating the maize or wheat weights as geographically invariant.

Table B8 reports errors separately for each held-out region. The pooled main-text RMSE is the square root of the observation-weighted mean of squared regional RMSEs. Regions with more observations therefore have more influence. Regional centering removes one overall mean, while within-region location differences remain. The training predictions used to estimate weights include fixed effects, so the held-out score assesses a different prediction task.

B.4.2 Component bias and anomaly tracking

Table B9 separates average level bias from within-location co-movement. Raw simulated yields exceed observed yields on average in every crop, by 0.13 to 0.77 log

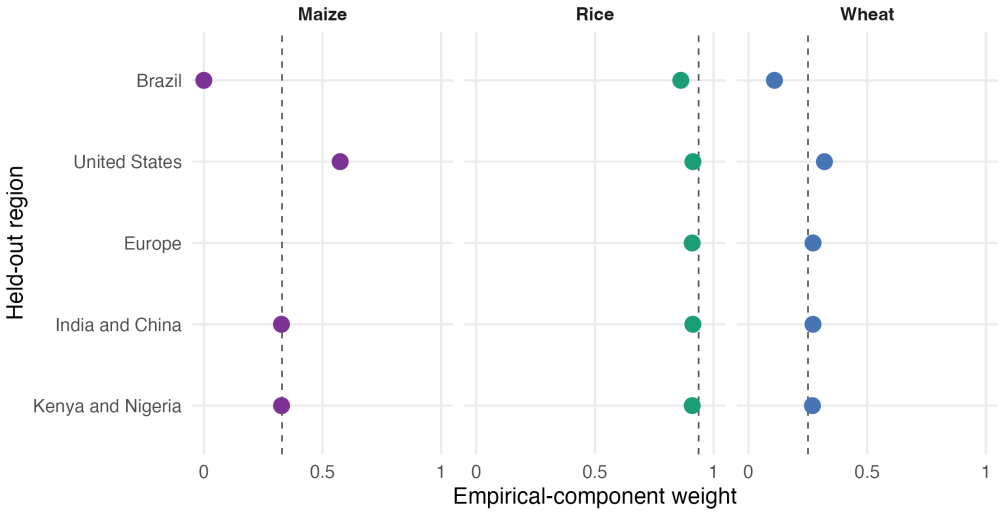


Figure B1: Empirical-Component Weight with Each Region Held Out. Each point re-estimates the four-component hybrid without the named region; dashed lines mark the full-sample weights in Table 2.4. Values are reported in Table B7. Maize has no European observations in the estimation sample.

Table B7: Empirical-Component Weight with Each Region Held Out

Held-out region	Maize	Rice	Wheat
Brazil	0.000	0.862	0.109
United States	0.574	0.913	0.320
Europe	—	0.910	0.272
India and China	0.327	0.913	0.272
Kenya and Nigeria	0.327	0.910	0.269
Full estimation sample	0.329	0.937	0.251

Notes: Each row re-estimates the empirical model, the PBM calibrations, and the four-component constrained weights without the named region and reports the empirical component's weight. The last row is the full-sample weight in Table 2.4. Europe comprises the European Union member states in the sample. A dash marks a region with no maize observations in the estimation sample.

Table B8: Prediction Errors within Each Held-Out Region

Crop	Region	Test N	Empirical	EPIC	pDSSAT	Unconstrained	Constrained
Maize	Brazil	171,478	0.964	0.974	0.867	0.888	0.930
Maize	United States	71,382	0.490	0.416	0.375	0.541	0.453
Maize	India and China	4,396	0.722	0.593	0.523	0.608	0.609
Maize	Kenya and Nigeria	568	0.408	0.501	0.426	0.448	0.447
Rice	Brazil	126,287	0.692	0.688	0.701	0.694	0.691
Rice	United States	2,088	0.346	0.192	0.185	0.350	0.327
Rice	Europe	128	0.319	0.297	0.308	0.302	0.312
Rice	India and China	4,568	0.897	0.662	0.668	0.857	0.868
Rice	Kenya and Nigeria	415	0.801	0.723	0.731	0.786	0.784
Wheat	Brazil	30,432	0.506	0.515	0.513	0.511	0.505
Wheat	United States	69,492	0.387	0.389	0.356	0.353	0.360
Wheat	Europe	2,155	0.522	0.521	0.465	0.447	0.477
Wheat	India and China	544	0.450	0.500	0.413	0.386	0.417
Wheat	Kenya and Nigeria	46	0.519	0.538	0.559	0.550	0.544

Notes: RMSE in log-yield units. Each fold refits slopes outside the named region and uses its training-sample ensemble weights. Observations and predictions are centered once within the held-out region. Main-text RMSE pools squared errors across test observations before taking the square root; it does not average these fold RMSEs. The table includes representative components and both fitted combinations. Maize has no eligible European fold.

points. This raw-level comparison measures simulation bias; it is distinct from the calibrated predictions compared in Table 2.6, because calibration and the fixed effects absorb constant level differences. Additive log-yield differences that are constant within a season-location or common within a season-year are absorbed by the calibration fixed effects. Varying errors can still affect calibrated predictions and weights. The correlations show how well each component tracks year-to-year deviations within locations, and all are weak.

Table B9: Level Bias and Within-Location Correlation of Component Predictions

	Maize			Rice			Wheat		
Model	Bias	Pear.	Spear.	Bias	Pear.	Spear.	Bias	Pear.	Spear.
Empirical	-0.000	0.05	0.04	+0.000	0.20	0.10	+0.000	-0.03	-0.03
EPIC	+0.754	0.16	0.13	+0.533	0.11	0.11	+0.261	0.06	0.07
pDSSAT	+0.639	0.10	0.07	+0.500	0.01	0.03	+0.127	0.14	0.16
LPJmL	+0.771	0.16	0.15	+0.654	0.15	0.11	+0.428	0.11	0.15

Notes: Bias is the mean predicted log yield minus the mean observed log yield on each crop's estimation sample. For the PBMs, the prediction is the raw simulated yield before calibration, so the bias measures how far simulated levels lie from observed levels; the empirical bias is zero by construction. Pear. and Spear. are Pearson and Spearman correlations between location-demeaned predicted and observed log yields; the empirical series uses only the estimated weather contribution, excluding fixed effects. Additive log-yield differences in the season-location and season-year fixed-effect span do not mechanically affect the historical fitted components or their weights; errors outside that span can affect both. The correlations describe how well each component tracks year-to-year deviations within locations.

B.4.3 Location bootstrap

The location bootstrap in Table 2.4 draws locations with replacement and applies their draw counts as frequency weights in both stages. It measures variation across location compositions rather than geographic transfer. The maize empirical weight is comparatively stable, with a 95% interval of 0.28 to 0.37. Rice and wheat have wider intervals: the lower tail of the rice distribution assigns substantial weight to the PBMs, and the wheat empirical weight ranges from 0.00 to 0.33 around a bootstrap median of 0.10.

B.5 Projection Sensitivity and Component Agreement

Main-text link: Section 2.6.3, Projected yield changes.

B.5.1 Reduced ensemble for SSP3-7.0

pDSSAT has no SSP3-7.0 simulation in the ISIMIP3b archive. The primary projection therefore re-estimates the empirical climate response, PBM calibration slopes, and ensemble weights on historical observations from the three components available under that scenario. Table B10 reports these weights. Like the four-component hybrid, the three-component combination is used as a historically estimated combination rule, not as the best predictor for every crop.

Table B10: Three-Component Weights Used for the SSP3-7.0 Projections

Crop	Empirical	EPIC	LPJmL	Historical observations
Maize	0.318	0.514	0.169	260,902
Rice	0.908	0.010	0.082	137,398
Wheat	0.534	0.219	0.247	104,219

Notes: Weights of the empirical model and the calibrated EPIC and LPJmL components, re-estimated on historical data together with the empirical coefficients and PBM calibrations, because pDSSAT has no SSP3-7.0 simulations. They are non-negative and sum to one. The crop-specific samples require finite values for these three components but not for pDSSAT, so they are larger than the four-component samples behind Table 2.4. The SSP1-2.6 projections use the four-component weights.

Table B11 compares this specification with two alternatives: rescaling the four-component weights after dropping pDSSAT, and keeping the four-component weights while assigning pDSSAT a projected change of zero. Rescaling the weights gives results close to the primary specification for maize, but differs by about 2.0 percentage points for rice and 4.5 points for wheat. Assigning pDSSAT zero change

produces a much smaller wheat loss because it attaches pDSSAT's large historical wheat weight to no change at all.

Table B11: Sensitivity of SSP3-7.0 Projections to the Treatment of the Missing pDSSAT Simulations

Crop	Re-estimated weights	Rescaled weights	Zero pDSSAT change
Maize	−16.61	−17.13	−17.13
Rice	−47.79	−49.75	−49.65
Wheat	−10.74	−15.23	−4.73

Notes: Entries are median SSP3-7.0 yield changes (%) for 2096–2100 relative to 2015, pooling three climate models. Re-estimated weights (the primary specification) refit the empirical coefficients, PBM calibrations, and weights using the three components with SSP3-7.0 simulations. Rescaled weights drop pDSSAT from the four-component weights and rescale the rest to sum to one. Zero pDSSAT change keeps the four-component weights and assigns pDSSAT a projected change of zero. Projection assumptions are stated in Section 2.5.4.

A separate calibration check sets the PBM calibration slopes to one when constructing projected changes, keeping the empirical response, ensemble weights, and projection records fixed (Table B12). Under SSP3-7.0, this gives median losses of 19.4% for maize, 49.3% for rice, and 18.0% for wheat, compared with 16.6%, 47.8%, and 10.7% after calibration. The difference is largest for wheat, where the unscaled loss exceeds the empirical model's 16.8%. The calibrated specification uses the scale of the components on which the historical weights were estimated. A slope below one can reflect a scale mismatch as well as noise or differences in weather forcing and aggregation; this sensitivity exercise does not distinguish those explanations or establish which projection scale transfers better to future climate.

B.5.2 Disagreement across model components

Figure B2 classifies each projected location by whether the empirical response and raw EPIC and LPJmL simulated changes agree on the sign of the end-of-century change. This comparison reports agreement on direction, while the main map reports

Table B12: Sensitivity of Hybrid Projections to PBM Rescaling

Crop	Scenario	Calibrated (%)	Unscaled (%)	Difference (pp)
Maize	SSP1-2.6	-2.4	-2.9	-0.5
Maize	SSP3-7.0	-16.6	-19.4	-2.7
Rice	SSP1-2.6	-8.7	-9.2	-0.5
Rice	SSP3-7.0	-47.8	-49.3	-1.5
Wheat	SSP1-2.6	-1.3	-2.9	-1.5
Wheat	SSP3-7.0	-10.7	-18.0	-7.2

Notes: Medians across the same finite subnational-year-climate-model records for 2096–2100 relative to 2015. Unscaled projections set PBM calibration slopes to one while retaining the fitted weights and empirical component. Difference is unscaled minus calibrated, calculated before rounding. Both scenarios have 38,347 maize records, 31,664 rice records, and 31,664 wheat records. These are unweighted medians, not global production changes.

the magnitude of the weighted hybrid projection. The PBM changes in this diagnostic are shown before statistical calibration; they are not the calibrated components used in the hybrid.

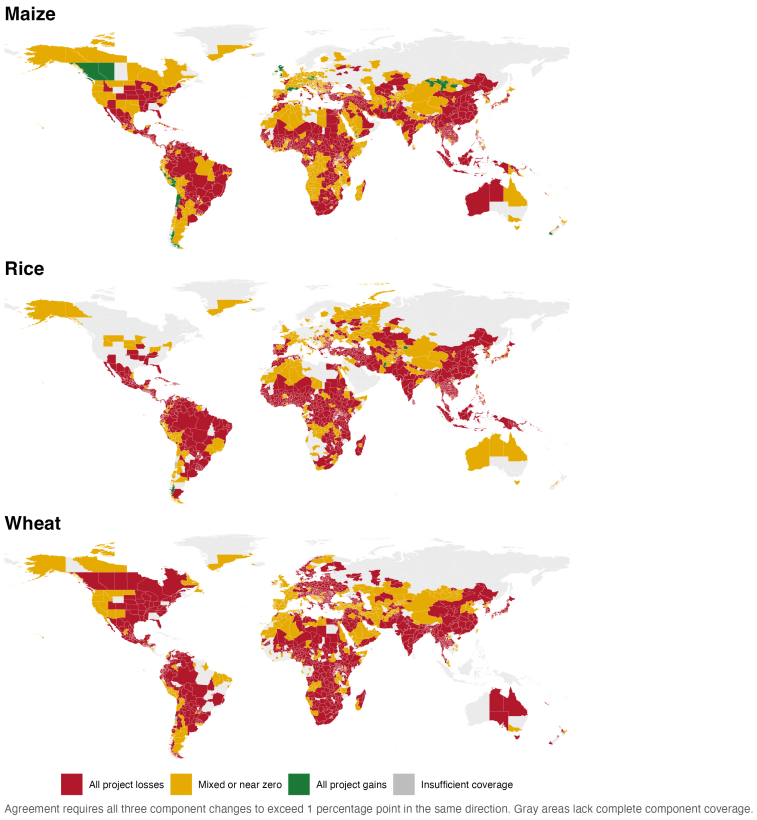


Figure B2: Agreement on the Direction of Projected Yield Changes under SSP3-7.0. Each location is classified by whether the empirical response and raw EPIC and LPJmL simulated changes agree on the sign of the 2096–2100 change relative to 2015, averaged over three climate models. Red marks agreement on losses larger than one percent, green agreement on gains larger than one percent, yellow mixed or near-zero changes, and gray insufficient component coverage. PBM changes are shown before statistical calibration. pDSSAT has no SSP3-7.0 simulations.

B.6 Regional and Country Projection Results

Main-text link: Section 2.6.3, Projected yield changes.

All summaries use 2015 as the baseline and the end-of-century window 2096–2100. Regional results report medians across finite ADM1-year-GCM records, pooling the three retained climate models. Country results first aggregate ADM1 yield levels using crop harvested-area weights and then calculate the change in the resulting country yield; the country table averages those changes across years and climate models. The calibration sample includes finer administrative units, while future PBM outputs are available at ADM1; the results should therefore not be interpreted as estimates of within-ADM1 heterogeneity.

B.6.1 Regional summary

Table B13 reports median changes by continent and crop under SSP3-7.0. Maize hybrid losses are largest in Asia and North America, while Europe has a small loss. Rice losses appear in every region and are largest in North America and Oceania, although Oceania contains relatively few projected units. Wheat losses are smaller than rice losses throughout the regional summaries, but remain negative in each region. These contrasts show what a single median pooled across all projection records conceals.

Table B13: Median Yield Changes by Region under SSP3-7.0, 2096–2100

Maize

Region	Empirical model (%)	Hybrid (%)	ADM1 units
Africa	−36.1	−18.0	663
Asia	−38.4	−21.9	872
Europe	−1.4	−3.7	494
North America	−40.5	−20.5	331
Oceania	−41.5	−17.0	35
South America	−30.6	−15.5	227

Rice

Region	Empirical model (%)	Hybrid (%)	ADM1 units
Africa	−53.5	−50.6	592
Asia	−49.3	−46.5	746
Europe	−18.9	−17.9	323
North America	−78.5	−75.5	216
Oceania	−73.1	−70.0	27
South America	−55.5	−52.4	213

Wheat

Region	Empirical model (%)	Hybrid (%)	ADM1 units
Africa	−22.5	−15.4	541
Asia	−18.3	−11.9	643
Europe	−10.3	−6.3	630
North America	−21.1	−14.0	127
Oceania	−11.9	−7.6	16
South America	−18.2	−13.0	181

Notes: Entries are medians of the percentage change in yield relative to 2015 across finite ADM1-by-year-by-climate-model projection records for 2096–2100. Records from the three retained climate models are pooled without harvested-area weighting. The hybrid uses the three-component weights re-estimated without pDSSAT. Management and atmospheric CO_2 are held at 2015 levels. ADM1 counts describe projection coverage, not the historical estimation sample.

B.6.2 Countries with the largest empirical-model losses

Table B14 ranks ten country–crop pairs by the losses projected by the empirical model. It shows how the hybrid changes those losses. The ranking is specific to the empirical projection; the hybrid may rank these countries differently. All ten combinations are rice, and Haitian rice has the largest empirical loss. The hybrid moderates every loss in this ranking, by 1.4 to 2.4 percentage points, because the empirical component carries most of the rice weight.

Table B14: Country–Crop Pairs with the Largest Empirical-Model Losses under SSP3-7.0

Country	Crop	Empirical model (%)	Hybrid (%)
Haiti	Rice	−94.9	−93.5
Guinea-Bissau	Rice	−93.8	−92.1
Sierra Leone	Rice	−93.1	−91.5
Ghana	Rice	−90.9	−89.1
Cuba	Rice	−90.0	−88.5
Senegal	Rice	−88.2	−86.0
Dominican Republic	Rice	−87.4	−85.4
Guinea	Rice	−87.2	−85.3
Brazil	Rice	−85.6	−83.2
Jamaica	Rice	−85.4	−83.8

Notes: The ten country–crop pairs are selected and ranked by the empirical model's projected loss, not by the hybrid. Each entry first aggregates ADM1 yield levels to the country with crop harvested-area weights, then averages the change relative to 2015 over 2096–2100 and three climate models. Management and atmospheric CO_2 are held at 2015 levels.

Appendix C

ECONOMIC VALUE OF RECREATIONAL CRABBING AND CLAMMING IN THE MID-ATLANTIC REGION

This appendix documents the construction and sensitivity of each activity's recreational day value. It reports bid responses and survey-design checks, then parametric specifications and estimator sensitivity, followed by income construction and sample coverage. The supporting evidence explains how the values are obtained and which respondents they represent.

C.1 Bid Responses and Survey-Design Sensitivity

C.1.1 Bid-Response Distribution

Table C1 supplies the denominators behind Figure 3.3: respondents assigned each cost increment, those accepting it, and the resulting acceptance rate. The cells show both the price response and the number of observations supporting each point.

C.1.2 Response Certainty

The certainty diagnostics describe how confidence varies with the assigned cost and the yes/no answer. Table C2 reports the shares with certainty of 10 and greater than 7 at each bid. Figure C1 uses scores of 9 or 10 to show the pattern across bids separately for yes and no answers. Confidence in a yes generally weakens as the extra cost rises, while confidence in a no strengthens.

Table C3 documents the continuous-adjustment comparison in Figure 3.4. It reports the sample size and binary lower-bound mean and the corresponding

Table C1: Acceptance of Additional Trip Costs

	Clamming			Crabbing		
Bid	N	Yes	% yes	N	Yes	% yes
\$5	165	145	87.9%	534	500	93.6%
\$15	161	123	76.4%	487	397	81.5%
\$25	178	123	69.1%	487	333	68.4%
\$50	181	85	47.0%	492	224	45.5%
\$75	184	44	23.9%	496	151	30.4%
\$150	181	34	18.8%	543	109	20.1%
\$300	177	9	5.1%	492	40	8.1%
Total	1,227	563	45.9%	3,531	1,754	49.7%

Note: Bids are randomly assigned increases in the out-of-pocket cost of the reported trip. Accepting means the respondent would still have taken that trip at the higher cost. N is the number of respondents assigned each bid. Total N is the sum of bid-cell sample sizes; total % accepting is the pooled yes share.

mapped-response index, together with the dollar and percentage difference. Shared respondent bootstrap draws preserve overlapping activity records when calculating the difference's interval. Because the adjusted measure requires a certainty score, the comparison also excludes 28 clammers and 61 crabbers with missing certainty. Its difference is therefore not a pure within-sample adjustment effect.

C.1.3 Prior-Question Order and Bid Comparisons

Table C4 uses 984 respondents with valid bids and yes/no answers for both activities and consistent observed order. Panel A reports lower-bound means separately when each module came first or second. All seven current bids are observed in every original order group. Panel B models binary acceptance with current-bid categories, a second-module indicator, and the absolute preceding bid centered at \$88.57 and scaled by \$100. The prior-bid term is zero for first-module answers; its coefficient describes prior-price sensitivity among second-module answers.

Table C2: Reported Certainty by Bid and Valuation Response

	Clamming				Crabbing			
	No		Yes		No		Yes	
Bid	$c = 10$	$c > 7$	$c = 10$	$c > 7$	$c = 10$	$c > 7$	$c = 10$	$c > 7$
\$5	35%	55%	50%	81%	36%	61%	55%	82%
\$15	31%	69%	38%	71%	45%	73%	41%	73%
\$25	39%	65%	36%	72%	40%	69%	36%	70%
\$50	41%	77%	41%	77%	42%	74%	41%	75%
\$75	40%	80%	30%	61%	46%	81%	31%	68%
\$150	58%	84%	30%	58%	51%	83%	43%	80%
\$300	63%	86%	14%	57%	62%	87%	20%	52%

Note: Share of respondents in each bid-and-response cell reporting certainty 10 ($c = 10$) or certainty greater than 7 ($c \in \{8, 9, 10\}$) on the 10-point follow-up slider. Percentages use respondents in each bid-and-response cell who reported certainty; some certainty scores are missing. At low bids, yes responses are more often high-certainty than no responses; at high bids the pattern reverses. Certainty screens and continuous p_{adj} adjustments in the main text are sensitivity analyses, not validated corrections for hypothetical bias.

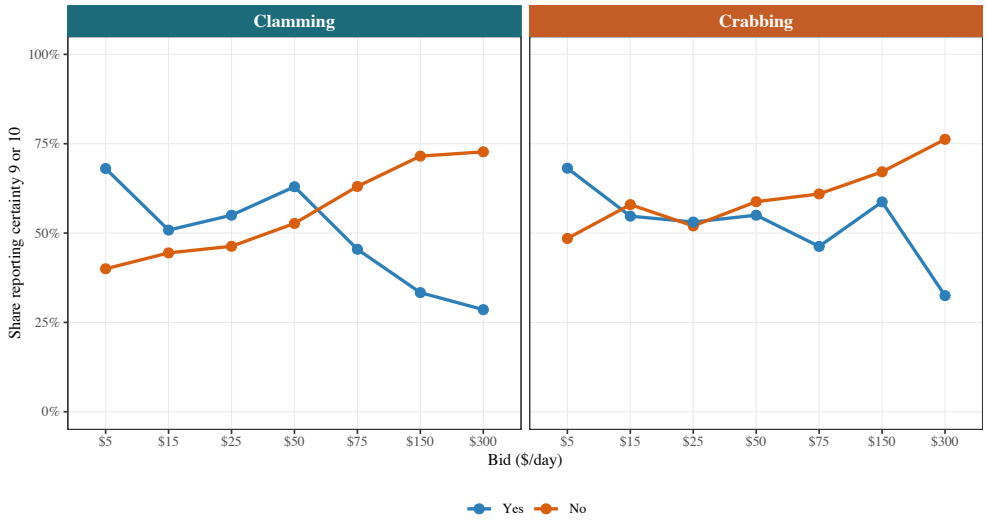


Figure C1: High-Certainty Answers by Additional Trip Cost

Note: Lines show the share reporting certainty of 9 or 10 within each bid and answer group. The underlying samples contain 1,199 clammers and 3,470 crabbers with a valid bid, valuation answer, and certainty score. Each share uses the respondents giving the indicated yes or no answer at that bid as its denominator. These scores differ from the certainty thresholds tabulated in Table C2.

Table C3: Lower-Bound Day-Value Indices with Continuous Certainty Mapping

	Unadj. N	Unadj. WTP [95% CI]	Adj. N	Adj. WTP [95% CI]	Difference [95% CI]	% change
Clamming	1227	\$58.38 [51.62, 65.52]	1199	\$64.97 [58.87, 71.09]	\$6.60 [2.85, 9.92]	11.3
Crabbing	3531	\$65.91 [61.68, 70.71]	3470	\$72.73 [68.77, 77.00]	\$6.82 [5.52, 8.16]	10.3

Note: Unadjusted Turnbull uses binary yes/no responses. Certainty-adjusted Turnbull replaces each response with the continuous acceptance probability p_{adj} described in the main text. Difference = mapped − full-sample binary. The mapped samples omit 28 clammers and 61 crabbers with missing certainty, so the difference includes a sample change. A mapped-response index is not a calibrated WTP mean without additional assumptions. Confidence intervals for the difference use 1,000 respondent-clustered bootstrap resamples (shared respondent draws across activities).

Randomized module order defines the comparison independently of the current bid, preserving a common offered cost range in both groups. The interpretation remains conditional on completion of both valuation questions; the checks do not correct for possible selection into completion.

C.2 Parametric Estimates and Estimator Sensitivity

C.2.1 Parametric Model Coefficients

Table C5 gives the full specifications behind the linear-bid summaries in Table 3.2. Columns separate binary and mapped responses by activity, using the activity-order specification. The table shows how the assigned cost and participant characteristics enter each model, including fishing-method controls for crabbing. The characteristic coefficients describe conditional associations; the randomly assigned bid provides the price variation for valuation.

Lifetime activity categories (never, 1–5, 6–25, 26–50, 51–100, and more than 100 days) are coded as 0, 3, 15.5, 38, 75.5, and 150 days. The six hours-per-day categories are coded as 1, 2.5, 3.5, 4.5, 5.5, and 7 hours. These assigned values summarize

Table C4: Valuation Responses by Randomized Module Order

Panel A: Lower-bound means among dual-activity respondents			
Activity / module order	N	Value ($/day)	95% CI
Clamming: First	504	$63.56	[51.91, 76.96]
Clamming: Second	480	$49.17	[40.08, 57.31]
Crabbing: First	480	$62.24	[50.87, 74.57]
Crabbing: Second	504	$68.11	[56.70, 80.35]
Panel B: Binary logit, dual-activity respondents			
Activity / regressor	Coefficient	HC3 SE	p-value
Clamming: Second module	-0.344	(0.164)	0.036
Clamming: Prior bid (+$100)	0.239	(0.114)	0.036
Crabbing: Second module	0.308	(0.163)	0.058
Crabbing: Prior bid (+$100)	0.199	(0.132)	0.133

Both panels use the same 984 respondents with valid bids and binary answers for both activities and consistent observed module order. Panel A uses 1,000 within-order respondent bootstrap draws; all seven current bids occur in every original order group. It does not group respondents by the rank of their current and preceding bids.
Each Panel B model includes current-bid categories, a second-module indicator, and the preceding bid centered at the mean of the seven offered bids ($88.57) and divided by 100. The preceding-bid term equals zero when the activity comes first. The second-module coefficient is evaluated at that reference prior bid; the prior-bid coefficient describes a $100 increase when the activity comes second. Coefficients are log-odds changes, not dollar WTP effects. HC3 standard errors appear in parentheses; each model has $N = 984$. These checks concern completed dual-activity responses and do not isolate a psychological mechanism.

survey categories rather than directly observed continuous quantities. Location and gear questions allow multiple selections; each indicator compares selection with nonselection, conditional on the other controls. Household income uses reported category values, with predictions for non-reporters and a separate imputation indicator.

C.2.2 Log Bid and Log Income

Table C6 varies the functional form of the assigned bid and household income while retaining the activity-order control set. Income enters in thousands of dollars in the linear specification. The alternatives log the bid, income, or both, and evaluate median willingness to pay at the corresponding estimation-sample means.

Table C5: Bid Acceptance Models with Participant and Order Controls

	Clamming		Crabbing	
	Binary	Mapped	Binary	Mapped
Constant	0.458	0.372	0.160	0.105
	(0.419)	(0.338)	(0.232)	(0.185)
Bid ($/day)	-0.019***	-0.015***	-0.018***	-0.014***
	(0.002)	(0.001)	(0.001)	(0.001)
Other activity first (=1)	-0.376***	-0.305***	0.112	0.072
	(0.144)	(0.116)	(0.107)	(0.087)
Income ($1,000)	0.003*	0.002**	0.002***	0.002***
	(0.001)	(0.001)	(0.001)	(0.001)
Lifetime clamming days	0.003**	0.002**		
	(0.001)	(0.001)		
Lifetime crabbing days			0.001	0.001*
			(0.001)	(0.001)
Age	-0.006	-0.005	-0.004	-0.003
	(0.005)	(0.004)	(0.003)	(0.002)
Male (=1)	0.003	-0.045	0.272***	0.177**
	(0.185)	(0.147)	(0.099)	(0.080)
Boat owner (=1)	0.036	0.079	0.096	0.069
	(0.153)	(0.121)	(0.089)	(0.072)
Hours/day: clamming	0.260***	0.205***		
	(0.067)	(0.052)		
Hours/day: crabbing			0.212***	0.170***
			(0.031)	(0.024)
Indian River Bay (=1)	0.045	0.044	-0.059	-0.038
	(0.202)	(0.160)	(0.099)	(0.080)
Rehoboth Bay (=1)	0.111	0.069		
	(0.217)	(0.169)		
Delaware Bay (=1)			-0.071	-0.063
			(0.104)	(0.084)
Pot gear (=1)			0.157*	0.138**
			(0.086)	(0.069)
Trotline gear (=1)			0.289**	0.238**
			(0.121)	(0.095)
Income imputed (=1)	-0.010	0.013	-0.173	-0.176
	(0.312)	(0.245)	(0.178)	(0.149)
N	1178	1156	3379	3324

Binary models use the yes/no valuation answer; mapped models use p_{adj} in fractional logit. All four specifications include the activity-order indicator. HC3 robust standard errors appear in parentheses. Income includes predicted values and a separate imputation indicator. Location and gear indicators allow multiple selections. Covariate coefficients describe associations.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The rows distinguish cost sensitivity from income sensitivity. Logging income alone leaves the medians nearly unchanged. Logging the bid reduces them to about $38 for clamming and $45 for crabbing, whether income enters in levels or logs. The check therefore locates the main functional-form sensitivity in the price response. The linear-bid row supplies a reference for these alternatives. Table C7 compares AIC on the same samples and control set; the log-bid form fits better by this criterion. Neither form is selected by proximity to the nonparametric mean.

Table C6: Median Day Values under Alternative Cost and Income Specifications

Specification	Clamming median WTP [95% CI]	Crabbing median WTP [95% CI]
Linear bid, linear income (reference)	$60.25 [52.70, 68.37]	$73.23 [68.22, 78.76]
Log bid, linear income	$38.31 [33.85, 42.76]	$45.48 [42.61, 48.55]
Linear bid, log income	$60.24 [53.48, 69.65]	$73.24 [68.36, 79.69]
Log bid, log income	$38.28 [33.46, 43.01]	$45.48 [42.55, 48.77]

Note: Binary logit models with the activity-order covariate set used in the main text. Median willingness to pay is evaluated at estimation-sample means. Confidence intervals use 1,000 Krinsky–Robb draws with HC3 covariance. The first row is the linear-bid reference specification.

Table C7: Fit of Linear- and Log-Bid Binary Models

Activity / bid form	N	AIC
Clamming: linear bid	1178	1237.9
Clamming: log bid	1178	1182.6
Crabbing: linear bid	3379	3497.2
Crabbing: log bid	3379	3196.5

Each pair uses the same observations, activity-order indicator, and participant controls as Table C5; only the current-bid transformation changes. Lower AIC indicates better in-sample fit after penalizing parameter count. This comparison does not establish out-of-sample performance or select the preferred nonparametric lower-bound mean.

C.2.3 Models without Prior Exposure

Table C8 reports the linear-bid models without a preceding activity module. Its binary estimates are conditional medians; its fractional estimates are 50-percent thresholds of the mapped response. The sample restriction changes participant composition, and overlapping intervals alone would not establish equivalence with the full-sample models.

Table C8: Parametric Values without a Preceding Activity Bid

Activity / statistic	N	Value ($/day)	95% CI
Clamming: Binary median	716	$68.41	[58.71, 79.99]
Clamming: Mapped threshold	699	$69.91	[60.10, 81.20]
Crabbing: Binary median	2,702	$71.55	[65.87, 77.19]
Crabbing: Mapped threshold	2,655	$73.52	[68.11, 79.06]

The sample retains respondents whose activity module came first or whose order record indicates no other module. Models use the participant controls in Table C5 without its order indicator. Binary results are conditional WTP medians; mapped results are 50-percent thresholds of the certainty-adjusted response. Values use sample-mean covariates and 1,000 Krinsky–Robb draws with HC3 covariance. Excluding prior exposure also changes sample composition.

C.2.4 Certainty as a Covariate

Table C9 examines a third use of the certainty follow-up. It retains the binary answer, adds the reported score as a logit covariate, and evaluates the median first at mean certainty and then at certainty of 10. Other covariates remain at estimation-sample means. This conditional prediction differs from the main text's threshold screen, which removes less-certain respondents.

The reference models use 1,178 clammers and 3,379 crabbers; adding certainty reduces these samples to 1,156 and 3,324, respectively. The comparison therefore combines a change in specification with the exclusion of missing certainty scores.

Certainty is reported after the valuation answer, so its coefficient describes a conditional association rather than a causal effect on willingness to pay. Adding certainty changes the median little at the mean score. Predicting at certainty of 10 lowers the medians by a few dollars. The certainty coefficient is small and imprecisely estimated for clamming and modestly negative for crabbing. The table supports a narrow conclusion about this covariate specification; it does not replace the comparisons based on response mapping and sample restriction.

Table C9: Certainty as a Covariate in the Binary Logit

	Clamming median WTP [95% CI]	Crabbing median WTP [95% CI]
Linear bid, no certainty covariate	\$60.25 [52.70, 68.37]	\$73.23 [68.22, 78.76]
With certainty, at mean certainty	\$59.49 [52.45, 67.78]	\$73.38 [68.25, 79.31]
With certainty, at certainty = 10	\$57.80 [49.20, 67.09]	\$70.26 [64.29, 76.68]
Certainty coefficient (HC3 SE)	-0.024 (0.042)	-0.040 (0.024)

Note: Binary logit models with the activity-order covariate set. Rows 2–3 add reported response certainty (1–10) as a covariate and evaluate median willingness to pay at the estimation-sample mean of the other covariates. Confidence intervals use 1,000 Krinsky–Robb draws. The certainty coefficient is from the model underlying rows 2–3.
Note: The reference models use 1,178 clammers and 3,379 crabbers. Models with certainty use 1,156 and 3,324, respectively; the comparison includes the exclusion of respondents with missing certainty. The reference intervals are those reported in the main table. Certainty is recorded after the valuation answer, so this is a descriptive specification check. HC3 covariance is used for the Krinsky–Robb draws.

C.2.5 Zero-Increment Interpolation

The preferred Turnbull estimate uses $t_0 = 0$ but leaves acceptance at zero unrestricted, giving the first interval a contribution of $t_1\hat{p}_1$. Because the respondents took the reported trip, an alternative sets acceptance at a zero increment to one. Table C10 linearly interpolates between $(0, 1)$ and $(t_1, \hat{p}_1)$ on $[0, t_1]$, while leaving all subsequent intervals unchanged.

That change adds about $0.30 per clamming day and $0.16 per crabbing day. The first-interval convention therefore accounts for little of the preferred value. This check concerns the lower endpoint; it supplies no information on the unobserved willingness-to-pay tail above the largest offered bid.

Table C10: Sensitivity of Lower-Bound Mean Day Values to the Zero-Cost Convention

	Turnbull lower bound [95% CI]	Zero-increment variant [95% CI]	Difference [95% CI]
Clamming	$58.38	$58.68	$0.30
	[51.62, 65.52]	[51.99, 65.70]	[0.19, 0.44]
Crabbing	$65.91	$66.07	$0.16
	[61.68, 70.71]	[61.55, 70.57]	[0.11, 0.22]

Note: The Turnbull lower bound uses $t_0 = 0$ and observed acceptance at each bid. The zero-increment variant sets acceptance at a zero cost increase to one, because the trip was taken, and replaces the contribution on $[0, t_1]$ with linear interpolation between $(0, 1)$ and $(t_1, \hat{p}_1)$; remaining intervals follow the Turnbull construction. Confidence intervals use 1,000 respondent bootstrap resamples.
The reference intervals reproduce the main table. Intervals for the alternative and the difference use separately generated paired bootstrap draws; the difference is calculated within each draw.

C.3 Data Construction and Sample Coverage

C.3.1 Income Imputation

Table C11 describes the model used to predict household income for 55 clamming and 177 crabbing valuation records without reported income (4.5 and 5.0 percent of the respective samples). Income remains missing for 20 clamming and 70 crabbing records; these records are excluded from models requiring income. The OLS model uses income reporters, with category midpoints as the dependent variable. Predictors include zip-code average adjusted gross income from the 2022 IRS Statistics of Income zip-code data, education, gender, age, and employment status. The income-category values are $25,000, $62,500, $87,500, $125,000, and $175,000, with the last assigned

to the open top category. Predictions retain some non-reporters in full-covariate models, which also include an imputation indicator. Model intervals condition on these predicted incomes and do not propagate first-stage imputation uncertainty.

Table C11: Income Imputation Regression

Variable	Coefficient	HC3 SE	t-stat	p-value
Constant	22,371.0***	(4,911.4)	4.55	<0.001
Zip-code average AGI ($)	0.192***	(0.023)	8.19	<0.001
Education (1–5)	14,408.4***	(740.5)	19.46	<0.001
Male (=1)	14,748.4***	(2,068.9)	7.13	<0.001
Age	462.5***	(76.6)	6.04	<0.001
Employment: Part-time	-25,143.1***	(3,820.6)	-6.58	<0.001
Employment: Retired	-34,346.7***	(2,187.5)	-15.70	<0.001
Employment: Student	-10,303.9	(12,196.9)	-0.84	0.398
Employment: Other	-31,861.5***	(6,236.6)	-5.11	<0.001
N (reporters)	3,371			
R^2	0.230			
Adj. R^2	0.229			
RMSE	$44,346			

Dependent variable: household income category midpoint (dollars), with the open top category (over $150,000) coded at $175,000. The 3,371 unique income reporters include dual-activity participants once. Zip-code average AGI is 2022 IRS total adjusted gross income divided by tax returns. Education ranges from 1 (less than high school) to 5 (graduate/professional); full-time employment is the reference category. HC3 robust standard errors appear in parentheses. Predictions impute missing income.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C.3.2 Sample Composition on Observables

Table C12 describes the people represented by the valuation estimates. It reports age, gender, income, and boat ownership for each final activity sample, then compares single- and dual-activity participants and pooled Delaware and out-of-state respondents. These are licensed participants who completed the survey and activity questions, rather than a random sample of Delaware adults.

Respondents are mostly male, relatively high income, and have high rates of boat ownership. Within each activity, single- and dual-activity participants are similar on age, gender, and income, while boat ownership is higher among dual-activity participants. Out-of-state respondents are more likely than Delaware residents to report household income of at least $75,000. The chapter's preferred estimates use the observed valuation samples without reweighting on these characteristics.

Table C12: Characteristics of the Valuation Samples

Sample	N	Mean age	Male (%)	Household income $\geq \$75,000$ (%)	Boat owner (%)
All clamming respondents	1227	57.5	81.2	74.4	63.9
Clamming, dual-activity	984	57.1	82.2	74.0	68.6
Clamming, single-activity	243	59.0	77.0	76.1	44.6
All crabbing respondents	3531	56.6	80.2	73.8	55.1
Crabbing, dual-activity	984	57.1	82.2	74.0	68.6
Crabbing, single-activity	2547	56.4	79.4	73.7	49.9
All unique respondents	3774	56.7	80.0	74.0	54.4
Pooled, Delaware residents	2006	57.1	81.5	69.5	56.8
Pooled, out of state	1768	56.3	78.3	78.9	51.7

Note: Contingent-behavior (CB) samples require a non-missing bid and yes/no response. Dual-activity respondents appear in both activity samples. Income uses reported category midpoints among reporters. The samples are licensed recreational participants who completed the valuation questions. Delaware-resident shares in the CB samples are 48% (clamming) and 54% (crabbing). Statistics exclude missing answers to each characteristic; N counts all valuation respondents in the row. In the full clamming sample, valid counts are 1,205 for age, 1,210 for gender, 1,152 for reported income, and 1,218 for boat ownership. Corresponding crabbing counts are 3,449, 3,474, 3,284, and 3,511. Subgroup statistics likewise use available answers. Income is household income; the threshold is based on reported income categories.